

Supplemental Materials

Molecular Biology of the Cell

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Ensembl ID	Description	Gene Name	OMIM number	Curator's Note	PubMed IDs	Localisation	First order	Second order	In SCGSv1	Predicted in SCGSv1 paper	Predicted in CiliaCarta
ENSG00000125257	ATP binding cassette subfamily C member 4(ABCC4)	ABCC4		Also known as leakytail. ATP-binding cassette transporter which transports prostaglandin E2 and other molecules across cellular membranes. Mutations in ABCC4 cause ciliogenesis defects. ABCC4-mediated PGE2 signalling acts through a ciliary G-protein-coupled receptor, EP4, to upregulate cAMP synthesis and increase anterograde IFT, thereby promoting ciliogenesis (PMID: 25173977). Zebrafish abcc4 mutants exhibited loss of connecting cilia and outer segment in photoreceptor cells, leading to mislocalization of opsins in the cell bodies of photoreceptors (PMID: 30685088). Specific missense associated with polydactyly in pigs (PMID: 32228435)	PMID: 25173977; PMID: 30685088; PMID: 32228435	Membrane		x			x
ENSG00000099204	actin binding LIM protein 1(ABLIM1)	ABLIM1		Involved in ciliation-related actin dynamics and LR patterning (PMID: 22684256). Asymmetrically expressed in both the node and left lateral plate mesoderm (LPM), similar to Nodal (PMID: 20487527)	PMID: 22684256; PMID: 20487527	actin cytoskeleton		x			x

ENSG00000173210	actin binding LIM protein family member 3(ABLIM3)	ABLIM3		Involved in ciliation-related actin dynamics and LR patterning (PMID: 22684256)	PMID: 22684256	actin cytoskeleton		x			x
ENSG00000130234	angiotensin I converting enzyme 2(ACE2)	ACE2		ACE2 receptor protein robustly localizes within the membrane of the motile cilia of airway epithelial cells, which likely represents the initial or early subcellular site of SARS-CoV-2 viral entry during host respiratory transmission	PMID: 33116139	Motile cilium membrane	x				
ENSG00000138071	ARP2 actin related protein 2 homolog(ACTR2)	ACTR2		Also known as ARP2. Target of miR-129-3p which regulates actin dynamics to regulate ciliation in cells and in zebrafish (PMID: 22684256)	PMID: 22684256	actin cytoskeleton		x			x
ENSG00000151694	ADAM metallopeptidase domain 17(ADAM17)	ADAM17		Also known as tumor necrosis factor (TNF)- α -converting enzyme (TACE). Metalloprotease, cleaves Notch receptor when glycosylated by GALNT11 to control formation of motile and non-motile cilia at the node. Heterotaxy gene (PMID: 24226769). Cleaves ACE2, found in respiratory cilia membrane (PMID: 15983030)	PMID: 24226769; PMID: 15983030	Cytoplasm		x			
ENSG00000173157	ADAM metallopeptidase with thrombospondin type 1 motif 20(ADAMTS20)	ADAMTS20		Secreted metalloproteases ADAMTS9 and ADAMTS20 have a non-canonical role in ciliary vesicle growth during ciliogenesis (PMID: 30814516)	PMID: 30814516	periciliary vesicles	x				
ENSG00000163638	ADAM metallopeptidase with thrombospondin type 1 motif 9(ADAMTS9)	ADAMTS9		Causes NPHP-related ciliopathy (PMID: 30609407). Has a non-canonical role in ciliary vesicle growth during ciliogenesis (PMID: 30814516)	PMID: 30609407; PMID: 30814516	periciliary vesicles	x				

ENSG00000138031	adenylate cyclase 3(ADCY3)	ADCY3		Localized to the plasma membrane of neuronal cilia (PMID: 17924533, PMID: 23392690)	PMID:17924533, PMID:23392690	Ciliary Membrane	x		x		x
ENSG00000173175	adenylate cyclase 5(ADCY5)	ADCY5	600293	AC5 and AC6 concentrate in the primary cilium, and mutation of a previously undescribed cilium-targeting motif in AC5 suppresses its ciliary location, as well as its capacity to inhibit Hh signalling. Mutated in familial dyskinesia	PMID: 26092933	Ciliary membrane	x				x
ENSG00000174233	adenylate cyclase 6(ADCY6)	ADCY6	600294	AC5 and AC6 concentrate in the primary cilium, and mutation of a previously undescribed cilium-targeting motif in AC5 suppresses its ciliary location, as well as its capacity to inhibit Hh signalling. Mutated in lethal contracture syndrome	PMID: 26092933	Ciliary membrane	x				x
ENSG00000164199	adhesion G protein-coupled receptor V1(ADGRV1)	ADGRV1		GPR98. USH2C OMIM 602851. Mutation causes Usher syndrome type IIC	PMID:15671307	Connecting Cilium, Stereocilium	x		x		x

ENSG00000084693	ATP/GTP binding protein like 5(AGBL5)	AGBL5	615900	CCP5. Tubulin deglutamylase. Knockdown in zebrafish leads to increase in glutamylation of axonemal microtubules (PMID: 24743595). Knockdown can rescue cilia defects in other zebrafish cilia mutants (PMID: 24743595). Biallelic missense mutations cause autosomal recessive retinitis pigmentosa (OMIM:615900) (PMID: 26355662; PMID: 26720455; PMID: 27764769; PMID: 27842159). Causes structural defects in sperm flagella and spermatids with supernumerary centrioles leading to male infertility in mutant mice (c)	PMID: 24743595; PMID: 26355662; PMID: 26720455; PMID: 27764769; PMID: 27842159; PMID: 26355662; PMID: 26720455; PMID: 27764769; PMID: 27842159	No published localisation studies					
ENSG00000173467	anterior gradient 3, protein disulphide isomerase family member(AGR3)	AGR3		AGR3 expression is restricted to ciliated cells in the airway epithelium. Required for Regulation of Ciliary Beat Frequency in the Airway.	PMID: 25751668	ER		x			x
ENSG00000135541	Abelson helper integration site 1(AHI1)	AHI1	608894	Mutated in Joubert syndrome (type 3) OMIM 608894, PMID:16453322. At mother centriole/basal body. Interacts with B9 complex of transition zone proteins including B9D1, B9D2, MKS1, TCTN1, TCTN2 PMID:22179047. Role in RAB8-dependent ciliary membrane targeting PMID:19625297.	PMID:16453322 PMID:22179047 PMID:19625297	Ciliary membrane	x		x		x

ENSG00000129221	aryl hydrocarbon receptor interacting protein like 1(AIPL1)	AIPL1	604392	Mutated in LCA. Localises to photoreceptor cilium, interacts with EB1	PMID: 25799540	Photoreceptor connecting cilium	x				
ENSG00000140057	adenylate kinase 7(AK7)	AK7	615364	Ultrastructural defects in 9+2 cilia, leading to PCD phenotype in mutant mice PMID:18776131. Role in ATP production to power ciliary beating. Reduced expression levels in PCD patients PMID:20537283. 1 single possibly pathogenic variant found in AK7 in 1 PCD patient PMID: 22801010. Homozygous missense mutation L673P in adenylate kinase 7 (AK7) leads to primary male infertility and multiple morphological anomalies of the flagella but not to primary ciliary dyskinesia. Mutations lead to spermatogenic failure in humans	PMID:18776131 PMID:20537283 PMID:22801010; PMID: 30517620	Sperm flagella and motile cilium	x		x		x
ENSG00000165695	adenylate kinase 8(AK8)	AK8		Hydrocephalus due to dysfunctional motile cilia in KO mouse PMID: 21746835	PMID:21746835; PMID: 24740601	Sperm flagella axoneme	x		x		x
ENSG00000142208	AKT serine/threonine kinase 1(AKT1)	AKT1		Neddylation inhibitor MLN4924 suppresses cilia formation by modulating AKT1. Regulates a Rab11-Effector Switch Required for Ciliogenesis. PMID: 31204173	PMID: 30850948	Cytoplasm/membrane		x			x

ENSG00000116127	ALMS1, centrosome and basal body associated protein(ALMS1)	ALMS1	606844	Mutated in Alstrom syndrome (type 1) OMIM 606844, PMID: 11941369, PMID: 11941370. Also mutated in 1 LCA family PMID:21901789. siRNA knockdown leads to shortened cilia and inability of cilia to respond to mechanical stimuli PMID:17206865.	PMID: 11941369, PMID: 11941370, PMID:15855349, PMID:21901789, PMID:17206865	Basal Body, Cilium	x		x		x
ENSG00000073331	alpha kinase 1(ALPK1)	ALPK1		missense pathogenic variant in five families leads to ROSAH syndrome, an ocular multisystem autosomal dominant disorder.	PMID: 30967659	Basal body of connecting cilium	x				
ENSG00000176248	anaphase promoting complex subunit 2(ANAPC2)	ANAPC2		Regulates ciliary length and disassembly of the primary cilium.	PMID: 25139956	Basal body	x				
ENSG00000107890	ankyrin repeat domain 26 (ANKRD26)	ANKRD26		Ankrd26 ^{-/-} mice display hyperphagia and have defects in primary cilia in regions of the central nervous system that control appetite and energy homeostasis (PMID: 24633808). With TALPID3 selectively orchestrates FBF1 localization and cilia gating (PMID: 32366837)	PMID: 24633808; PMID: 30824690; PMID: 32366837	Centriole distal appendage	x				
ENSG00000168096	ankyrin repeat and sterile alpha motif domain containing 3(ANKS3)	ANKS3		Interacts with ANKS6 through sterile alpha motif (SAM), localised to renal cilia in mice	PMID: 26327442	Renal cilia	x				

ENSG00000165138	ankyrin repeat and sterile alpha motif domain containing 6(ANKS6)	ANKS6	615370	Ciliopathy gene NPHP16. Found in Inv compartment of cilia, where it undergoes phosphorylation PMID: 29395339	PMID: 23793029, PMID: 29395339	Ciliary tip	x					x
ENSG00000131620	anoctamin 1(ANO1)	ANO1		TMEM16A. Ca2+-activated Cl- channel .Regulates primary ciliogenesis.	PMID: 24694595	Primary cilia membrane	x					
ENSG00000047617	anoctamin 2(ANO2)	ANO2		Localises to cilia of olfactory epithelium.	PMID: 19561302	Cilia of olfactory epithelium.	x					x
ENSG00000168374	ADP ribosylation factor 4(ARF4)	ARF4		Involved in ciliary targeting of rhodopsin and other proteins with VxPx motif. PMID:19153612. Targets proteins to cilia in association with ASAP1, RAB11 and RAB8 PMID:22983554. Dispensable for ciliogenesis	PMID:19153612, PMID:22983554	Trans-Golgi Network		x		x		x
ENSG00000147256	Rho GTPase activating protein 36	ARHGAP36		Patched1-ArhGAP36-PKA-Inversin axis determines the ciliary translocation of Smoothened for Sonic Hedgehog pathway activation.	PMID: 30598432	Mother centriole		x				

ENSG00000169379	ADP ribosylation factor like GTPase 13B(ARL13B)	ARL13B	608922	Mutated in Joubert Syndrome (type 8) OMIM 608922, PMID: 18674751. In proximal ciliary membrane PMID:20231383. Role in IFT and therefore ciliogenesis PMID:20530210. Necessary for normal Shh PMID:21976698 and INPP5E signalling PMID:23150559 and normal expression of Nodal and Cerl2 PMID:22554696.	PMID:18674751, PMID:20231383, PMID:20530210, PMID:21976698, PMID: 23150559, PMID:22554696	Basal Body, Cytosol	x		x		x
ENSG00000213465	ADP ribosylation factor like GTPase 2(ARL2)	ARL2		Regulates cilia stability and development of outer segments in rod photoreceptor neurons.	PMID: 30446707	Photoreceptor connecting cilium	x				
ENSG00000102931	ADP ribosylation factor like GTPase 2 binding protein(ARL2BP)	ARL2BP	615407	Needed for normal photoreceptor cilia doublets and outer segment structure.	PMID: 29718757	Photoreceptor inner segment		x			x
ENSG00000138175	ADP ribosylation factor like GTPase 3(ARL3)	ARL3		interactor of RP2 PMID: 12417528 and PDEdelta PMID:10518933. Involved in trafficking of vesicles from the Golgi to the cilium PMID:20106869, especially; farnesylated cargo PMID:22002721 in association with PDEdelta and RPGR PMID:23559067; and myristoylated cargo in association with UNC119 and RP2 PMID:22085962 ARL3 Mutations Cause Joubert Syndrome by Disrupting Ciliary Protein Composition PMID: 30269812	PMID:12417528, PMID:10518933, PMID:20106869, PMID:22002721, PMID:23559067, PMID:22085962; PMID: 30269812	Cytosol		x	x		x

ENSG00000113966	ADP ribosylation factor like GTPase 6(ARL6)	ARL6	608845	Mutated in Bardet-Biedl syndrome (type 3) and retinitis pigmentosa (type 55) OMIM 608845, PMID:19956407, PMID:15314642. At the ciliary gate, regulates Wnt signalling PMID:20207729. Functions with BBSome to coat proteins for trafficking to the cilium PMID:20603001	PMID:19956407, PMID:15314642, PMID:20207729, PMID:20603001	Basal Body, Cilium, Cytosol, Transition Zone	x		x		x
ENSG00000118690	armadillo repeat containing 2(ARMC2)	ARMC2		Required for sperm flagellum formation. Bi-allelic Mutations in ARMC2 Lead to Severe Astheno-Teratozoospermia Due to Sperm Flagellum Malformations in Humans and Mice.	PMID: 30686508	Not reported					
ENSG00000169126	armadillo repeat containing 4(ARMC4)	ARMC4	615408	PCD with randomisation of body axis PMID: 23849778. Causes outer dynein arm defects in motile cilia PMID: 24203976	PMID: 23849778; PMID: 24203976	Motile cilium axoneme and basal body	x			x	x
ENSG00000135931	armadillo repeat containing 9(ARMC9)	ARMC9	617612	Ciliopathy gene JBTS30 PMID:28625504. Interacts with coiled-coil domain containing 66 (CCDC66) and TOG array regulator of axonemal microtubules 1 (TOGARAM1) PMID: 32453716. Regulator of b-tubule length PMID: 30217954	PMID:28625504; PMID: 32453716; PMID: 30217954	Primary cilium basal body	x				
ENSG00000153317	ArfGAP with SH3 domain, ankyrin repeat and PH domain 1(ASAP1)	ASAP1		Targets proteins, such as rhodopsin and polycystin-1, to cilia in association with ARF4, RAB11 and RAB8 PMID:19153612, PMID:21775626. Provides a platform to regulate Arf4- and Rab11-Rab8-mediated ciliary cargo trafficking PMID:22983554	PMID: 19153612, PMID:21775626, PMID:22983554	Trans-Golgi Network		x	x		x
ENSG00000137343	alpha tubulin acetyltransferase 1(ATAT1)	ATAT1		Major alpha-tubulin K40 acetyltransferase alphaTAT1. Promotes rapid ciliogenesis and efficient mechanosensation.	PMID: 21068373; PMID: 23720746	Microtubule cytoskeleton		x			x

ENSG00000144848	autophagy related 3(ATG3)	ATG3		OFD1 at the centriolar satellites is rapidly degraded by autophagy upon serum starvation. In autophagy-deficient Atg5 or Atg3 null mouse embryonic fibroblasts, OFD1 accumulates at centriolar satellites, leading to fewer and shorter primary cilia and a defective recruitment of BBS4 (Bardet-Biedl syndrome 4) to cilia.	PMID: 24089205	Cytoplasm. Discrete punctae along the ciliary axoneme and base of the ciliary axoneme	x					x
ENSG00000057663	autophagy related 5(ATG5)	ATG5	604261	OFD1 at the centriolar satellites is rapidly degraded by autophagy upon serum starvation. In autophagy-deficient Atg5 or Atg3 null mouse embryonic fibroblasts, OFD1 accumulates at centriolar satellites, leading to fewer and shorter primary cilia and a defective recruitment of BBS4 (Bardet-Biedl syndrome 4) to cilia. Mutations cause spinocerebellar ataxia 25	PMID: 24089205	Cytoplasm. Discrete punctae along the ciliary axoneme and base of the ciliary axoneme	x					x
ENSG00000166454	ATM interactor(ATMIN)	ATMIN		Regulates ciliogenesis. Interacts with DYNLL1 (PMID: 25294941)	PMID: 25294941	Nucleus (in discrete foci in G1)		x				x
ENSG00000172238	atonal bHLH transcription factor 1(ATOH1)	ATOH1		Controls Primary Cilia Formation to Allow for SHH-Triggered Granule Neuron Progenitor Proliferation.	PMID: 30695697	Nucleus		x				
ENSG00000105675	ATPase H+/K+ transporting alpha subunit(ATP4A)	ATP4A		ATP4a and canonical Wnt signaling are required in the epidermis for expression of foxj1, a transcriptional regulator of motile ciliogenesis.	PMID: 25848696	Cell membrane		x				

ENSG00000130638	ataxin 10(ATXN10)	ATXN10	611150	Mutated in 1 NPHP family (splice-site mutation). Forms complex with NPHP5 and 6 at the basal body PMID: 21565611 Mutated in spinocerebellar ataxia 10	PMID:21565611	No data			x		x
ENSG00000087586	aurora kinase A(AURKA)	AURKA		kinase at the basal body of cilia. Phosphorylates HDAC6 to promote cilia disassembly	PMID: 17604723	Centrosome, basal body, spindle pole	x				x
ENSG00000108641	B9 domain containing 1(B9D1)	B9D1	611951	Mutated in Meckel-Gruber syndrome (type 9) OMIM 614144 although mutations only found in 1 family PMID:21493627. Interacts with B9D2 and MKS1 at the transition zone PMID:22179047	PMID:21493627, PMID:21763481, PMID:22179047	Transition Zone	x		x		x
ENSG00000123810	B9 protein domain 2(B9D2)	B9D2	614145	Mutated in Meckel-Gruber syndrome (type10) OMIM 611951 although mutations only found in 1 family PMID:21763481. Interacts with B9D2 and MKS1 at the transition zone PMID:22179047	PMID:21493627, PMID:21763481, PMID:22179047	Transition Zone	x		x		x

ENSG00000204463	BAG cochaperone 6(BAG6)	BAG6		Microtubule-binding protein that regulates ciliogenesis by modulating the cell cycle and interacting with γ -tubulin. Colocalizes with intracellular microtubules and by co-immunoprecipitation we demonstrated that it can interact with α -tubulin. Assembly of primary cilia in RPE-1 cells is significantly inhibited upon the depletion of BAG6 (PMID: 31838060)	PMID: 31838060	Intracellular microtubules		x				
ENSG00000214413	BBSome interacting protein 1(BBIP1)	BBIP1	613605	Component of BBSome, implicated in BBS (BBS18) PMID: 24026985, PMID: 32055034	PMID: 19081074; PMID: 24026985; PMID: 32055034	Basal body, Cilium	x					x
ENSG00000119636	basal body orientation factor 1(BBOF1)	BBOF1		Required to maintain cilia orientation in multiciliated cells	PMID: 23900544	Basal body	x				x	x
ENSG00000174483	Bardet-Biedl syndrome 1(BBS1)	BBS1	209901	Mutated in Bardet-Biedl syndrome (type 1) OMIM 209901 PMID: 12118255. Forms part of BBSome, mediates interaction with RABIN8 (RAB3IP), important for RAB8-mediated membrane protein trafficking PMID: 20603001, PMID:17574030, PMID:22072986	PMID:12118255, PMID: 20603001, PMID:17574030, PMID:22072986	Basal Body, Cilium	x			x		x

ENSG00000179941	Bardet-Biedl syndrome 10(BBS10)	BBS10	610148	Mutated in Bardet-Biedl syndrome (type 10) OMIM 610148 PMID:16582908. Chaperonin-like protein. Forms complex with other chaperonin-like BBS proteins to facilitate BBSome assembly PMID:20080638. At basal body of differentiating preadipocytes. Role in targeting of Wnt and Hh receptors to this cilium PMID:19190184	PMID:16582908, PMID:20080638	Basal body	x		x		x
ENSG00000181004	Bardet-Biedl syndrome 12(BBS12)	BBS12	610683	Mutated in Bardet-Biedl syndrome (type 12) OMIM 610683 PMID:17160889. Forms complex with other chaperonin-like BBS proteins to facilitate BBSome assembly PMID:20080638. At basal body of differentiating preadipocytes. Role in targeting of Wnt and Hh receptors to this cilium PMID:19190184.	PMID:17160889, PMID:20080638, PMID:19190184.	Basal body	x		x		x
ENSG00000125124	Bardet-Biedl syndrome 2(BBS2)	BBS2	606151	Mutated in Bardet-Biedl syndrome (type 2) OMIM 606151 PMID:11285252. Forms part of BBSome, important for RAB8-mediated membrane protein trafficking PMID: 20603001, PMID:17574030, PMID:22072986	PMID:11285252, PMID: 20603001, PMID:17574030, PMID:22072986	Basal Body, Cilium	x		x		x
ENSG00000140463	Bardet-Biedl syndrome 4(BBS4)	BBS4	600374	Mutated in Bardet-Biedl syndrome (type 4) OMIM 600374 PMID:11381270. Forms part of BBSome, important for RAB8-mediated membrane protein trafficking PMID: 20603001, PMID:17574030, PMID:22072986	PMID:11381270, PMID: 20603001, PMID:17574030, PMID:22072986	Basal Body, Cilium	x		x		x
ENSG00000163093	Bardet-Biedl syndrome 5(BBS5)	BBS5	603650	Mutated in Bardet-Biedl syndrome (type 5) OMIM 603650 PMID:15137946. Forms part of BBSome, mediates binding to phospholipids, predominantly phosphatidylinositol 3-phosphate PMID: 20603001, PMID:17574030,	PMID:15137946, PMID: 20603001, PMID:17574030, PMID:22072986	Basal Body	x		x		x

				PMID:22072986								
ENSG00000138686	Bardet-Biedl syndrome 7(BBS7)	BBS7	607590	Mutated in Bardet-Biedl syndrome (type 7) OMIM 607590 PMID:12567324. Forms part of BBSome important for RAB8-mediated membrane protein trafficking PMID: 20603001, PMID:17574030, PMID:22072986	PMID:12567324, PMID: 20603001, PMID:17574030, PMID:22072986	Basal Body, Cilium	x		x		x	
ENSG00000122507	Bardet-Biedl syndrome 9(BBS9)	BBS9	607968	Mutated in Bardet-Biedl syndrome (type 7) OMIM 607968 PMID:16380913. Forms part of BBSome, important for RAB8-mediated membrane protein trafficking. Complex-organizing subunit of BBSome. PMID: 20603001, PMID:17574030, PMID:22072986	PMID:16380913, PMID: 20603001, PMID:17574030, PMID:22072986	Cilium	x		x		x	
ENSG00000166352	chromosome 11 open reading frame 74(C11orf74)	C11orf74		C11ORF74, interacts with the IFT-A complex via the IFT122 subunit and is accumulated at the distal tip in the absence of an IFT-A subunit IFT139, suggesting that at least a fraction of C11ORF74 molecules can be transported towards the ciliary tip by associating with the IFT-A complex, although its majority might be out of cilia at steady state (PMID: 30476139).	PMID: 30476139	Ciliary tip, out of cilia.	x				x	
ENSG00000168014	C2 calcium dependent domain containing 3(C2CD3)	C2CD3	615948	C2cd3 is an essential regulator of intracellular transduction of the Hedgehog signal (PMID: 19004860). Regulates primary cilium biogenesis, recruiting centriolar distal appendage proteins. C2cd3 is essential for ciliary vesicle docking (PMID: 24469809). Promotes centriolar elongation (PMID: 24997988)	PMID: 19004860, PMID: 24469809, PMID: 24997988.	Basal Body. Centrioles and centriolar satellites in microtubules.	x		x		x	

ENSG00000156172	chromosome 8 open reading frame 37(C8orf37)	C8orf37	614477	Mutations are associated with cone-rod dystrophy, retinitis pigmentosa (PMID: 22177090) and Bardet Biedl syndrome (PMID: 27008867). Studies suggest C8orf37 is involved with primary cilia function and implicated in the development of BBS (PMID: 27008867).	PMID: 22177090, PMID: 27008867.	Basal body.	x		x		x
ENSG00000151067	calcium voltage-gated channel subunit alpha1 C(CACNA1C)	CACNA1C		CACNA1C, also called CaV1.2Serves as a modifier gene to cystic kidney phenotype. Plays an important role in Akt and Erk activities through cytosolic calcium and CaM kinase II. CaV1.2 might be a secondary calcium-permeable channel to the polycystins complex (PMID: 24925129). CaV1.2 expression is regulated through Wnt signaling and plays an essential sensory role in primary cilia for cellular homeostasis (PMID: 24700505).	PMID: 24925129, PMID: 24700505	Primary cilia of epithelial cells. Cytoplasm of Pkd cells.	x				
ENSG00000158428	ciliogenesis associated TTC17 interacting protein(CATIP)	CATIP		It is involved in primary ciliogenesis in human cells and modulates actin polymerization, its depletion reduces the number of primary cilia without changing their length, suggesting it would be involved in cilia initiation rather than in cilia elongation. C2orf62 in association with TTC17 probably act on both motile and primary cilia formation (PMID, 24475127)	PMID: 24475127	Cytoplasm of spermatocytes, round and elongating spermatid, and Leydig cells. Nucleus and F-actin zone in cell membrane.		x			

ENSG00000105974	caveolin 1(CAV1)	CAV1		CAV1 regulates primary cilium length by modulating apical actin cytoskeleton, its silencing results in an increased size of cilia. α isoform is necessary for normal ciliary length, through mechanisms involving GTPase RhoA, ROCK and DIA1 (PMID 30718762). CAV1 membrane microdomains establishment by KIF13B and NPHP4, is essential for Sonic hedgehog-induced ciliary accumulation of Smoothened (SMO) (PMID 28134340)	PMID: 30718762, PMID: 28134340	Cell membrane (caveolar and non-caveolar flat membrane domains). Transition zone.	x					x
ENSG00000100211	chibby family member 1, beta catenin antagonist(CBY1)	CBY1		Inhibits beta-catenin-mediated transcriptional activation by competing with Tcf/Lef. Cby1 is required for recruitment of Arl13b to the primary cilium. Promotes Ahi1 localisation to the centriole and is probably important for assembly and function of transition zone components (PMID: 25103236). Cby1 may associate with membranes to promote the recruitment of FAM92 proteins for their higher-order assembly and subsequent membrane remodelling to promote ciliogenesis (PMID: 27528616). Because of its proximity to the membrane vesicle, it is possible that Cby proteins may be involved in targeting and/or attachment of the vesicles to the centriole, controlling intraflagellar transport and protein sorting of the ciliary membrane (PMID: 22911743).	PMID: 25103236, PMID: 27528616, PMID: 22911743.	Distal end of mother centrioles and transition zone.	x		x			x

ENSG00000048342	coiled-coil and C2 domain containing 2A(CC2D2A)	CC2D2A	612013	It is essential for the assembly of subdistal appendages. Protein is required for cilia formation and predicted to be involved in calcium-dependent membrane targeting. (PMID 24947469). Interacts with CEP290 in the basal body (PMID 24947469; PMID: 18950740). CC2D2A associates with NINL to provide a docking point for incoming transport vesicles at the entrance of ciliary compartments (PMID: 26485645). Required for vesicle fusion (PMID: 29281629). Play a role in Ca2+ regulated signalling pathways (PMID: 18513680)	PMID: 24947469; PMID: 26485645; PMID: 29281629	Transition zone between the basal body and ciliary axoneme.	x		x		x
ENSG00000167131	coiled-coil domain containing 103(CCDC103)	CCDC103	614677	Participates in microtubule stabilization that facilitates dynein arm attachment and ciliary motility (PMID: 28790179). Contributes to axonemal stability. A potential role, is to modify the doublet microtubule to provide a high affinity binding site for the docking complex and needed for dynein attachment (PMID: 25572396) Binds microtubules and stabilises them against cold depolymerization. It is also an outer dynein arm assembly factor (PMID: 31858719, PMID: 22581229)	PMID: 28790179, PMID: 25572396, PMID: 31858719, PMID: 22581229	Axoneme. In zebrafish: Cytoplasm and axoneme.	x		x		x
ENSG00000105479	coiled-coil domain containing 114(CCDC114)	CCDC114	615038	Subunit of the outer dynein arm (ODA) docking complex. It is responsible for axonemal microtubule attachment .	PMID: 25192045, PMID: 23261303	Basal body and axoneme.	x		x		x

ENSG00000244607	coiled-coil domain containing 13(CCDC13)	CCDC13		Required for primary cilia formation and promotes the localisation of protein BBS4 to both centriolar satellites. Promotes genome stability and maintaining a stable microtubule array	PMID: 24816561.	Basal Body and centrosome.	x				x
ENSG00000198003	coiled-coil domain containing 151(CCDC151)	CCDC151	615956	Ensures correct attachment of outer dynein arm (ODA) complexes to microtubules and is required for proper motile and non-motile functions, controls intraflagellar transport (IFT)-dependent dynein arm assembly and regulates primary cilia length in mammals. Required for the correct establishment of left-right asymmetry. When mutated causes PCD by disruption of the outer dynein arm docking complex formation (PMID: 25192045)	PMID: 24067530, PMID: 25224326; PMID: 25192045	Outer dynein arm docking complex of motile cilia	x			x	x
ENSG00000117477	coiled-coil domain containing 181(CCDC181)	CCDC181		CCDC181 is a structural component of the axonemal structure of sperm flagella as well as motile cilia. It might be a component of the apparatus necessary for the generation of ciliary motility. C-terminal region interacts with microtubules of the axoneme and Hook1, acting as a linker protein for phosphoprotein phosphatase 1 (PP1) to tack the phosphatase to the axonemal structure of cilia and flagella. Pronostic factor for prostate cancer.	PMID: 28282151, PMID: 28283191.	Proximal portion of the ciliary axoneme. Basal half of motile cilia.	x			x	x

ENSG00000160050	coiled-coil domain containing 28B(CCDC28B)	CCDC28B	610162	Reported as second site modifier of the ciliopathy Bardet-Biedl syndrome. It is a positive regulator of mTOR complex 2 and interacts with SIN1 and Kinesin 1 to regulate cilia length <i>in vivo</i> (PMID: 23727834, PMID: 29445114).	PMID: 23727834, PMID: 29445114	Pericentriolar regions, base of cilia.	x		x		x
ENSG00000145075	coiled-coil domain containing 39(CCDC39)	CCDC39	613798	CCDC39 regulates centriole/deuterosome amplification and multiciliogenesis. CCDC39/40 protein complex might define the binding site of radial spokes along peripheral microtubules and provide anchoring sites for inner dynein arms (PMID: 29317443). It is essential for assembly of inner dynein arms and the dynein regulatory complex (PMID: 21131972, PMID: 21131974)	PMID: 29317443, PMID: 21131972, PMID: 21131974	Ciliary axoneme.	x		x		x
ENSG00000141519	coiled-coil domain containing 40(CCDC40)	CCDC40	613799	CCDC40 is required for assembly of motile cilia. It is responsible for CCDC39 axonemal recruitment. It is important for cytoplasmic pre-assembly, axonemal targeting, and/or transport of axonemal components CCDC39, GAS11 and DNALI1. Necessary for assembly of inner dynein arms, the dynein regulatory complex and plays a role in the establishment of the left-right patterning (PMID: 21131974)	PMID: 21131974	Ciliary axoneme, apical cytoplasm.	x		x		x

ENSG00000139537	coiled-coil domain containing 65(CCDC65)	CCDC65	611088	This gene encodes a protein that is required for proper cilia motility and regulation of dynein activity (PMID: 24094744). Assembly and maintenance of the protein components of the nexin-dynein regulatory complex (N-DRC) and thus the regulation of cilia motion. Maintenance of the integrity of N-DRC required for regulation of dynein-driven microtubule sliding (PMID: 23991085, PMID: 29167384).	PMID: 23991085, PMID: 24094744, PMID: 29167384	Cytoplasm, axoneme, basal body.	x			x	x
ENSG00000180376	coiled-coil domain containing 66(CCDC66)	CCDC66		Organisation of spindle pole and centriolar satellite. It is involved in cilium formation, satellite organization, BBS4 recruitment to the primary cilium and ciliary trafficking. Overexpression induces formation of microtubule bundles, disruption of centriolar satellite organization and inhibits ciliogenesis (PMID: 28235840, PMID: 31582766). Interacts with ARMC9 and TOGORAM1 (PMID: 32453716). When mutated in dogs causes late onset retinal degeneration (PMID: 33273526)	PMID: 28235840, PMID: 31582766; PMID: 32453716; PMID: 33273526	Centrosome, centriolar satellites and the primary cilium, along the axoneme.	x				
ENSG00000162004	coiled-coil domain containing 78(CCDC78)	CCDC78		Centriole amplification mediated by deuterosome in multi-ciliated cells. Recruitment of Cep152 to the deuterosome required for centriole biogenesis.	PMID: 24075808	Deuterosome, centrioles	x				x

ENSG00000152669	cyclin O(CCNO)	CCNO	607752	Regulator of deuterosome formation and function for the amplification of centrioles in MCCs, for docking of centrioles to the apical cell membrane (PMID: 25712475). Cyclin O is required to generate the basal bodies necessary for multiple cilia formation by promoting a process of centriole assembly and maturation (PMID: 25048963)	PMID: 25712475, PMID: 25048963	Apical cytoplasm.		x				x
ENSG00000103540	centriolar coiled-coil protein 110	CCP110		Suppressor of ciliogenesis by acting simply as a passive physical barrier and by capping the growing ends of microtubules to prevent their elongation. Regulation the assembly of subdistal appendages (SDA) core components. Has a role in regulating centrosome duplication and separation (PMID: 26965371). Regulates the length of centrioles and prevents aberrant formation of primary cilia, components to a CP110 scaffold may be needed to facilitate depolymerization (PMID: 21620453). Studies suggest that CP110 might control discrete functions in the centrosome cycle and constitutes a set of controls that govern the switch from centriolar to basal body function (PMID: 17719545).	PMID: 26965371, PMID: 21620453, PMID: 17719545	Distal ends of both centrioles.	x		x			x
ENSG00000166226	chaperonin containing TCP1 subunit 2(CCT2)	CCT2		Roles in the folding of β -propeller proteins, such as G β 1, in addition to its major target proteins, actin and tubulin. Interacts with chaperonin type BBSs, including BBSs 6, 10, and 12. Mutations in CCT2 could lead to genetic retinal diseases (PMID: 27645772)	PMID: 27645772	Basal body or the base of the connecting cilium	x			x		x

ENSG00000079335	cell division cycle 14A(CDC14A)	CDC14A		CDC14A activity in human and mouse is needed for the formation of the kinocilia in the organ of corti. Proline-directed phosphatase. CDC14A has the potential to counteract kinases that control the actin network during ciliogenesis because it is associated with actin filaments and in addition plays a role in ciliogenesis. Contributes to cilia length control by regulating the actin nucleator Arp2 at the basal body. CDC14A has additional functions in endocytosis and transport of myosin Va marked vesicles to the basal body (PMID: 30467237). Plays a role in mitotic timing, cytokinesis, meiosis, DNA repair, and transcriptional repression (PMID: 23027426).	PMID: 30467237, PMID: 23027426	Centrosome during interphase, spindle and mibbody during mitosis, actin cytoskeleton, basal body.	x					x
ENSG00000117399	cell division cycle 20(CDC20)	CDC20		Neuronal differentiation, ciliary formation or DNA damage repair. Studies suggest that is involved in neuronal connectivity and plasticity (PMID: 25850036). Potential activity in quiescent and differentiated cells, is recruited to the basal body of the primary cilium, where in complex with APC it negatively regulates ciliary length, by destabilizing axonemal microtubules (PMID: 25139956).	PMID: 25850036, PMID: 25139956	Centrosome of postmitotic neurons, basal body.	x					
ENSG00000164287	cell division cycle 20B(CDC20B)	CDC20B		Specifically expressed in maturing multiciliated cells during the phase of centriole multiplication; it has been established the corresponding CDC20B protein as an essential regulator of centriole-deuterosome disengagement. Plays an essential and conserved role of in vertebrate multiciliogenesis. Function of promoting deuterosome	PMID: 30405130	Possible locations are: perideuterosomal region, multiciliary locus, deuterosome.						

				assembly and/or activity, may contribute to activation of Separase. Involved in the production of centrioles (PMID: 30405130).							
ENSG00000107736	cadherin related 23(CDH23)	CDH23	605516	Essential tip link component required for hair-cell mechanotransduction (PMID: 15057246). Encodes a protein of 3,354 amino acids with a single transmembrane domain and 27 cadherin repeats (PMID: 11138009). Studies suggest that Cdh23 are required for establishing and/or maintaining the proper organisation of the stereocilia bundle (PMID: 12121736).	PMID: 15057246, PMID: 11138009, PMID: 12121736	Near the tips of hair bundles.	x		x		x
ENSG00000185324	cyclin dependent kinase 10(CDK10)	CDK10	603464	Interacts with Cyclin M (CycM), encoded by FAM58A to form an active protein kinase. CDK10/CycM is a key repressor of ciliogenesis through the regulation of RhoA and modulation of actin dynamics (PMID: 27104747). Important for normal regulation of ciliogenesis (PMID: 29130579).	PMID: 27104747, PMID: 29130579	Base of primary cilia, basal body.	x				
ENSG00000156345	cyclin dependent kinase 20(CDK20)	CDK20		Positively and negatively controls both ciliary length and the activity of the Hh pathway, regulates a broad array of processes at levels ranging from the subcellular to the organismal (PMID: 28817564).	PMID: 28817564, PMID: 20159594	Cilium, nucleus, cytoplasm.	x			x	x

ENSG00000008086	cyclin dependent kinase like 5(CDKL5)	CDKL5	300203	Regulate cilium length, has a potential role in ciliogenesis, facilitates dendritic spine and excitatory synapse formation (PMID: 29420175).	PMID: 29420175,	Basal body, cilium, centrosome, nucleus.	x				
ENSG00000111276	cyclin dependent kinase inhibitor 1B(CDKN1B)	CDKN1B		Also known as p27Kip1. Multifunctional protein that plays crucial roles in cell architecture and migration by regulating rearrangements of the actin cytoskeleton and microtubules. p27Kip1 abrogation impairs one of the earliest events of ciliogenesis, docking of the Ehd1-associated preciliary vesicles to the distal appendages of the basal body (PMID: 32423824)	PMID: 32423824	Nucleus, cytoplasm	x				
ENSG00000143126	cadherin EGF LAG seven-pass G-type receptor 2(CELSR2)	CELSR2		Has been shown to be required for ciliogenesis (PMID: 27894351). Lack of cadherins Celsr2 and Celsr3 impairs ependymal ciliogenesis, leading to fatal hydrocephalus (PMID: 29420175). Whereas inactivation of Celsr2, Celsr3, Fzd3, and Vangl2 perturbs the organization of ependymal multicilia at the single-cell level (PMID: 25024228).	PMID: 29420175, PMID: 27894351, PMID: 25024228	Cell membrane.	x				
ENSG00000008300	cadherin EGF LAG seven-pass G-type receptor 3(CELSR3)	CELSR3		Lack of cadherins Celsr2 and Celsr3 impairs ependymal ciliogenesis, leading to fatal hydrocephalus (PMID: 29420175). controls the function of ependymal cilia (PMID: 29327175). Whereas inactivation of Celsr2, Celsr3, Fzd3, and Vangl2 perturbs the organization of ependymal multicilia at the single-cell level (PMID: 25024228).	PMID: 29420175, PMID: 29327175, PMID: 25024228	Cell membrane.	x				x

ENSG00000117724	centromere protein F(CENPF)	CENPF	243605	Centriolar disease gene implicated in severe human ciliopathy and MCPH-related phenotypes (PMID: 25564561; PMID: 26820108; PMID: 31488893; PMID: 31953238).	PMID: 25564561; PMID: 26820108; PMID: 31488893; PMID: 31953238	Subdistal appendages and the distal end of the mother centriole, basal body, axoneme.	x				x
ENSG00000151849	centromere protein J(CENPJ)	CENPJ	609279	Acts as a scaffold for the cilium disassembly complex, which includes Nde1, Aurora A, and OFD1, which is recruited to the ciliary base for timely cilium disassembly, plays a role in NPC proliferation, cilia formation, controlling the cilium axoneme and BB structure, regulates centriole duplication and elongation (PMID: 30626697). Promotes the extension of the centriolar cylinder, presumably via its ability to recruit tubulin to the nascent structure, studies propose that functions as a scaffold for tubulin addition (PMID: 19481458).	PMID: 30626697, PMID: 19481458	Centrosome.	x		x		x
ENSG00000116198	centrosomal protein 104(CEP104)	CEP104	616690	Has an important role in regulating microtubule length in cilia by facilitating tubulin addition to ciliary microtubules and might also be involved in regulating other factors, possibly by interacting with Nek1. Cep104 is involved in the stabilization of the ciliary axoneme or in cilia formation and elongation, mutations cause ciliopathies (PMID: 28017521). Regulates ciliary assembly by interacting with centriolar proteins that suppress cilia formation, detaches from these proteins at the onset of cilia formation and moves to the tip of the nascent cilium (PMID: 23970417).	PMID: 31412255, PMID: 28017521, PMID: 23970417	Axoneme, centriole.	x		x		x

ENSG00000168944	centrosomal protein 120(CEP120)	CEP120	613446	Is essential for the recruitment of C2CD3 and Talpid3 to the nascent centrioles, and is responsible for promoting the assembly of the centriole appendages that are needed for subsequent ciliogenesis. It is a daughter centriole-enriched protein that plays an essential role in centriole elongation (PMID: 30988386). Plays a key role in the duplication and maturation of centrioles, which in principle could impact the production of a functional basal body (PMID: 29847808).	PMID: 30988386, PMID: 29847808	Centriole, centrosome.	x					x
ENSG00000100629	centrosomal protein 128(CEP128)	CEP128		It encodes a subdistal appendage protein that regulates TGF- β /bone morphogenetic protein (BMP) signaling and RAB11 trafficking at the primary cilium. It has no major effect on centrosome structure but leads to increased ciliation frequency (PMID: 29514088).	PMID: 29514088	Centrosome, centriole.	x					
ENSG00000141577	centrosomal protein 131(CEP131)	CEP131		Contributor to ciliogenesis in certain tissues in the developing vertebrate embryo, involved in assembling the cilium and in determination of the length of the axoneme (PMID: 19254375). The Atoh1-Cep131-CS integrity axis would act as a gatekeeper during GNP's ciliogenesis, as it would provide cells an essential machinery (i.e., the CS) for building the primary cilia (PMID: 30695697).	PMID: 19254375, PMID: 30695697,	Centrosome, basal body.	x		x			x

ENSG00000174799	centrosomal protein 135(CEP135)	CEP135	611423	CEP135/Bld10 promotes the assembly and stability of the centriolar microtubule wall. Induces the formation of protofilament-based ring-like structures, binding directly to tubulin, protofilaments, and microtubules (PMID: 27477386). Required for centriole conversion to centrosome (PMID: 26595382). Plays an important role in the assembly of centriole/basal bodies and stabilisation of microtubules, maintaining basal body integrity (PMID: 23810536, PMID: 23115304, PMID: 22898782). Knockdown of CEP135 leads to disorganised interphase and mitotic spindle microtubules (PMID: 22521416)	PMID: 27477386, PMID: 26595382, PMID: 23810536, PMID: 23115304, PMID: 22898782, PMID: 22521416	Centrioles (walls and lumen), centrosome, basal body, microtubule barrel of centrioles.	x		x		x
ENSG00000135315	centrosomal protein 162(CEP162)	CEP162		Promotes and restricts transition zone (TZ) assembly before ciliogenesis. Interacts with core TZ components CEP290 and mediates their association with MTs. Loss of CEP162 arrests ciliogenesis at the stage of TZ assembly; gain of CEP162 at cilia tips promotes ectopic assembly of TZ components	PMID: 23644468	Centriole distal end, axoneme.	x				x
ENSG00000110274	centrosomal protein 164(CEP164)	CEP164	614848	Assembly of primary cilia and promotes vesicular docking and trafficking that initiates cilia membrane biogenesis (PMID: 17954613, PMID: 23253480). CEP164 promotes accumulation of GTP-Rab8 at the centrosome, providing a molecular link connecting the M-centriole to components of the machinery that initiate ciliary membrane biogenesis (PMID: 23253480). Recruitment of active tau tubulin kinase 2 (TTBK2) to centrioles, leading to removal of CP110 and recruitment of intraflagellar	PMID: 17954613, PMID: 23253480, PMID: 24982133, PMID: 23150559; PMID: 32367404	Distal appendages of mature centrioles.	x		x		x

				transport proteins (PMID: 24982133). Mutations in CEP164 are related to nephronophthisis and Joubert syndrome, it may facilitate the docking of the INPP5E–PDE6D complex to the centrosomal distal appendage/transition fiber (PMID: 23150559). Mutations also reported in a BBS/PCD case (PMID: 32367404)								
ENSG00000143702	centrosomal protein 170(CEP170)	CEP170		Required for anchoring the subdistal appendages (SDA) complex in the centrifuge to the cytoplasmic microtubule network. It is an FHDC1 buinding protein to keep this connection (PMID: 29742020). Recruits microtubule-depolymerizing factor KIF2A to disassemble cilium (PMID: 31197141).	PMID: 29742020, PMID: 31197141	Subdistal appendages of centrioles.	x					
ENSG00000174007	centrosomal protein 19(CEP19)	CEP19	615586	Recruitment of ciliary vesicles to the centrioles during early ciliogenesis, which suggest it is a regulator for membrane trafficking pathways . CEP19 recruits and associates with RABL2 to the basal body, promoting cilia assembly via its interactions with the FT-B complex (PMID: 28659385). CEP19-RABL2B complex plays a critical role as a regulator to initiate intraflagellar transport (PMID: 28625565, PMID: 28428259).	PMID: 28659385, PMID: 28625565, PMID: 28428259	Basal body: subdistal appendages of mother centriole.	x					x
ENSG00000126001	centrosomal protein 250(CEP250)	CEP250	609689	Interact with proteins implicated in rhodopsin trafficking (NEK2 and NINL). Rrecruitment of several proteins to centriole proximal ends, including Girdin (Girders of actin filaments) and proteins that associate with the distal region of the mother centriole (NEK2, ninein, CEP170, p150glued and KIF2A) (PMID: 30421101). C-NAP1 anchors rootletin filaments that hold	PMID: 30421101, PMID: 28005958, PMID: 28100636	Proximal end of centrioles.	x		x			x

				duplicated centrosomes together(PMID: 28100636). Mutations are associated to Usher Syndrome and retinitis pigmentosa (PMID: 28005958).							
ENSG00000198707	centrosomal protein 290(CEP290)	CEP290	610142	Assembly and stabilisation of the primary cilium as well as centrosomal function (PMID: 30478281, PMID: 22563985). Interacts with a centriolar satellite protein PCM-1 and is implicated in molecular pathways related to BBS4 function. CEP250 binds PCM-1 and are involved in ciliary targeting of Rab8 required to promote ciliogenesis. Interacts with components of both dynein and kinesin motor machineries (PMID: 18772192). Regulation of intracellular protein trafficking by association with microtubule-based transport protein RPGR (PMID: 16632484). CEP250 plays an important role in the formation and stabilisation of the primary cilium as well as centrosomal function, stabilises the linkage of axoneme microtubules to the ciliary membrane to maintain the integrity of the ciliary diffusion barrier. CEP290 is critical for positioning the molecular components that are necessary for correct processing of Shh signalling in the primary cilium (PMID: 30478281)	PMID: 30478281, PMID: 18772192, PMID: 16632484, PMID: 30478281, PMID: 22563985.	Centriolar satellites. Centrosome and basal body. Y-link junctions of the ciliary transition zone	x		x		x

ENSG00000135837	centrosomal protein 350(CEP350)	CEP350		Also called CAP350. Facilitates the localization of CEP19 to the centriole (PMID: 28659385). CEP250 is necessary for recruitment of FOP and provides a docking site for it at the centrosome. Both CEP250 and FOP form a complex and are required for microtubules anchoring to the centrosome (PMID: 16314388). CEP350-FOP complex promotes recruitment of both the CEP19-RABL2B complex and TTBK2 to the mother centriole (PMID: 28625565). Centriole growth is regulated by CAP350, which is also required early during the procentriole assembly process when the first microtubule is polymerised (PMID: 19052644)	PMID: 28659385, PMID: 16314388, PMID: 28625565, PMID: 19052644	Centrosome (distal end of centrioles), spindle.	x					
ENSG00000106477	centrosomal protein 41(CEP41)	CEP41	610523	Regulation of tubulin glutamyl posttranslational modifications by mediating ciliary transport of TTLL6 between the basal body and cilium, therefore, it is also required for localising TTLL6 to the cilium. CEP41 is also involved in ciliary structural formation and motility through glutamylation and its loss of CEP41 has been identified as a cause of Joubert syndrome (PMID: 22246503). CEP41 regulates cilia disassembly and angiogenesis through AURKA activation by modulating axonemal tubulin glutamylation (PMID: 31885126)	PMID: 22246503, PMID: 31885126	Basal body, centriole.	x		x			x

ENSG00000011523	centrosomal protein 68(CEP68)	CEP68	Connection of the two centrosomes of a cell during interphase, required for cilia and mitotic spingle formation. Rootletin binds CEP68 modulating the thickness of rootletin filaments and promoting rootletin filament formation. Rootletin and CEP68 rings are the anchoring point of filaments that radiate into the cytoplasm (PMID: 29463719). CEP68, C-NAP1 and Rootletin, cooperate in centrosome cohesion (PMID: 18042621). Interacts with CEP215 to retain it on the peripheral pericentriolar material, inhibiting centriole separation following disengagement (PMID: 25503564)	PMID: 29463719, PMID: 18042621, PMID: 25503564.	Centrioles, centrosome linker filaments	x					
ENSG00000112877	centrosomal protein 72(CEP72)	CEP72	Regulation of the centrosomal localisation of γ-tubulin complex components, CG-NAP, and Kiz. Cep72 provides centrosomal microtubule-nucleation activity and has critical roles in spindle-pole formation, which is needed for proper tension generation between sister chromatids during mitosis (PMID: 19536135). Recruitment of Cep290 to centriolar satellites by interaction with PCM1. Regulates ciliary recruitment and relocalisation of BBS4 from centriolar satellites to primary cilia during ciliogenesis, which affects assembly and recruitment of the BBSome (PMID: 22767577).	PMID: 19536135, PMID: 22767577	Centriolar satellites.	x		x			x

ENSG00000148019	centrosomal protein 78(CEP78)	CEP78	617110	Centrosome overduplication by interaction with the N-terminal catalytic domain of Plk4 (PMID: 27246242). Regulation of centrosome homeostasis by inhibiting the final step of the enzymatic reaction catalyzed by EDD-DYRK2-DDB1Vpr BP (PMID: 28242748). Mutations in CEP78 cause cone-rod degeneration associated with hearing loss (PMID: 27588451)	PMID: 27246242, PMID: 27588451, PMID: 28242748	Mature centrioles (centriolar walls)	x			x	x
ENSG00000173588	centrosomal protein 83(CEP83)	CEP83	615847	Recruitment of distal appendages proteins like SCLT1 and CEP89 (PMID: 23348840). CEP83 phosphorylation mediated by TTBK2, regulates docking of membrane vesicles to mother centrioles, promoting CP110 removal and cilia initiation. Mutation have been associated with infantile nephronophthisis (PMID: 31455668).	PMID: 23348840, PMID: 31455668,	Centrioles (distal ends)	x		x		x
ENSG00000121289	centrosomal protein 89(CEP89)	CEP89		Also calles CEP13 and CCDC123. Assembly of primary cilium and membrane docking (PMID: 23348840). It interacts with the centriolar satellite proteins PCM-1 and OFD1 for cilia formation, and with Cep290 for the recruitment of membranes to the centrosome. CEP123 is required for the formation of a ciliary vesicle at the distal end of the mother centriole during primary ciliogenesis (PMID: 23789104)	PMID: 23348840, PMID: 23789104.	Distal appendages of mother centriole, centriolar satellite.	x		x		x

ENSG00000182504	centrosomal protein 97(CEP97)	CEP97	Recruitment and stabilisation of CP110 to centrosomes. Both CP110 and CEP97 complexes regulate basal body function, centrosome functions and inhibit cilia assembly (PMID: 17719545). Loss of CEP97 causes premature cilia formation or abnormal centriole elongation (PMID: 30375385). CEP97 ubiquitylation and degradation play a key role in the removal of the CEP97–CP110 complex from the mother centriole for initiating ciliary axoneme extension (PMID: 30404837).	PMID: 17719545, PMID: 30375385, PMID: 30404837.	Distal end of mother centrioles.	x		x		x
ENSG00000147400	centrin 2(CETN2)	CETN2	Synthesis and accumulation of CETN2 is required for centriole duplication (PMID: 12176356). Regulation of primary ciliogenesis through controlling CP110 levels and localisation (PMID: 25753040). Stabilisation of connecting cilium axoneme structure. CETN2 is required for photoreceptor survival by regulating SPATA7 CC localization, distal connecting cilium/axoneme microtubule stability, and outer segment disc orientation (PMID: 30647131)	PMID: 12176356, PMID: 25753040, PMID: 30647131	Centrioles	x				x
ENSG00000153140	centrin 3(CETN3)	CETN3	CETN3 is also required for photoreceptor survival by regulating SPATA7 CC localization, distal connecting cilium/axoneme microtubule stability, and outer segment disc orientation (PMID: 30647131). CETN3 restrains centriole assembly by inhibiting Mps1 kinase activity at centrosomes and phosphorylation of both the specific substrate CETN2 (PMID: 26354417)	PMID: 30647131, PMID: 26354417.	Centrioles, pericentriolar regions.	x				x

ENSG00000163885	cilia and flagella associated protein 100(CFAP100)	CFAP100		Also called CCDC37 or MIA1. Formation of the modifier of inner arms (MIA) complex in flagella, which is required for normal ciliary motility. MIA1 affects the structure and stability of I1 dynein (PMID: 23569216)	PMID: 23569216	Axoneme.	x		x		x
ENSG00000188931	cilia and flagella associated protein 126(CFAP126)	CFAP126		Also called Flattop. Required for cilia formation. Regulator for BB docking and positioning in mono- and multiciliated cell types that acquire planar cell polarity. Fltp interacts with Dlg3 to surround the basal body and anchor this organelle asymmetrically to the apical cortex (PMID: 25296022)	PMID: 25296022	Next to basal body, microtubules plus end, adjacent to the apical plasma membrane	x			x	x
ENSG00000160401	cilia and flagella associated protein 157(CFAP157)	CFAP157		Interacts with tubulin and CEP350, it targets proteins to the centrosome and is therefore required for ciliogenesis. Downstream effector of FOXJ1 that acts during spermiogenesis potentially suppressing the formation of supernumerary axonemes and ensures a correct ultrastructure (PMID: 27965440)	PMID: 27965440	Basal body	x				x
ENSG00000156206	cilia and flagella associated protein 161(CFAP161)	CFAP161		Also called C15orf26. Studies in zebrafish suggest it is essential for cilia motility (PMID: 24094744)	PMID: 24094744	No data?				x	x

ENSG00000163075	cilia and flagella associated protein 221(CFAP221)	CFAP221		Mouse homologue Pcdp1 causes PCD and brain defects in mice when mutated (PMID: 18039845; PMID: 25073043). Interacts with CFAP54 and SPEF2 (PMID: 32704025). Identified as a cause of PCD in humans in 2020 (PMID: 31636325)	PMID: 18039845; PMID: 25073043; PMID: 31636325	Central pair apparatus of motile cilia	x					x
ENSG00000159079	cilia and flagella associated protein 298(CFAP298)	CFAP298	615408	Also known as C21orf59 or kur) Mutated in primary ciliary dyskinesia and regulates dynein arm assembly required for motile cilia function (PMID: 24094744). Involved in cilia-associated developmental processes, recruitment of outer dynein arms required for initiating cilia motility. Studies suggest it has a role in the establishment of cilia polarity (PMID: 26904945)	PMID: 24094744, PMID: 26904945	Cytoplasm, basal body	x					x
ENSG00000137691	cilia and flagella associated protein 300(CFAP300)	CFAP300	618058	(c11orf70) Involved in the cytoplasmic assembly, maturation, intraflagellar transport (IFT)-dependant assembly of axonemal dyneins required for cilia motility in multiciliated cells. Plays a role within protein folding and trafficking pathways required for the cytoplasmic preassembly of dynein arm motors and subsequent import into the motile ciliary axoneme (PMID: 29727692, PMID: 29727693)	PMID: 29727692, PMID: 29727693	Axoneme, cytoplasm.	x					x
ENSG00000163001	cilia and flagella associated protein 36(CFAP36)	CFAP36		Interacts with Arl3 and may regulate its ciliary localisation and function (PMID: 26455799)	PMID: 26455799	Transition zone.	x					x

ENSG00000160226	cilia and flagella associated protein 410(CFAP410)	CFAP410	603191	C21orf2. It may play roles in cilia formation and/or maintenance (PMID: 21289087). Mutated in both syndromic and non-syndromic retinal degeneration. It is part of a ciliary functional complex that includes NEK1, C21orf2 and retinal ciliopathy protein SPATA7, where C21orf2 might represent a substrate of NEK1	PMID: 21289087, PMID: 26167768	Basal body.	x		x		x
ENSG00000197748	cilia and flagella associated protein 43(CFAP43)	CFAP43	617558	Formation of flagellar axonemes during spermatogenesis. Contains WD repeat domains, which are highly enriched in the intraflagellar transport proteins of sperm flagella (PMID: 28552195). Essential for beating, but not rotating left-right organiser (LRO) or immotile cilia. Involved in the regulation of the beating frequency of tracheal cilia (PMID: 31884020)	PMID: 31884020, PMID: 28552195	Ciliary axoneme.	x			x	x
ENSG00000165164	cilia and flagella associated protein 47(CFAP47)	CFAP47		Hemizygous mutations in CFAP47 can induce X-linked multiple morphological abnormalities of the flagella and asthenoteratozoospermia in humans and mice. Expressed in spermatozoa, but exact localisation not known (PMID: 33472045)	PMID: 33472045	No data				x	
ENSG00000166596	cilia and flagella associated protein 52(CFAP52)	CFAP52		Also called WDR16. Wdr16 can be considered a differentiation marker of kinocilia-bearing cells. In the brain, it appears to be functionally related to water homeostasis or osmoregulation. It might be speculated that Wdr16 has some role in cilia-mediated cell polarity (PMID: 17394468). CFAP52 protein has been proposed to play a role in cilia-related signal transduction processes and	PMID: 17394468, PMID: 25469542, PMID: 32111882	Cytosol (kinocilia-bearing cells).		x		x	x

				mutations have been known as a cause of either heterotaxy or situs inversus without Primary Ciliary Dyskinesia, suggesting a role in human laterality patterning (PMID: 32111882, PMID: 25469542).							
ENSG00000172361	cilia and flagella associated protein 53(CFAP53)	CFAP53	614759	Also called CCDC11. Assembly and function of primary and motile cilia, providing recruitment sites for proteins destined for the centrosome and cilium. Involved in left-right patterning and maintaining centriolar satellite organization of PCM-1 and Cep290 at centriolar satellites (PMID: 26538025).	PMID: 26538025,	Centriolar satellites.	x				x
ENSG00000188596	cilia and flagella associated protein 54(CFAP54)	CFAP54		Member of C1d complex. CFAP54 is required C1d assembly mechanisms, therefor, it is necessary for proper assembly and motility of mammalian cilia and flagella. Role in spermatogenesis (PMID: 26224312)	PMID: 26224312	Central microtubule apparatus (axoneme).	x				x
ENSG00000243710	cilia and flagella associated protein 57 (CFAP57)	CFAP57		Also known as WDR65. Required for the asymmetric targeting of a subset of inner dynein arms in Chlamydomonas. In normal human nasal epithelial cells, CFAP57 localizes throughout the ciliary axoneme. Homozygous nonsense variant found in one PCD case (PMID: 32764743)	PMID: 32764743	Throughout the ciliary axoneme in human nasal epithelial cilia	x			x	

ENSG00000120051	cilia and flagella associated protein 58(CFAP58)	CFAP58	619144	Bi-allelic Loss-of-function variants cause flagellar axoneme and mitochondrial sheath defects and asthenoteratozoospermia in humans and mice (PMID: 32791035). Exhibits similar localization to Odf2/Cenexin proteins and is required for the elongation of the primary cilium and sperm midpiece via modulation of the Notch signaling pathway. (PMID: 31904090)	PMID: 32791035; PMID: 31904090	Along the entire sperm flagellum, predominantly concentrated in the mid-piece.	x			x	x
ENSG00000181378	cilia and flagella associated protein 65(CFAP65)	CFAP65		Biallelic mutations in CFAP65 lead to severe asthenoteratospermia due to acrosome hypoplasia and multiple malformations of the sperm flagella (PMID: 31413122; PMID: 31501240; PMID: 31571197)	PMID: 31413122; PMID: 31501240	No data				x	x
ENSG00000105792	cilia and flagella associated protein 69(CFAP69)	CFAP69	617949	CFAP69 is necessary for flagellum assembly/stability. Mutations cause multiple morphological abnormalities of the flagella and primary male infertility (PMID: 29606301). Regulates olfactory transduction kinetics in the olfactory sensory neurons of mice (PMID: 28495971)	PMID: 29606301, PMID: 28495971	Midpiece of the sperm flagellum.	x			x	x
ENSG00000156042	cilia and flagella associated protein 70(CFAP70)	CFAP70		Regulator component of the outer dynein arm in motile cilia and flagella, regulating ciliary motility and length (PMID: 30158508)	PMID: 30158508	Ciliary axoneme at the base of outer dynein arm.	x			x	x

ENSG00000101421	charged multivesicular body protein 4B(CHMP4B)	CHMP4B		Subunit of the endosomal sorting complexes required for transport (ESCRT). Localized to primary cilia in mammalian cells. Knockdown of CHMP4B interfered with cilium assembly and also caused fragmentation of preexisting cilia. Morpholino (MO)-mediated CHMP4B depletion in zebrafish embryos induced characteristic phenotypes of ciliary defects such as curved body axis, hydrocephalus, otolith malformation, and kidney cyst (PMID: 31914703)	PMID: 31914703	Ciliary axoneme	x				
ENSG00000112144	ciliogenesis associated kinase 1(CILK)	CILK		Also known as ICK. Regulator of ciliary transport through regulation of BBS proteins and IFT-B localisation. Loss of ICK causes accumulation of IFT-A, IFT-B, and BBSome components at the ciliary tips. It is a Kif3a kinase essential for proper ciliogenesis (PMID: 24797473, PMID: 25243405). Involved in the negative control of ciliary length; abnormal ICK function results in the mislocalisation of key mediators for Shh (PMID: 24853502, PMID: 25243405). Anterograde trafficking of CILK is essential for retrograde IFT (PMID: 32732286)	PMID: 24797473, PMID: 24853502; PMID: 25243405; PMID: 32732286	Basal body, ciliary tip, nucleus.	x				x
ENSG00000163507	cell proliferation regulating inhibitor of protein phosphatase 2A(CIP2A)	CIP2A		Promotes primary cilia disassembly through the activation of Aurora A kinase. Depletion increases ciliated cells and cilia length in retinal pigment epithelium cells. CIP2A enhances NEK2 kinase activity to facilitate centrosome separation. It plays a role in c-MYC stabilization and metabolic reprogramming toward the glycolytic pathway (PMID: 29491003)	PMID: 29491003	Transition zone of primary cilia.	x				

ENSG00000073464	chloride voltage-gated channel 4(CLCN4)	CLCN4		Ion channel gene required for ciliogenesis. Mutations cause defects in cilia structure. Potential new candidate in human ciliopathies (PMID: 26546361)	PMID: 26546361	Membrane subdomains at the base of primary cilium in human fibroblasts and mIMCD3 cells). Location is species- and tissue-specific	x				
ENSG00000165376	claudin 2(CLDN2)	CLDN2		Responsible for the Na ⁺ and H ₂ O permeability of the tight junctions and despite being a ciliary protein, it is not necessary for ciliogenesis. Claudin-2 is associated to acetylated α -tubulin (PMID: 22143774)	PMID: 22143774	Along the whole cilium.	x		x		x
ENSG00000163646	clarin 1(CLRN1)	CLRN1		Regulation of localization of Pcdh15a to the apical aspect of hair cells, contributing to the assembly and/or function of the mechanotransduction apparatus. Clarin-2 constitutes a docking site for proteins destined for the cilium and may be a modulator during late ciliogenesis. Modulation of vesicle recycling, distribution and/or assembly of diverse components necessary for normal hair cell function. Mutations in the clarin-1 gene are associated with Usher syndrome type 3 (PMID: 25365995). Vesicle transport and stereocilia/synaptic development and/or maintenance (PMID: 19539019)	PMID: 25365995, PMID: 19539019	Stereocilia and below the cuticular plate in the transition zone ciliary subcompartment. Microtubules in UB/OC-1 cells.	x			x	

ENSG00000103351	clusterin associated protein 1(CLUAP1)	CLUAP1		Cilia protein required for cilia formation. Key regulator of hedgehog signaling and as an intraflagellar transport B complex protein involved in anterograde and retrograde cilia transport. Deletion of Cluap1 causes a total loss of cilia within the developing embryo in mice (PMID: 23351563, PMID: 31554018, PMID: 23742838). Role in cilia assembly and maintenance, intracellular transport processes and cytoskeletal rearrangement by its interactions with TRIP6, which is suggested to be an actin and migration-modulating factor (PMID: 29615496)	PMID: 23351563, PMID: 29615496, PMID: 31554018, PMID: 23742838	Tip and base of cilia, axoneme of primary cilia. Cytoplasmic IFT spot. Centrosome in nonciliated cells (distal appendages of mother centrioles).	x		x		x
ENSG00000198515	cyclic nucleotide gated channel alpha 1(CNGA1)	CNGA1	123825	Subunit of the cyclic nucleotide-gated (Cng) channel localised to the outer segment. Mutations in Cnga1 which have been associated with retinitis pigmentosa in humans (PMID: 25319694)	PMID: 25319694	Outer segment plasma membrane. Rab11-positive recycling endosomes (mediated by Numb).		x			x
ENSG00000183862	cyclic nucleotide gated channel alpha 2(CNGA2)	CNGA2		Principal subunit of heteromeric olfactory CNG channels and homomeric CNGA2 channels in nonmotile cilia on olfactory sensory neurons (ORNs). N-terminus of CNGA2 are necessary for PIP3 regulation. In absence of CNGB1, it does assemble and form functional heteromeric channels in ORNs (PMID: 17032767)	PMID: 17032767, PMID: 25500808	Plasma membrane, cilium	x		x		x

ENSG00000132259	cyclic nucleotide gated channel alpha 4(CNGA4)	CNGA4		Required for ciliary targeting of the olfactory CNG channel. Cooperation in mediating feedback modulation by Ca2+-CaM. In absence of CNGB1, it does assemble and form functional heteromeric channels in olfactory sensory neurons (PMID: 16980309)	PMID: 16980309, PMID: 16782012	Cytoplasm, cilium.	x		x		x
ENSG00000070729	cyclic nucleotide gated channel beta 1(CNGB1)	CNGB1	600724	It endows the olfactory CNG channel with a variety of biophysical properties tailored to the specific requirements of olfactory transduction and is required for ciliary targeting of the olfactory CNG channel. Cooperation in mediating feedback modulation by Ca2+-CaM (PMID: 16980309). Key regulators of disk size by mediating a physical connection between disk rims and the OS plasma membrane Loss of Cnqb1 disruptes links between the disk and plasma membranes (PMID: 23178122)	PMID: 16980309, PMID: 16782012, PMID: 23178122	Cytoplasm, cilium.	x		x		x
ENSG00000197603	ciliogenesis and planar polarity effector 1(CPLANE1)	CPLANE1	614571	Also known as JBTS17. Most common mutated gene in Joubert syndrome and the related oral-facial-digital syndrome type VI (OFDVI). (PMID: 29321670). JBTS17 contributes to mitotic progression by interacting with LIS1, its depletion disrupts chromosome alignment and spindle pole orientation, resulting in mitotic delay (PMID: 31004438). JBTS17 acts upstream of NPHP1 in the assembly of the NPHP1-4-8 complex in the cilia transition zone, it is required for efficient SHH signaling and for establishing cell polarity (PMID: 25877302).	PMID: 29321670, PMID: 31004438, PMID: 25877302	Kinetochore. Cilia transition zone	x		x		x

ENSG00000132881	ciliogenesis and planar polarity effector 2(CPLANE2)	CPLANE2	Also called RSG1. Regulator of ciliogenesis, including apical trafficking of basal bodies and the cytoplasmic localisation and axonemal dynamics of retrograde intraflagellar transport proteins IFT-A protein IFT43 (PMID: 24192041). Component of the CPLANE complex and acts at increasing the efficiency of primary cilia initiation in a late step. RSG1 activity promotes assembly of the microtubule axoneme after the mother centriole has docked onto the ciliary vesicle and accelerates the rate of elongation of the young axoneme. Only gene identified to date in which null mutations cause the production of fewer primary cilia of normal length and without a detectable defect in ciliary trafficking (PMID: 29038301)	PMID: 24192041, PMID: 29038301	Mother centriole inside the CEP164 ring. Transition zone.	x				x
ENSG00000130545	crumbs 3, cell polarity complex component(CRB3)	CRB3	Epithelial polarity, tight junction formation and ciliogenesis. CRB3-CLPI isoform interacts with importin β -1 to be delivered to spindle poles during cell division and ciliogenesis, and this delivery may be important for centrosome maintenance (PMID: 17646395). It modulates intraflagellar transport and cilia elongation together with Par3/Par6/aPKC polarity complex (PMID: 28882989, PMID: 15324661)	PMID: 17646395, PMID: 28882989, PMID: 15324661	Cilium, cilia membrane and a membrane compartment at the mitotic spindle poles	x		x		x

ENSG00000106113	corticotropin releasing hormone receptor 2(CRHR2)	CRHR2	Mammalian G-protein coupled receptor (GPCR) with high affinity for corticotropin-releasing hormone (CRH) and the CRH family of peptides, involved in neurotransmitters signaling. Component of ciliogenesis pathways. Studies suggest that GPCRs may regulate endocytic membrane trafficking, playing sensory or signalling roles in the ciliary vesicle or procilium at the earliest stages of ciliogenesis both in the developing brain and in other tissues.	PMID: 26167768	Base of primary cillium	x				
ENSG00000058453	ciliary rootlet coiled-coil, rootletin(CROCC)	CROCC	Structural component of the ciliary rootlet. Coiled-coil interaction in the extended α -helical rod domain of rootletin is a key feature in rootlet formation (PMID: 12427867). Interaction between the rootlet and intermediate filaments may contribute substantially to the role as support structure for the cillium. Candidate gene for human retinal degeneration (PMID: 15870283). interacts with the large coiled-coil protein, C-NAP1, to connect parental centrioles and maintain centrosome cohesion. Positive regulation of ciliogenesis (PMID: 23070519). CHE-10/rootletin maintains ciliary integrity by modulating the assembly, motility, and flux of IFT particles, which are critical for axoneme length control; and is essential for stabilizing ciliary transition zones and basal bodies (PMID: 24094853)	PMID: 12427867, PMID: 15870283, PMID: 23070519, PMID: 24094853	Rootlets. Proximal ends of centrioles in nonciliated cells	x		x		x

ENSG00000141551	casein kinase 1 delta(CSNK1D)	CSNK1D		Also called CKI-delta. Required for microtubule nucleation at the Golgi and maintenance of Golgi integrity. It mediates primary ciliogenesis by multiple mechanisms involving its centrosomal function and another dependent on its interaction with AKAP450 at the Golgi, where it is important for maintaining Golgi organization and polarized trafficking of multiple factors that mediate ciliary transport. Required for proper positioning of centriolar satellite proteins (PMID: 24648492)	PMID: 24648492	Centrosome, trans-Golgi network, cytoplasmic vesicles.	x				x
ENSG00000104218	centrosome and spindle pole associated protein 1(CSPP1)	CSPP1	611654	CSPP proteins may be involved in cytoskeletal organization and for correct localisation of NPHP8 to the tip of the basal body that is forming the transition zone (PMID: 20519441). Interacts with CEP104 at microtubules for the regulation of axoneme length and Hedgehog pathway stimulation (PMID: 31412255). CSPP1 is required for localization of axonemal proteins, such as ARL13B and AC3; loss of CSPP1 is a cause of Joubert syndrome (PMID: 24360808).	PMID: 20519441, PMID: 31412255, PMID: 24360808, PMID: 24360803	Transition zone, basal body and axoneme of primary cilia (centrioles and midbody of cycling cells; centrioles and cilia axoneme in resting cells)	x				
ENSG00000168036	catenin beta 1(CTNNB1)	CTNNB1		Component of the intercentrosomal linker and a key regulator of mitotic centrosome separation (PMID: 18086858). Required for Kupffer's vesicle (KV) and cilia formation, and organ laterality. Ctnnb1/2 activity in dorsal forerunner cells is required for cilia formation by controlling the expression of a set of transcription factors (PMID: 22535411). Interacts with TCF family members, activating Wnt target genes (PMID: 15852005). Ciliary	PMID: 18086858, PMID: 22535411, PMID: 15852005, PMID: 30008323	Centrosome: proximal and distal regions of the centriole, and between centrioles.	x		x		x

				deconstruction initiates the formation of metastasising melanoma by promoting WNT/ β -catenin signaling (PMID: 30008323).							
ENSG00000044090	cullin 7(CUL7)	CUL7	609577	Mutated in M3 syndrome. Fibroblast with CUL7 homozygous mutations showed significantly fewer cilia than wild-type. Nevertheless, cilia could be restored by overexpression of wild type CUL7, suggesting that 3M syndrome is indeed a ciliopathy (PMID: 27173435).	PMID: 27173435	No data?					
ENSG00000083799	CYLD lysine 63 deubiquitinase(CYLD)	CYLD		Deconjugation of the polyubiquitin chain from centrosomal protein Cep70 and promotes ciliogenesis by inhibiting the tubulin deacetylase activity of HDAC6. Regulation of the docking of basal bodies to the apical plasma membrane. CYLD is also required for proper organization of basal bodies and axenomes (PMID: 25342559). CYLD may control the centriolar satellites proteostasis by preventing the proteasomal degradation of PCM1 (PMID: 31067453).	PMID: 25342559, PMID: 31067453	Basal body/centrosome, centriolar satellites.	x				x
ENSG00000205795	cystin 1(CYS1)	CYS1		Cystin may not involved in a ciliogenic pathway, but would rather play a novel role in ciliary function. It bounds to the axonemal membrane and may function as part of a molecular scaffold that stabilizes microtubule assembly within the ciliary axoneme (PMID: 11854326).	PMID: 11854326	Ciliary membrane, axoneme, primary apical cilia.	x				x

ENSG00000123977	dynein assembly factor with WD repeats 1(DAW1)	DAW1		Also known as ODA16. Adaptor between intraflagellar transport proteins and outer arm dynein during ciliary assembly, needed for efficient dynein transport from the cytoplasm to the flagellar compartment. (PMID: 20568242, PMID: 18852297, PMID: 28298440)	PMID: 20568242, PMID: 18852297, PMID: 28298440	Flagellar matrix, peribasal body region in Chlamydomonas.	x					x
ENSG00000146038	doublecortin domain containing 2(DCDC2)	DCDC2	605755	Functional role in ciliary signaling . DCDC2 overexpression increases ciliary length and enhances the Kif3a mediated translocation of Smo to the cilium, resulting in overactivation of the Shh pathway in neurons (PMID: 21698230). Regulation of microtubule nucleation, stability and length in hair cells (PMID: 25601850). Overexpression inhibits β -catenin-dependent Wnt signaling (PMID: 25557784)	PMID: 21698230, PMID: 25601850, PMID: 25557784	Neurites, ciliary axoneme, cytoplasm, cellular microtubule network.	x		x			x
ENSG00000118197	DEAD-box helicase 59(DDX59)	DDX59	615464	Mutations in DDX59 cause an autosomal-recessive form of orofacioidigital syndrome, nevertheless, normal ciliogenesis is observed.	PMID: 23972372	Nucleus, cytoplasm.		x				
ENSG00000131504	diaphanous related formin 1(DIAPH1)	DIAPH1	602121	Promotes both actin polymerization and EB1-dependent microtubule (MT) stability, both actin nucleating and microtubule-stabilizing properties of DIAPH1 are important for their cilia functions. Depletion of DIAPH1 decreased ciliogenesis, cilia length, and reduced trafficking within cilia. Conversely, targeting of DIAPH1 specifically to the	PMID: 33124112	Ends of microtubules, actin		x				

				basal body causes an increase in cilia length and increased trafficking within cilia. Thought to regulate cilia maintenance in part by regulating vesicular trafficking to the base of the primary cilia (PMID: 33124112)							
ENSG00000162946	disrupted in schizophrenia 1(DISC1)	DISC1	605210	Regulation of the formation and/or maintenance of primary cilia, and establish subtype-specific targeting of dopamine receptors to the ciliary surface. Interacts with centriolar and ciliary components (PMID: 20531939). Ciliogenesis and intraflagellar transport regulation through MIPT3 interaction. DISC1 is involved in plus-end and minus-end microtubular transport (PMID: 28125008)	PMID: 20531939, PMID: 28125008.	Near nucleus, basal body, transition zone, microtubules, and mitochondria	x		x		x
ENSG00000008226	DLEC1 cilia and flagella associated protein(DLEC1)	DLEC1		Interacts with BBS protein complex subunits, as well as α - and β -tubulin. Expression enhances primary cilia formation and cilia length in A549 lung adenocarcinoma cells. Elongating spermatids in Dlec1-/- mice show head deformation, a shortened tail, and abnormal manchette organization. These phenotypes were similar to those of various intraflagellar transport (IFT)-associated gene-deficient sperm (PMID: 33144677)	PMID: 33144677	No data				x	

ENSG00000154099	dynein axonemal assembly factor 1(DNAAF1)	DNAAF1	613190	Dynein heavy chain assembly factor. Interacts with the intraflagellar transport protein IFT88, mediating the delivery of dynein arm components directly to the intraflagellar transport complexes (PMID: 29228333). Required for the stability of the ciliary architecture, cytoplasmic preassembly of dynein-arm complexes, including outer and inner dynein arms (PMID: 27543293, PMID: 19944405). Important for cilia beat generation. Loss of LRRC50 affects the assembly of proximally and distally located ODA complexes. Candidate for human cystic kidney disease (PMID: 19944400)	PMID: 29228333, PMID: 27543293, PMID: 19944400, PMID: 19944405	Cytoplasm. Basal body of primary cillium. Ciliary axonemes.	x		x		x
ENSG00000165506	dynein axonemal assembly factor 2(DNAAF2)	DNAAF2	612517	Preassembly protein that functions a link between the early- and late-phase complexes requiered for proper assembly of the IDA and ODA in motile cilia, providing a mechanism for amplification for rapid preassembly, allowing simultaneous fabrication of thousands of motor units for delivery to axoneme microtubules during ciliogenesis (PMID: 29358401). Required for dynein heavy chains folding (PMID: 22387996)	PMID: 29358401, PMID: 22387996	Cytoplasm		x	x		x
ENSG00000167646	dynein axonemal assembly factor 3(DNAAF3)	DNAAF3	614566	Needed for assembly of multiple axonemal inner and outer dynein arms into complexes beefore being transported into cilia, critical for nodal cilia motility. Loss prevents correct assembly of inner and outer dynein arms, abolishing motility of respiratory cilia and giving rise to classical primary ciliary dyskinesia with defective left-right organ asymmetry and male infertility (PMID:	PMID: 22387996, PMID: 29323929	Cytoplasm.		x	x		x

				22387996, PMID: 29323929)							
ENSG00000256061	dynein axonemal assembly factor 4(DNAAF4)	DNAAF4	608706	Cytoplasmic axonemal dynein assembly factor (DYX1C1) that probably acts together with DNAAF2 at an early step of cytoplasmic ODA and IDA assembly (PMID: 23872636). Involved in ciliary and basal body biogenesis (PMID: 23650548)	PMID: 23872636, PMID: 23650548	Cytoplasm, centrosome.	x		x		x
ENSG00000164818	dynein axonemal assembly factor 5(DNAAF5)	DNAAF5	614864	Preamssembly and/or stability of both outer and inner axonemal dynein arms, required for ciliary motility. Interacts with DNAI2 in the initial step of outer dynein arms assembly (PMID: 25232951)	PMID: 25232951	Granular localisation throughout the cytoplasm		x	x		x
ENSG00000080572	dynein axonemal assembly factor 6(DNAAF6)	DNAAF6		Also known as PIH1D3. Involved in the cytoplasmatic preassembly of both outer (ODA) and inner (IDA) dynein arms of cilia and sperm flagella. Loss-of-function mutations cause X-linked primary ciliary dyskinesia with ciliary and sperm immotility in humans (PMID: 28041644, PMID: 28176794). Interacts with Dnaic2 (ODA IC2), Hsp70 and Hsp90, and may be required to maintain the stability of the IC1-IC2 complex (PMID: 24421334). Mutations lead to absence of outer and inner	PMID: 28041644, PMID: 28176794, PMID: 24421334; PMID: 32170493	Cytoplasm in sperm, trans-Golgi network in ciliated epithelia cells		x		x	x

				dynein arms in sperm flagella, male infertility and primary ciliary dyskinesia (PMID: 32170493)						
ENSG00000114841	dynein axonemal heavy chain 1(DNAH1)	DNAH1	603332	Formation of the inner dynein arms; its absence is deleterious for the organization and biogenesis of the axoneme. Knockout causes asthenozoospermia and respiratory disease due to decrease in the beat frequency of tracheal cilia (PMID: 11371505)	PMID:11371505	Inner surface of the peripheral microtubule doublets of the motile axoneme.	x		x	x
ENSG00000197653	dynein axonemal heavy chain 10(DNAH10)	DNAH10		Component of outer dynein arm (PMID:15937072, PMID: 11175280, PMID: 29843777). Heterotaxy candidate gene, involved in left-right patterning regulaton by affecting the notochord. Participates in protozoan flagellar motility (PMID: 29843777)	PMID: 11175280, PMID: 15937072, PMID: 29843777	Motile cilia axoneme, attached to the peripheral microtubule doublets	x		x	x
ENSG00000105877	dynein axonemal heavy chain 11(DNAH11)	DNAH11	603339	Encodes a ciliary outer dynein arm protein. Required for bending of the proximal ciliary region during the effective and recovery strokes of the ciliary beat cycle. Mutations are a common cause of primary ciliary dyskinesia and situs inversus (PMID: 26909801, PMID: 29467202, PMID: 12142464)	PMID: 26909801, PMID: 29467202, PMID: 12142464	Proximal region of ciliary axonemes.	x		x	x

ENSG00000187775	dynein axonemal heavy chain 17(DNAH17)	DNAH17	618643	Sperm-specific axonemal outer dynein arm heavy chain. Mutations cause isolated male infertility due to asthenozoospermia (PMID: 31178125) and flagella instability (PMID: 31658987)	PMID: 31178125; PMID: 31658987	Colocalizes with alpha-tubulin along the sperm axoneme	x				x
ENSG00000183914	dynein axonemal heavy chain 2(DNAH2)	DNAH2		Component of axonemal inner dynein arm-f, essential for ciliary beat (PMID: 30060180). DNAH2 mutation can cause severe sperm flagella defects that damage sperm motility (PMID: 30811583).	PMID: 30060180, PMID: 30811583.	Axoneme.	x		x		x
ENSG00000039139	dynein axonemal heavy chain 5(DNAH5)	DNAH5	603335	Component of ciliary outer dynein arms, mutations can cause primary ciliary dyskinesia (PMID: 30471717, PMID: 11062149). Cilia with complete axonemal DNAH5 deficiency are immotile (PMID: 15750039). Essential for correct assembly of DNAI2 in type 1 and 2 ODA complexes (PMID: 18950741)	PMID: 11062149, PMID: 30471717, PMID: 15750039, PMID: 18950741.	Along entire axoneme in respiratory cells. Reduced in distal part of the cilia when DNAH9 is mutated. In sperm cells, localises only to the proximal part of the flagellum.	x		x		x
ENSG00000115423	dynein axonemal heavy chain 6(DNAH6)	DNAH6		Putative inner dynein arm component. Required for ciliogenesis and motile cilia function mediating airway clearance and left/right patterning. Mutations can cause heterotaxy and ciliary dysfunction similar to primary ciliary dyskinesia by disruption of cilia ultrastructure (PMID: 26918822)	PMID: 26918822	Ciliary axoneme.	x		x		x

ENSG00000118997	dynein axonemal heavy chain 7(DNAH7)	DNAH7		Axonemal inner arm component of human cilia. Absent in primary ciliary dyskinesia (PMID: 11877439)	PMID: 11877439	Along entire axoneme.	x				x
ENSG00000007174	dynein axonemal heavy chain 9(DNAH9)	DNAH9	603330	Component of ciliary outer dynein arms and is required in cilia motility and ODA stability. In individuals with pathogenic variants in DNAH11, DNAH9 can reside in the place of DNAH11. Mutations confer a loss of ODAs in the distal portion of the ciliary axoneme and affects the ciliary beat pattern (PMID: 30471717). Essential in distal axonemal assembly and docking of ODAs type 2, which occurs via direct interaction of DNAH9 with the docking complex protein CCDC114 (PMID: 30471718)	PMID: 30471717, PMID: 30471718	Distal region of ciliary axonemes in respiratory cells. In sperm cells, it localises the entire length of the sperm tail.	x				x
ENSG00000122735	dynein axonemal intermediate chain 1(DNAI1)	DNAI1	604366	Axonemal outer dynein intermediate-chain protein. Mutations result in primary ciliary dyskinesia with situs solitus or situs inversus (PMID: 11231901). DNAI and DNAI2 form a complex essential for the stability of both proteins during early stages of outer dynein arms assembly (PMID: 29113992)	PMID: 11231901, PMID: 29113992.	Ciliary axoneme.	x		x		x
ENSG00000171595	dynein axonemal intermediate chain 2(DNAI2)	DNAI2	605483	Outer dynein arm intermediate chain protein. DNAI and DNAI2 form a complex essential for the stability of both proteins during early stages of outer dynein arms assembly (PMID: 29113992). Essential for axonemal assembly of the ODA heavy chains DNAH5 and DNAH9 (PMID: 20707998)	PMID: 29113992, PMID: 18950741, PMID: 20709053, PMID: 20707998	Entire length of axoneme	x		x		x

				18950741). Required for ODA formation in Kupffer's Vesicle cilia and controls the establishment of left-right body axis and kidney formation through regulation of ciliary motility.(PMID: 20709053, PMID: 20707998)							
ENSG00000187726	DnaJ heat shock protein family (Hsp40) member B13(DNAJB13)	DNAJB13	617091	Role in the proper formation and function of ciliary and flagellar axonemes in humans, preserving the integrity of the central complex of motile cilia and flagella (PMID: 27486783)	PMID 27486783	Co-localises with α -tubulin along the length of cilia and flagella. Radial spokes, annulus during spermiogenesis.	x				x
ENSG00000119661	dynein axonemal light chain 1(DNAL1)	DNAL1	610062	Encodes an outer dynein arm light chain protein. Candidate gene for primary ciliary dyskinesia. Regulatory function for DNAH5 activity in outer dynein arms of sperm flagella, respiratory cilia, and ependymal cilia (PMID: 15845866). Necessary for the function of the cilia that produce the nodal flow essential for the determination of left-right asymmetry (PMID: 21496787)	PMID: 15845866, PMID: 21496787	Associated with A-tubule of the outer doublets	x		x		x
ENSG00000163879	dynein axonemal light intermediate chain 1(DNALI1)	DNALI1		Light chain of axonemal inner dynein arms protein. Formation of complexes with the dynein heavy chains, actin, and caltractin/centrin molecules required in ciliary motility (PMID: 9284741)	PMID: 9284741	Ciliary axoneme.	x		x		x

ENSG00000107554	dynamin binding protein(DNMBP)	DNMBP		Role in ciliogenesis and tubulogenesis, acting as a guanine nucleotide exchange factor that regulates Cdc42 activity. Depletion results in an absence of cilia, with impaired apical polarisation. Controller of left-right patterning (26895965). TUBA/Cdc42/Par6A complex is required to ensure the formation of a single apical domain during enterocyte polarization (PMID: 30408122).	PMID: 26895965, PMID: 30408122.	Apical-most margin of cell junctions, cytoplasm	x				
ENSG00000166171	deleted in primary ciliary dyskinesia homolog (mouse)(DPCD)	DPCD		Reported situs inversus, hydrocephalus, and suppurative rhinosinusitis in Dpcd/Poll-/- mice (PMID: 21746835). Potential role in the formation or function of ciliated cells. Mutations in DPCD may be responsible for PCD (PMID: 14630615)	PMID: 21746835, PMID: 14630615	No data			x		x
ENSG00000092964	dihydropyrimidinase like 2(DPYSL2)	DPYSL2		Neuronal protein that guides neurite growth. Also called CRMP-2, is a GSK-3 β substrate and a microtubule-binding protein. Required for primary cilium biogenesis.	PMID: 23185275	Centrosome/Basal body. Primary cilium formation during cilium formation.	x		x		x
ENSG00000157856	dynein regulatory complex subunit 1(DRC1)	DRC1	615288	Structural component of the nexin-dynein regulatory complex (N-DRC), critical for regulation of the dynein motors. Its integrity is essential for the regulation of ciliary beat. Mutations are a cause for an autosomal recessive variant of primary ciliary dyskinesia (PMID: 23354437). DRC1 and DRC2 form a subcomplex in the cytoplasm prior to assembly into the axoneme. DRC1 and	PMID: 23354437, PMID: 29167384, PMID: 31960620, PMID: 25411337.	Cytoplasm, axoneme. N-DRC starting from the linker domain, spanning across the base plate, and ending near the hole on the inner junction.	x		x		x

				DRC2 coassemble to form a central scaffold needed for assembly of the N-DRC and its attachment to the outer doublet (PMID: 29167384). Loss of function of DRC1 may weaken the N-DRC and its attachment to the peripheral microtubules, leading to axonemal microtubule disorganization (PMID: 31960620).							
ENSG00000171962	dynein regulatory complex subunit 3(DRC3)	DRC3		Component of N-DRC, important in flagellar motility (PMID: 27261005). Studies suggest it is required for other N-DRC components to assemble into the N-DRC. Its location suggests that DRC3 could be important for transducing regulatory signals from the radial spokes through the base plate of the N-DRC to the dynein motor (PMID: 25564608).	PMID: 27261005, PMID: 25564608.	Axoneme. L1 projection of the nexin linker region of N-DRC that connects to inner dynein arm.	x			x	x
ENSG00000159625	dynein regulatory complex subunit 7(DRC7)	DRC7		Central linker subunit that helps connect the N-DRC to the outer dynein arms. Loss of DRC7 causes defects in sperm storage and fertility (PMID: 31659045). Essential for the correct assembly of the N-DRC and flagella; stability of the N-DRC is essential for proper axoneme elongation. DRC7 is essential to localize DRC5 in the correct place, and that DRC5 regulates flagellar motility (PMID: 31961863)	PMID: 31659045, PMID: 31961863.	Entire axoneme. Central region of the N-DRC linker including the outer dynein/inner dynein linker.	x			x	x
ENSG00000184845	dopamine receptor D1(DRD1)	DRD1		Role in motile ciliary beating in ependymal cells by regulation of catecholamines. Mediation of cellular responses to distinct endogenous catecholamines, which suggests that primary cilia play a fundamental role in enhancing the selectivity of integrated catecholamine signaling (PMID: 23936473)	PMID: 23936473	Ciliary surface, plasma membrane	x		x		x

ENSG00000149295	dopamine receptor D2(DRD2)	DRD2		Regulates hormone secretion and proliferation. Plays a key role in dopaminergic regulation (PMID: 21673453)	PMID: 21673453	Ciliary surface, plasma membrane	x		x		x
ENSG00000169676	dopamine receptor D5(DRD5)	DRD5		Ciliary mechanosensory protein in vascular endothelial cells. DR5 activation increases cilia length in a concentration-dependent manner and restores cilia function in the mutant cells (PMID: 21709211)	PMID: 21709211	Primary cilia, ciliary surface.	x		x		x
ENSG00000107404	dishevelled segment polarity protein 1(DVL1)	DVL1	601365	Essential for the docking of basal bodies to the apical plasma membrane through the activation of the Rho GTPase. DVL and Rho are required for the planar polarization of basal bodies that underlies directional beating of cilia (PMID: 18552847). DVL is a central component of WNT signalling. PTEN regulates multicilia formation and cilia disassembly by controlling the phosphorylation of DVL (PMID: 26399523).	PMID: 18552847, PMID: 26399523	Basal Body	x		x		x
ENSG00000187240	dynein cytoplasmic 2 heavy chain 1(DYNC2H1)	DYNC2H1	603297	Also known as DHC2. Component of interflagellar transport dynein assembly complex, that regulates retrograde trafficking from the ciliary tip to the base (PMID: 30429209, PMID: 9971742). Directly involved in the contacts and translocation of the dynein complex along microtubules via its large motor domain (PMID: 19442771). Role in Golgi	PMID: 30429209, PMID: 9971742, PMID: 19442771, PMID: 8666668.	Golgi, axoneme, peri-basal body pool	x		x		x

				organisation and function (PMID: 8666668).								
ENSG00000138036	dynein cytoplasmic 2 light intermediate chain 1(DYNC2LI1)	DYNC2LI1	617083	Component of the dynein-2 microtubule motor protein complex with a role in the retrograde transport of cargo in primary cilia. Mutations have been associated to Ellis-van Creveld syndrome (PMID: 28857138, PMID: 26130459). Essential for dynein-2 complex stability, cilium function, Hedgehog regulation, skeletogenesis and primary cilia length (PMID: 26077881),	PMID: 28857138, PMID: 26130459, PMID: 26077881	Cytoplasm, centrosomes, around basal body, transition zone.	x			x	x	
ENSG00000088986	dynein light chain LC8-type 1(DYNLL1)	DYNLL1		Control of retrograde intraflagellar trafficking of ciliary proteins through interaction with WDR34 and ciliogenesis by serving as dynein-1 subunits (PMID: 25294941, PMID: 30649997).	PMID: 25294941, PMID: 30649997	Base of cilium, ciliary axoneme.	x				x	
ENSG00000264364	dynein light chain LC8-type 2(DYNLL2)	DYNLL2		Protein existing within cytoplasmic dynein 2. Interacts with DYNLL1 and WDR34 to control retrograde intraflagellar transport. Studies suggest that DYNLL2 may be able to function in place of DYNLL1. (PMID: 25294941, PMID: 30649997)	PMID: 25294941, PMID: 30649997	Base of cilium, ciliary axoneme.	x					

ENSG00000125971	dynein light chain roadblock-type 1(DYNLRB1)	DYNLRB1	Required for the correct assembly of the dynein complex. It has been implicated in controlling its association with cargo molecules (PMID: 25205765). Component of the WDR34 subcomplex (DYNLL1/DYNLL2 and DYNLRB1/DYNLRB2), crucial for dynein-2 function in retrograde ciliary protein trafficking (PMID: 30649997). DYNLRB1 is essential for dynein-mediated axonal transport of lysosomes, late endosomes and signaling endosomes (PMID: 32088381)	PMID: 25205765, PMID: 30649997, PMID: 32088381	Base of cillium, ciliary axoneme.	x					x
ENSG00000168589	dynein light chain roadblock-type 2(DYNLRB2)	DYNLRB2	Component of the WDR34 subcomplex (DYNLL1/DYNLL2 and DYNLRB1/DYNLRB2), crucial for dynein-2 function in retrograde ciliary protein trafficking (PMID: 30649997, PMID: 32088381).	PMID: 30649997, PMID: 32088381	Base of cillium, ciliary axoneme.	x					x
ENSG00000146425	dynein light chain Tctex-type 1(DYNLT1)	DYNLT1	Also known as TCTEX1. Light chain subunit of cytoplasmic dynein. Regulation of ciliary resorption and S-phase entry. Tctex-1 is not required for ciliogenesis, but is critical for cilium disassembly (PMID: 21394082). Component of dynein-2. Modulator of cilia length control; depletion is associated to increased cilia length (PMID: 21700358).	PMID:21700358, PMID:21394082	Axoneme, transition Zone between primary cilia and basal body.	x		x			x

ENSG00000089091	double zinc ribbon and ankyrin repeat domains 1(DZANK1)	DZANK1		Interaction partner of NINL. NINL-DZANK1 is involved in the proper assembly and folding of the cytoplasmic dynein 1 motor complex, which is essential for intracellular dynein-based transport (PMID: 26485514)	PMID: 26485514	Both centrioles of centrosome and basal body.	x				x
ENSG00000134874	DAZ interacting zinc finger protein 1(DZIP1)	DZIP1		Mediates membrane/basal-body interactions, functions in basal-body maturation and transition zone organisation. Required to recruit Fam92 and Cby at the basal body distal end, allowing proper transition zone assembly; both DZIP and FARM92 control the distal elongation of basal bodies and docking to the plasma membrane (PMID: 31821146, PMID: 32491167). Mediates the assembly of the BBSome-Dzip1-PCM1 complex in the centriolar satellites for ciliary translocation of the BBSome (PMID: 27979967). Necessary for recruiting IFT machinery to the distal appendages of the mother centrioles and for bridging IFT88 to the cargo that contains GLI3. Positive regulator of Shh (PMID: 23955340). Together with GSK3 β and Rab8, DZIP1 co-regulates ciliogenesis by promoting ciliary membrane assembly (PMID: 25860027).	PMID: 27979967, PMID: 32491167, PMID: 31118289, PMID: 23955340, PMID: 25860027.	Transition zone, mother centrioles, centriolar satellites, cytoplasm, nucleus.	x				x

ENSG00000158163	DAZ interacting zinc finger protein 1 like(DZIP1L)	DZIP1L	617570	Role in Hh signaling and ciliogenesis. It promotes ciliary bud formation by removing Cp110 from the distal end of the mother centriole, and is required for the integrity of the transition zone as it is implicated in maintenance of the periciliary diffusion barrier at the ciliary transition zone. Regulates the localisation of Cby to the appendages of mother centriole. (PMID: 29487109, PMID: 28530676)	PMID: 29487109, PMID: 28530676	Mother centriole and distal end of basal body, possibly at the site where the transition zone forms.	x					x
ENSG00000205250	E2F transcription factor 4(E2F4)	E2F4		Mediation of the transcriptional responses of Multicilin. Forms apical cytoplasmic organizing centres for multiciliogenesis, recruitment and accumulation of early regulators of centriole biogenesis to allow assembly and nucleation of deuterosomes (PMID: 28675157).	PMID: 28675157	Nucleus, cytoplasm (beneath primary cilium and in small foci or granules).		x				x
ENSG00000203965	EF-hand calcium binding domain 7(EFCAB7)	EFCAB7		Positive regulation of Hedgehog pathway. Adaptor to link the EVC-EVC2 complex to IQCE (PMID: 24582806).	PMID: 24582806	Base of cilia in the EVC zone	x					x
ENSG00000096093	EF-hand domain containing 1(EFHC1)	EFHC1	608815	Also known as RIB72. Encodes a non-ion channel protein myoclonin1. Modulates TRP mechanosensitive calcium channels. Mutations are linked to juvenile myoclonic epilepsy (PMID: 30810526). Regulation of Wnt signaling (PMID: 26783883). Interacts functionally with the system of linkages that hold the doublet microtubules and limits microtubule sliding (PMID: 12435737,	PMID: 30810526, PMID: 26783883, PMID: 12435737,	Ciliary axoneme.	x		x			x

				12435737).								
ENSG00000146648	epidermal growth factor receptor(EGFR)	EGFR		Suppression of ciliogenesis by directly phosphorylating the deubiquitinase USP8, which stabilises the trichoplein-Aurora A pathway (PMID: 29472535) and through the MEK and ERK pathway (PMID: 31430961). Contributes to cell cycle progression through inhibition of ciliogenesis (PMID: 29472535).	PMID: 29472535, PMID: 31430961.	Golgi, plasma membrane, clathrin coated vesicle membrane, lipid raft, cell junctions		x				
ENSG00000110047	EH domain containing 1(EHD1)	EHD1		Required for ciliogenesis by regulating activation and repression of the Sonic hedgehog (SHH) signalling pathway. Trafficking of SHH signaling components to the primary cilia; its activation leads to entry of SMO into cilia (PMID: 26884322). Role in ciliary vesicles assembly. Formation of membrane tubules that attach the intracellular developing cilium to the cell surface and create an extracellular membrane channel (EMC). Recruitment of PACSIN proteins to cellular membranes (PMID: 30683896). Responsible for removal of the capping protein CP110 from the m-centriole to facilitate ciliary growth. Function in early ciliogenesis by causing mother centriole reorganisation. Regulation of membrane tubulation required for ciliary vesicle formation from smaller distal appendage	PMID: 26884322, PMID: 30683896, PMID: 25686250	Ciliary pocket membrane, membrane tubules, pre-ciliary vesicles and distal appendage vesicles. Centrosome in the absence of Rab8	x					x

				vesicles. EHD1 and EHD3 bind to the Rab8 effector MICAL-L1 and affect membrane tubulo-vesicle formation and scission (PMID: 25686250).							
ENSG00000013016	EH domain containing 3(EHD3)	EHD3		Function in early ciliogenesis by causing mother centriole reorganisation. Regulation of membrane tubulation required for ciliary vesicle formation from smaller distal appendage vesicles. EHD1 and EHD3 bind to the Rab8 effector MICAL-L1 and affect membrane tubulo-vesicle formation and scission (PMID: 25686250).	PMID: 25686250	Ciliary pocket membrane, membrane tubules, pre-ciliary vesicles and distal appendage vesicles.	x				x
ENSG000000151023	enkurin, TRPC channel interacting protein(ENKUR)	ENKUR		Required for patterning the left/right axis in vertebrates. Mutations in humans cause situs inversus (PMID: 29257953). Potential role as a Ca2+ sensor that participates in the regulation of mammalian flagellar bending (PMID: 29733335, PMID: 15385169).	PMID: 29257953, PMID: 29733335, PMID: 15385169.	Motile cilia, axoneme.Near to developing acrosome in mouse spermatides.	x			x	x
ENSG000000165689	endosome associated trafficking regulator 1(ENTR1)	ENTR1		Interacts with the intraflagellar transport protein 88 (IFT88), involved in ciliogenesis and intraciliary transport. Required for ciliary localization of Polycystin-2, mediating the transport of Polycystin-2 from the ciliary base into the cilium (PMID: 27767179).	PMID: 27767179	Early and recycling endosomes, basal body (in the proximity interactome of several centriole distal appendage proteins)	x				

ENSG00000072840	EvC ciliary complex subunit 1(EVC)	EVC	604831	Positive regulator of the Hh signalling pathway, interacting with Smo and promoting Sufu/Gli3 dissociation and Gli3 ciliary traffic (PMID: 17660199, PMID: 23026747, PMID: 21356043, PMID: 22986504). Mutated in Ellis-van Creveld syndrome (PMID: 21356043). EVC1 and EVC2 are mutually required for localizing to primary cilia and also for maintaining their normal protein levels (PMID: 23026747).	PMID: 17660199, PMID: 23026747, PMID: 22986504, PMID: 21356043	Distal end of the basal body, primary cilia.	x		x		x
ENSG00000173040	EvC ciliary complex subunit 2(EVC2)	EVC2	607261	Positive regulator of the Hh signalling pathway, interacting with Smo and promoting Sufu/Gli3 dissociation and Gli3 ciliary traffic (PMID: 17660199, PMID: 23026747, PMID: 21356043, PMID: 22986504, PMID: 22981989). Mutated in Ellis-van Creveld syndrome (PMID: 21356043). EVC1 and EVC2 are mutually required for localizing to primary cilia and also for maintaining their normal protein levels (PMID: 23026747).	PMID: 17660199, PMID: 23026747, PMID: 22986504, PMID: 21356043, PMID: 22981989	Basal body, primary cilia, nucleus.	x		x		x
ENSG00000180104	exocyst complex component 3(EXOC3)	EXOC3		Also known as SEC6. Inhibition of ciliary shedding, blocks deciliation-associated down-regulation of claudin-2 and gp135 (PMID:19005211). Sec6/8 complex cooperates with cooperate with syntaxin 3 and SNAP-25, ceing a candidate effector for Rab8 at the rhodopsin transport carriers (RTC) docking site (PMID: 19454479)	PMID: 19005211, PMID: 19454479.	Primary cilia, domains in the vicinity of RTC fusion sites	x		x		x

ENSG00000131558	exocyst complex component 4(EXOC4)	EXOC4		Also known as SEC8. Sec6/8 complex cooperates with cooperate with syntaxin 3 and SNAP-25, ceing a candidate effector for Rab8 at the rhodopsin transport carriers (RTC) docking site (PMID: 19454479). Together with Dvl and Rho, Sec8 is required for regulating the docking and maintenance of basal bodies to the apical plasma membrane in multi-ciliated cells (PMID: 18552847).	PMID: 19454479, PMID: 18552847	Primary cilia, domains in the vicinity of RTC fusion sites. Apical cell surface	x		x		x
ENSG00000070367	exocyst complex component 5(EXOC5)	EXOC5		Also known as SEC10. Central and particularly important exocyst component required for exocyst formation and function. Regulation of primary ciliogenesis and cystogenesis through interactions with Par3 (PMID: 19297529, PMID: 30824539). Linker for other exocyst components, and interacts with polycystin-2, IFT20 and IFT88. Dysfunction may contribute to ciliopathies including autosomal dominant polycystic kidney disease ADPKD (PMID: 21490950).	PMID: 19297529, PMID: 30824539, PMID: 21490950	Entire length of primary cillium, and tight junction.	x		x		x
ENSG00000138190	exocyst complex component 6(EXOC6)	EXOC6		Also known as Sec15. Interacts with Rabin8 and is a downstream effector of Rab8. Rabin8-Rab8-Sec15 interaction is involved in the regulation of vesicular trafficking, tethering, and fusion for primary cilium formation. Disruption of function of Sec15 leads to a block of endosomal recycling and impaired ciliogenesis (PMID: 22433857, PMID: 23435566).	PMID: 22433857, PMID: 23435566	Along primary cillium	x		x		x

ENSG00000144036	exocyst complex component 6B(EXOC6B)	EXOC6B		Also called sec15b. Interacts with Rabin8 which is crucial for ciliogenesis by switching binding specificity from phosphatidylserine to Sec15. Depletion decreases the number of ciliated cells.	PMID:23435566	Cell plate, near plasma membrane.		x	x		x
ENSG00000106462	enhancer of zeste 2 polycomb repressive complex 2 subunit(EZH2)	EZH2	601573	Methyltransferase unit of PRC2. EZH2 suppresses genes functionally connected to the primary cilium, promoting loss of primary cilia; inhibition induces primary cilium formation in most melanoma cell. Loss of cilia	PMID: 30008323	Primary cilium.	x				
ENSG00000188343	family with sequence similarity 92 member A(FAM92A)	FAM92A	617273	Interact with Cby1 to promote ciliogenesis via regulation of membrane-remodeling processes, inducing the formation of membrane globule like structures, which contain Rab8 (PMID: 27528616). Mutated in nonsyndromic postaxial polydactyly. Role in limb development (PMID: 30395363).	PMID: 27528616, PMID: 30395363	Centrioles/basal body, distal appendages	x				x
ENSG00000153789	family with sequence similarity 92 member B(FAM92B)	FAM92B		Cby1-interacting partner to promote ciliogenesis through regulation of membrane-remodeling processes, inducing the formation of membrane tubule like structures, which contain Rab8 (PMID: 27528616).	PMID: 27528616	Centrioles/basal body, distal appendages	x				x

ENSG00000138286	family with sequence similarity 149 member B1(FAM149B1)	FAM149B1		Required for normal ciliary biology and that its deficiency results in impaired ciliary structure and a range of ciliopathy phenotypes in humans along the spectrum of Joubert syndrome.	PMID: 30905400	Not reported					
ENSG00000170264	family with sequence similarity 161 member A(FAM161A)	FAM161A	613596	Causative gene for retinitis pigmentosa (PMID: 20705278). Retinal ciliopathy protein that is part of connecting cilium complex. Interacts with CEP290, lebercin, OFD1 and SDCCAG8. These interactions suggest that FAM161A could be involved in protein transport and docking between the inner segments and outer plexiform layer (PMID: 22940612). Component of microtubule- organizing centers that associates with the intracellular microtubule network to increase the acetylation of α -tubulin. Involved in microtubule stabilization to maintain the microtubule tracks and/or in transport processes along cytoplasmic and ciliary microtubules (PMID: 24664697, PMID: 22791751).	PMID: 22940612, PMID: 24664697, PMID: 22791751, PMID: 20705278.	Connecting cilium, basal body, adjacent centriole, centrosome. Inner segments and outer plexiform layer	x		x		x
ENSG00000164556	family with sequence similarity 183 member B, pseudogene(FAM183BP)	FAM183BP		Required for ciliogenesis and motility of epidermal multiciliated cells and nephrostomes. Interaction partner of nucleoporin NUP93, which may function as scaffolding proteins at the ciliary base.	PMID: 30279523	Basal Body	x				

ENSG00000203780	fibronectin type III and ankyrin repeat domains 1(FANK1)	FANK1		Regulator of multiciliated cell differentiation in the mouse airway epithelium.	PMID: 29661797	Basal body, cytosol, nucleus	x				x
ENSG00000188878	Fas binding factor 1(FBF1)	FBF1		Transition fiber protein required for ciliogenesis and the ciliary import of assembled intraflagellar transport particles at the cilia base (PMID: 24231678). Functions in centriole duplication by recruiting HsSAS-6 and promotes centrosome separation (PMID: 30318703).	PMID: 24231678, PMID: 30318703.	Distal appendages of mother centriole, proximal end of centrioles, transition fibers.	x		x		x
ENSG00000068078	fibroblast growth factor receptor 3(FGFR3)	FGFR3	134934	Interacts with ICK to regulate ciliogenesis and control ciliary length (PMID: 30782830). FGFR3 signaling affects the elongation process of the primary cilia. Constitutively active FGFR3 perturbs cilia length, sorting and trafficking of intraflagellar transport 20 (PMID: 29040558). Mutations in FGFR3 lead to short primary cilia, and constitutive activation of FGFR3 signaling produces the skeletal disorders achondroplasia and thanatophoric dysplasia (PMID: 29360984).	PMID: 30782830, PMID: 29040558, PMID: 29360984.	Axoneme, cytoplasm.	x				
ENSG00000137460	FH2 domain containing 1(FHDC1)	FHDC1		Coordination of cytoskeletal dynamics during normal cilia assembly. Interacts with CEP170 to regulate cilia elongation and inhibition of cilia assembly. Overexpression of FHDC1 induces F-actin accumulation, microtubule stabilization, blocks cilia assembly and induces cilia resorption	PMID: 29742020	Cytoplasmic microtubules converging on the base of the cilia.		x			

				(PMID: 29742020).							
ENSG00000154803	folliculin(FLCN)	FLCN	607273	Part of the mechanosensory machinery of primary cilia. Tumor suppressor associated with Birt-Hogg-Dubé (BHD) syndrome. Down-regulation of mTORC1 signaling pathway through primary cilia, it recruits and associates with LKB1 and recruits the kinase to primary cilia for activation of AMPK (PMID: 27072130). FLCN expression levels are critically important for correct timing of ciliogenesis and intracellular transport events involved in ciliogenesis (PMID: 23784378).	PMID: 27072130; PMID: 23784378	Motile and non-motile cilia, centrosomes, mitotic spindle, cytoplasm, nucleus.	x				x
ENSG00000196924	filamin A(FLNA)	FLNA	300017	Role in the normal processes of ciliogenesis and basal body positioning. Interacts with meckelin to regulate levels of Wnt signalling. Knockdown causes a failure in ciliogenesis and aberrant migration of the centriole pair to the apical cell surface suggestion FLNA is essential in mediating basal body migration and docking at the apical cell surface (PMID: 22121117, PMID: 28973524).	PMID: 22121117, PMID: 28973524	Basal body, transition zone and apical cell surface.	x		x		x

ENSG00000137942	formin binding protein 1 like(FNBP1L)	FNBP1L		TOCA1. Target of miR-129-3p in ciliation-related actin dynamics, which induces ciliation in proliferating cells and also promoted cilia elongation (PMID: 22684256). Binds to Cdc42, leading to actin rearrangements, which are thought to be involved in processes such as endocytosis (PMID: 27129201).	PMID: 22684256, PMID: 27129201	Cytoskeleton, cell membrane, cytoplasm, vesicles, cell cortex		x				x
ENSG00000133393	FGFR1OP N-terminal like(FOPNL)	FOPNL		Regulation of integrity of the centriole distal end and membrane anchoring. Complex with OFD1 to initiate ciliogenesis and control cilia length (PMID: 26643951). Involved in cilia formation and regulating the interaction of PCM1 satellites with microtubules and motors (PMID: 20551181). Regulates the assembly and maintains the stability and dynamics of microtubules. It may inhibit tubulin assembly by sequestering soluble tubulin (PMID: 28694353).	PMID: 26643951, PMID: 20551181, PMID: 28694353	Pericentriolar satellites, centrosomes.	x		x			x
ENSG00000129654	forkhead box J1(FOXJ1)	FOXJ1		Transcription factor that controls motile cilia formation (PMID: 19011630). Activates the transcription of numerous known ciliary genes and specifically regulates motile cilia biogenesis (PMID: 30881373). Regulation of programs promoting basal body docking and axoneme formation. Role of FOXJ1 in the differentiation of motile ciliated cells is modulated by the transcription factor Not homeobox (NOTO) and the regulatory factor X (RFX)-family (PMID: 23822649)	PMID: 19011630, PMID: 30881373, PMID: 23822649	Nucleus		x	x			x

ENSG00000139445	forkhead box N4(FOXN4)	FOXN4		Downstream target of Multicilin required for multiciliated cells differentiation (MCC). Complements Foxj1 transcriptionally by co-activating Foxj1 target genes. Ensures robust gene expression during MCC differentiation, allowing large numbers of specialized organelles to be assembled.	PMID: 27864379	Nucleus		x				
ENSG00000105255	fibronectin type III and SPRY domain containing 1(FSD1)	FSD1		Anchors microtubules asters to interphase centrosomes by directly binding microtubules to mother centriole, and controls transition zone assembly during ciliogenesis	PMID: 30538248	Middle region of both centrioles	x					
ENSG00000188738	fibrous sheath interacting protein 2(FSIP2)	FSIP2	615796	Major structural protein of the fibrous sheath (FS) shown to bind to AKAP4, required for FS assembly. Regulates PKA anchoring to AKAP4. Involved in the control of sperm flagellum elasticity and signaling pathways that regulate sperm motility. Mutations in FSIP2 are a recurrent cause of multiple morphological abnormalities of the sperm flagella.	PMID: 30137358, PMID: 30867909.	Periaxonemal structures.	x					
ENSG00000163430	follistatin like 1(FSTL1)	FSTL1		Novel regulator of ciliogenesis in a non-cell autonomous manner due to its activity as a BMP4 inhibitor. BBS4 regulates the cellular trafficking of FSTL1.	PMID: 28852127	Cytoplasm (intracellular vesicles), secreted protein expected to be in the lumen of the Golgi and post-Golgi vesicles		x				

ENSG00000010361	fuzzy planar cell polarity protein(FUZ)	FUZ	610622	Critical regulator of cilia structure and function. Trafficking and localisation of retrograde, but not anterograde, intraflagellar transport proteins, governing their traffic from the cytoplasm to the basal body and then to the ciliary tip, representing a role in maintaining ciliary trafficking to the distal axoneme (PMID: 22778277).	PMID: 22778277	Cytosol		x	x		x
ENSG000000178234	polypeptide N-acetylglucosaminyltransferase 11(GALNT11)	GALNT11		Candidate disease gene in heterotaxy. Modifies Notch to establish an essential balance between motile and immotile cilia at the left-right organiser, necessary to determine laterality.	PMID: 24226769	Golgi		x			x
ENSG000000182687	galanin receptor 2(GALR2)	GALR2		Neuronal ciliary GPCR. Galanin receptor.	PMID: 26167768	Neuronal cilia membrane	x				
ENSG000000128310	galanin receptor 3(GALR3)	GALR3		Neuronal ciliary GPCR. Galanin receptor.	PMID: 26167768	Neuronal cilia membrane	x				x

ENSG00000270765	growth arrest specific 2 like 2(GAS2L2)	GAS2L2		Alignment of basal bodies. Mediation of the crosstalk between microtubules, actin filaments, and the basal body. Mutations cause poorly aligned cilia, hyperkinetic ciliary beat, and primary ciliary dyskinesia.	PMID: 30665704	Microtubules, basal body, basal feet, rootlets, actin filaments.	x				
ENSG00000141013	growth arrest specific 8(GAS8)	GAS8	605178	Role in ciliary motility. Implicated in the Hh signaling pathway by binding to Smo and regulating its trafficking into the cilium (PMID: 21659505). Mutations are associated with human disease as a cause of severe PCD phenotypes (PMID: 27472056).	PMID: 27472056, PMID: 21659505	Axoneme of motile cilia, base of primary cilia, vesicles in cytoplasm.	x		x		x
ENSG00000111087	GLI family zinc finger 1(GLI1)	GLI1		Hedgehog signaling pathway effector. GLI1 activation depends on cilia.	PMID: 21209912	Ciliary Tip	x		x		x
ENSG00000074047	GLI family zinc finger 2(GLI2)	GLI2		Hedgehog signaling pathway effector. Cilia or interflagellar transport proteins are required for GLI2 function and regulation (PMID: 21209912). Regulates autophagy-mediated Odf1 removal to control primary cilia formation and length (PMID: 30463852).	PMID: 21209912, PMID: 30463852	Ciliary Tip	x		x		x

ENSG00000106571	GLI family zinc finger 3(GLI3)	GLI3		Hedgehog signaling pathway effector. Loss of Polaris results in inefficient Gli3 processing. Inhibit Gli1-mediated transcription.	PMID: 21209912	Ciliary Tip	x		x		x
ENSG00000126603	GLIS family zinc finger 2(GLIS2)	GLIS2	608539	Bifunctional transcriptional regulator, either as an activator and a repressor (PMID: 11741991). Repressor of Hh signaling. Loss of function causes nephronophthisis-like phenotype (PMID: 27181777). Involved in epithelial-to-mesenchymal transition, fibrosis and apoptosis. Together with BBS1, GLIS2 (also called NPHP7) is required for normal ciliary motility, responsible for body asymmetry (PMID: 24069149).	PMID: 11741991, PMID: 27181777, PMID: 24069149.	Nucleus, cytoplasm and cilium.	x		x		x
ENSG00000205835	geminin coiled-coil domain containing(GMNC)	GMNC		Also known as GEMC1. Regulator of multiciliated cells differentiation, interacting with E2F4/5-DP1 and Multicilin. Candidate gene for human reduced generation of multiple motile cilia RGMC (PMID: 26933123). It transactivates the Mcldas and FoxJ1 upstream regulatory sequences, markers of ciliogenesis (PMID: 26882546).	PMID: 26933123, PMID: 26882546.	Nucleus		x			x

ENSG00000114349	G protein subunit alpha transducin 1(GNAT1)	GNAT1	139330	Subunit of transducin. Localises to photoreceptor outer segment. Requires RP2 for normal localisation	PMID: 26034134, PMID: 27481943	photoreceptor outer segment	x				x
ENSG00000078369	G protein subunit beta 1(GNB1)	GNB1		Subunit of transducin. Localises to photoreceptor outer segment. Requires RP2 for normal localisation	PMID: 26034135, PMID: 30970040	photoreceptor outer segment	x				x
ENSG00000173230	golgin B1(GOLGB1)	GOLGB1		Required for ciliogenesis and control of ciliary length, through the regulation of pericentrosomal location of the dynein-2 intermediate chain WDR34. Depletion results in cilia number and structure (fewer but longer cilia) (PMID: 28546340, PMID: 24046448).	PMID: 28546340; PMID: 24046448	Golgi stack		x			
ENSG00000179921	G protein-coupled bile acid receptor 1(GPBAR)	GPBAR		Also known as TGR5. Linked to ERK signaling, which is activated in ciliated cholangiocytes but inhibited in nonciliated cholangiocytes. Involved in bile acid-induced proliferation of nonciliated cells. Its localisation allows cholangiocyte functional responses to bile acid signaling.	PMID: 23578785	Apical plasma, ciliary, and nuclear membrane	x				

ENSG00000143147	G protein-coupled receptor 161(GPR161)	GPR161		GPR161 increases cAMP levels, establishing the basal cAMP gradient and activation of PKA. Repression of Shh signaling by determining the processing of Gli3 to its repressor form (PMID: 23332756).	PMID:23332756	Cilium	x		x		x
ENSG00000184194	G protein-coupled receptor 173(GPR173)	GPR173		Also known as SREB3. Neuronal ciliary GPCR which may play sensory or signalling roles in the ciliary vesicle or procilium during ciliogenesis	PMID: 26167768	Adjacent centriole and the basal body of the connecting cilium	x				
ENSG00000204882	G protein-coupled receptor 20(GPR20)	GPR20		Neuronal ciliary GPCR	PMID: 26167768	Adjacent centriole and the basal body of the connecting cilium	x				
ENSG00000172209	G protein-coupled receptor 22(GPR22)	GPR22		Regulation of Left-Right asymmetry through the Kupffer's vesicle (KV) and maintains cilia length and structure in KV.	PMID: 25335082	Cilium membrane, cell membrane	x				

ENSG00000123901	G protein-coupled receptor 83(GPR83)	GPR83	605472	Suggested to belong to NPY receptor family, which shares common determinants for ciliary trafficking	PMID: 24316073	Cilium membrane, cell membrane	x				x
ENSG00000181656	G protein-coupled receptor 88(GPR88)	GPR88		Regulation of D1 dopamine receptor (D1R) signaling to inhibit cAMP accumulation, without affecting B2AR-mediated response.	PMID: 23936473	Cilium.	x				
ENSG00000160360	G protein signaling modulator 1(GPSM1)	GPSM1		Also known as AGS3. Regulation of Gβγ signaling pathway to recruits phospho(T94)Tctex-1 to the transition zone. Stabilises the GDP-bound form of Gαi with IGF-1R to prevent its re-association with Gβγ and maintain it free.	PMID: 23954591	Base of the cilium, basal body.	x				
ENSG00000185974	G protein-coupled receptor kinase 1(GRK1)	GRK1	180381	Phosphorylation of opsins, playing a key role in the shutdown of the rod and cone cascades (PMID: 25422369). Peripherally associated proteins anchored to the membrane by lipid modifications. It inactivates rhodopsin through phosphorylation (PMID: 25550383).	PMID: 25422369, PMID: 25550383	Cone and rod outer segments.	x				x

ENSG00000173020	G protein-coupled receptor kinase 2(GRK2)	GRK2		Positive regulation of Sonic Hedgehog signaling, regulating phosphorylation events required for ciliary accumulation and activation of smoothened (PMID: 21695114). Modulates dynamic intracellular trafficking of Smo and contributes to the positive effect of Gas8 on Smo signaling. Modulates microtubule organisation through the phosphorylation of cytoplasmic tubulin or other proteins (PMID: 21659505).	PMID: 21695114, PMID: 21659505	Cell membrane, cytoplasm		x				x
ENSG00000082701	glycogen synthase kinase 3 beta(GSK3B)	GSK3B		Involved in primary cilia elongation, microtubule assembly, stability and dynamics. Modulates CRMP-2 ciliary localisation, and its ephosphorylated state is required for ciliary function (PMID: 23185275). Ciliary-maintenance signalling network, its disruption causes loss of cilia (PMID: 17450132). Phosphorylates Sufu and stabilises it against Shh signaling-induced degradation (PMID: 21317289).	PMID: 23185275, PMID: 17450132, PMID: 21317289	Cillium	x		x			x
ENSG00000173805	huntingtin associated protein 1(HAP1)	HAP1		HTT-HAP1-dynactin complex facilitates microtubule dependent vesicular transport. Regulates PCM1 trafficking and interacts with and huntingtin to control ciliogenesis. Depletion leads to dispersion of PCM1 from centrosomes and reduced ciliogenesis in cells. (PMID: 21985783).	PMID:21985783	Centrosome, pericentriolar satellites	x		x			x

ENSG00000113249	hepatitis A virus cellular receptor 1(HAVCR1)	HAVCR1		Also known as KIM1. Biomarker of proximal tubular injury. Depletion does not disturb cilia formation (PMID: 20375059). Modulates ciliary calcium signaling and plays a role in ciliary flow sensing (PMID: 18273441; PMID: 17205356).	PMID: 18273441, PMID: 17205356, PMID: 20375059	Apical membrane		x			x
ENSG00000094631	histone deacetylase 6(HDAC6)	HDAC6		Regulation of ciliary resorption in response to extracellular stress and it is also a downstream AurA effector. Destabilises the microtubular axoneme through deacetylation of tubulin, which promotes ciliary disassembly (PMID: 17604723, PMID: 24026243). Mediates an autophagy-dependent mechanism called ciliophagy (PMID: 24200693)	PMID: 17604723, PMID: 24026243, PMID: 24200693	Apical cell surface, cytoplasm.		x			x
ENSG00000100644	hypoxia inducible factor 1 alpha subunit(HIF1A)	HIF1A		Maintains the chondrocytes healthy under hypoxia (PMID: 31357086). Stabilised by USP8 through deubiquitination, which is essential for its ciliogenesis-promoting function (PMID: 24378640).	PMID: 24378640, PMID: 31357086	Nucleus, nuclear speckles, cytoplasm		x			x
ENSG00000275410	HNF1 homeobox B(HNF1B)	HNF1B	189907	HNF1B deficiency is associated with ciliary anomalies in cholangiocytes and causes renal cysts. Regulates bile duct formation (PMID: 22706971). Control of the transcription of several genes expressed in tubular epithelial cells and transcriptional activation of Tg737/Polaris. HNF1β is not directly necessary for ciliogenesis, nevertheless, it may regulate the expression of genes that play a crucial role in cilium function (PMID: 22706971, PMID: 15029248)	PMID: 22706971, PMID: 15029248	Nucleus.		x			

				15029248).								
ENSG00000117594	hydroxysteroid 11-beta dehydrogenase 1	HSD11B1		oxysterol synthase HSD11β2 participates in the production of Smoothed-activating cilia-localised oxysterols and promotes Hedgehog pathway activity.	PMID: 30340023	ER, microsome		x				
ENSG00000080824	heat shock protein 90 alpha family class A member 1	HSP90AA1		Effector protein in the FOXJ1 regulatory network (PMID: 30881373). Hsp90α shows chaperone activity that might facilitate protein assembly and disassembly during ciliogenesis. Phosphorylated Hsp90α is likely to be a component of the distal appendages in the mother centriole (PMID: 25359884).	PMID: 30881373, PMID: 25359884	Transition neck region, periciliary base,	x			x		x
ENSG00000109971	heat shock protein family A (Hsp70) member 8(HSPA8)	HSPA8		Also called HSC70. Function in the formation or stabilization of IFT-cargo complexes working with MRJ. It may bind directly to IFT88 or to one or more additional TPR containing proteins within the IFT particle, which in turn recruit MRJ bound cargo proteins (PMID: 19302411).	PMID: 19302411	Inner and outer segments, basal body	x		x			x

ENSG00000081870	heat shock protein family B (small) member 11(HSPB11)	HSPB11	Also known as IFT25. Not required in ciliogenesis of somatic cells. Component of IFT-B complex and might be involved in retrograde transport. Interacts with IFT27 and is indispensable for sperm flagellum axoneme formation, integrity and stability in mice, as well as contributing to the trafficking of signalling molecules via BBSome complex. (PMID: 28430876, PMID: 28964737, PMID: 30709917). Regulation of ciliary trafficking of Hh pathway components, including Patched-1, Smoothened, and Gli2 (PMID: 26023097, PMID: 22595669).	PMID: 28430876, PMID: 28964737, PMID: 26023097, PMID: 22595669, PMID: 30709917.	Distal portion of basal bodies, transition zone, centrioles and cilia	x		x		x
ENSG00000158748	5-hydroxytryptamine receptor 6(HTR6)	HTR6	Also known as 5-HT6. Regulates length, morphology and function of neuronal primary cilia through ARL13B, influences cilia and axon initial segment. Involved in cognition and Alzheimer's disease development (PMID: 28931427). Enable chemosensory functions in ciliated neurons to sense 5-HT in the immediate milieu (PMID: 10432491).	PMID: 10924708, PMID: 28931427, PMID: 10432491	Ciliary shaft, ciliary membrane of neural cilia.	x		x		x
ENSG00000197386	huntingtin(HTT)	HTT	Mutated in Huntington's Disease. Scaffolding protein required for vesicular transport via interactions with microtubules and motor protein cargo adaptors such as HAP1 and dynactin. Functional role in autophagy (PMID: 26160070). Essential for protein trafficking to the centrosome and normal ciliogenesis. Retrograde trafficking of PCM1 regulated by HTT and HAP1 to control ciliogenesis (PMID: 21985783).	PMID: 26160070, PMID: 21985783.	Centrosome, pericentriolar satellites, basal body. Tip of axoneme of photoreceptor cilium	x		x		x

ENSG00000157423	HYDIN, axonemal central pair apparatus protein(HYDIN)	HYDIN	610812	Candidate for hydrocephalus in humans and primary ciliary dyskinesia (PMID: 23022101). Interacts with the central pair proteins CPC1 and kinesin-like protein 1 (KLP1). Required for flagellar motility. Regulation of contact with the radial spokes dynein arm activity (PMID: 17296796). Lack of Hydin causes deficiency on ciliary orientation and reduced alignment of neighboring cilia, as it is involved in regulating the dynein arms during the transitions between effective and recovery strokes (PMID: 18250199).	PMID: 17296796, PMID: 23022101, PMID: 18250199.	Central pair apparatus, C2b projections.	x		x		x
ENSG00000198331	HYLS1, centriolar and ciliogenesis associated(HYLS1)	HYLS1		Required for intraflagellar transport trafficking and for centrioles to form cilia. Loss of HYLS-1 results in severe defects in axoneme assembly and transition zone organisation, due to impaired entry of IFT machinery to the cilia via DYF-19 (PMID: 28411189). Ciliogenesis mediation by regulating the formation of the ciliary gate; minor role in transition zone assembly. Required for distal centriole elongation and recruiting distal centriole proteins (PMID: 32509774). Essential for the apical targeting and/or anchoring of centrioles/basal body at the plasma membrane, contributin to basal body organisation (PMID: 19656802).	PMID: 28411189, PMID: 32509774, PMID: 19656802	Ciliary base, remanant in centriole. Outer centriole wall during centriole assembly.	x		x		x

ENSG00000163913	intraflagellar transport 122(IFT122)	IFT122	606045	Negative regulator of the Shh pathway, controlling the ciliary localization of Shh pathway activators and repressors. Loss of IFT122 causes accumulation of Gli2 and Gli3 at cilia tips. Function in retrograde intraflagellar transport (ITF) that facilitates complex A stability (PMID: 21209331). Required for transport of opsin, and the distal elongation of outer segments (PMID: 27681595). IFT122 connects the core and peripheral subcomplexes of the IFT-A (PMID: 29220510).	PMID: 21209331, PMID: 27681595, PMID: 29220510.	Basal Body, cilium, cytoplasm.	x		x		x
ENSG00000187535	intraflagellar transport 140(IFT140)	IFT140	614620	Component of IFT-A complex, mutated in skeletal ciliopathies with kidney disease (PMID: 23418020). Associates with IFT122 and IFT144 to a stable IFT-A core subcomplex; indispensable for dynein proteins retrograde transport into the cytoplasm and flagellar assembly. Required for development and maintenance of outer segments, ciliary/flagellar assembly and signal transduction (PMID: 29236364, PMID: 24619649).	PMID: 23418020, PMID: 29236364, PMID: 24619649.	Basal Body, cilium, cytoplasm.	x		x		x
ENSG00000138002	intraflagellar transport 172(IFT172)	IFT172	607386	Component of IFT complex B. Involved in the switching between the anterograde and retrograde transport (PMID: 29659833). Essential for ciliary growth and maintenance, and morphogenesis of the cerebral cortex. Mutation cause alterations in multiple intracellular signaling pathways (PMID: 31850339, PMID: 18930042). Mutations are associated with severe skeleto-renal disorders with retinal involvement (PMID: 25168386).	PMID: 29659833, PMID: 31850339, PMID: 18930042, PMID: 25168386.	Basal Body, Cilium, IFT	x		x		x

ENSG00000109083	intraflagellar transport 20(IFT20)	IFT20		Component of IFT complex B that functions as a link between IFT complex and kinesin II, binding to both IFT57 and KIF3B, making it essential for anterograde transport in cilia and flagella (PMID: 12821668). Required for ciliary assembly and localisation of polycystin-2 and opsin to cilia (PMID: 16775004). Essential for spermiogenesis and male fertility (PMID: 27682589).	PMID: 12821668, PMID: 16775004, PMID: 27682589	Basal Body, centrioles, Cilium, Golgi, IFT	x		x		x
ENSG00000128581	intraflagellar transport 22(IFT22)	IFT22		Also known as RABL5, component of IFT complex B. Controls the cellular levels of both IFT-A and IFT-B proteins and to regulate availability of particles participating in IFT (PMID: 30940671). Recruitment of BBSome to the basal body through BBS3 and regulation of BBSome entry into the cilia (PMID: 31953262).	PMID: 30940671, PMID: 31953262.	Basal body, cilium, cell body, IFT	x		x		x
ENSG00000100360	intraflagellar transport 27(IFT27)	IFT27	615870	Also known as RABL4, component of IFT complex B. Forms a complex with IFT25 and associates with IFT-B inside cilia, interacts with and stabilises the nucleotide-free form of ARL6 to promote the exit of BBSome out of cilia. Mutated in Bardet-Biedl syndrome (PMID: 25443296). Required for BBSome trafficking essential for hedgehog signaling. IFT25/IFT27 complex plays a role in transporting Gli2 to the ciliary tip (PMID: 25446516). Role in the cargo loading of the retrograde transport machinery (PMID: 24843028).	PMID: 25443296, PMID: 25446516, PMID: 24843028	Basal Body, Cilium, IFT	x		x		x

ENSG00000119650	intraflagellar transport 43(IFT43)	IFT43	614068	Component of IFT complex A. Mutations leads to abnormal ciliogenesis (PMID: 28973684). Interacts with IFT121 to regulate stability of IFT-A, and is essential for ciliogenesis. Involved in transporting ciliary precursors from the cytoplasmic pool to the peri-basal body region (PMID: 28207750)	PMID: 28973684, PMID: 28207750	Peri-basal body region, Cilium, IFT	x		x		x
ENSG00000118096	intraflagellar transport 46(IFT46)	IFT46		Component of IFT complex B, required for ciliogenesis and controls cilia number and length (PMID: 25722189). Plays a role in the transport of IFT proteins between the cytoplasm and cilia (PMID: 29915351). IFT46 is crucial for transport of ODA16 into the flagellum; it interacts with IFT52 to assembly a subcomplex which is a prerequisite for the assembly of IFT-B1 (PMID: 28701346).	PMID: 25722189, PMID: 29915351, PMID: 28701346	Basal Body, Cilium, IFT	x		x		x
ENSG00000101052	intraflagellar transport 52(IFT52)	IFT52	617094	Component of IFT complex B. Essential for anterograde complex integrity, biosynthesis and maintenance of cilia. Regulation of ciliogenesis and cilia length; it also plays a role as a docking site for complex B proteins attaching to the transition fibres of the basal bodies and is required for localization of the IFT-B complex proteins (PMID: 27466190). Recruits and interacts with IFT46 to assembly a subcomplex which is a prerequisite for the assembly of IFT-B1, it is also required to localise IFT46 to the basal body (PMID: 28302912).	PMID: 27466190, PMID: 28302912	Basal Body, Cilium, IFT	x		x		x

ENSG00000114446	intraflagellar transport 57(IFT57)	IFT57	606621	Component of IFT complex B. Required for effective transport, outer segment maintenance and ciliary elongation. Mediates the interaction between IFT20 and the IFT particle (PMID: 18492793). Trafficking of motility-related proteins and prevents IFT particle complex from degradation (PMID: 28104816).	PMID: 18492793, PMID: 28104816	Basal Body, Cilium, IFT	x		x		x
ENSG00000096872	intraflagellar transport 74(IFT74)	IFT74	608040	Component of IFT complex B. Interacts with IFT81 to stabilise IFT-B complex. This interaction is required for binding and transport of tubulin to regulate cilium length, required for ciliogenesis (PMID: 23990561). Essential for normal mouse spermatogenesis and sperm flagella formation by transporting cargo proteins and modulating axoneme and microtubule assembly (PMID: 31004481).	PMID: 31004481, PMID: 23990561	Basal Body, centrosomes, Cilium, IFT	x		x		x
ENSG00000068885	intraflagellar transport 80(IFT80)	IFT80	611177	Component of IFT complex B. Role in cilia formation and osteogenesis through regulating Hedgehog/Gli signal pathways. Silencing IFT80 led to either shortening or loss of cilia and the decrease of cilia marker protein Arl13b expression. Mutations cause Jeune asphyxiating thoracic dystrophy (JATD) and short rib polydactyly type III (SRPIII) (PMID: 22771375). Required for IFT mediated recruitment and transport of ciliary cargo, as well as for initiation of the ciliary axoneme (PMID: 29658880).	PMID: 22771375, PMID: 29658880	Basal Body, Cilium, IFT	x		x		x

ENSG00000122970	intraflagellar transport 81(IFT81)	IFT81	605489	Component of IFT complex B. Interacts with IFT74 to form a tubulin-binding module that is the main site of tubulin binding for IFT and is required for ciliogenesis. Both are essential for flagellar assembly and elongation (PMID: 27068536, PMID: 23990561).	PMID: 27068536, PMID: 23990561	Basal Body, Cilium, IFT	x		x		x
ENSG00000032742	intraflagellar transport 88(IFT88)	IFT88		Component of IFT complex B. Required for ciliary and flagellar assembly and may play a role in the attachment of the IFT particle to its cargo or to the anterograde IFT motor FLA10-kinesin-II (PMID: 11062270).	PMID: 11062270	Basal Body, Cilium, IFT	x		x		x
ENSG00000017427	insulin like growth factor 1(IGF1)	IGF1		IGF-1 signals are predominantly transduced through IGF-1 receptor (IGF-1R) on the cilia. Promotes cell cycle entry. Mediation of ciliary resorption through activation of IGF-1R (PMID: 23954591). IGF1 mediated signaling regulates ciliated cell differentiation (PMID: 28050756).	PMID: 23954591, PMID: 28050756	Secreted		x			
ENSG00000140443	nsulin like growth factor 1 receptor(IGF1R)	IGF1R		Activation of an AGS3-regulated Gβγ signaling pathway that subsequently recruits phospho(T94)Tctex-1 to the transition zone, leading to cilia resorption (PMID: 23954591). IGF1R mediated plays a role in regulating ciliated cell differentiation (PMID: 28050756).	PMID: 23954591, PMID: 28050756	Cell surface, primary cillium, ciliary membrane.	x				

ENSG00000204084	inositol polyphosphate-5-phosphatase B	INPP5B		Compensatory role in cilia development and regulation. Regulation of ciliary length (PMID: 23805271).	PMID: 23805271	Endosomes, cell membrane, outer segment of the photoreceptors, along primary cilia, basal body	x				
ENSG00000148384	inositol polyphosphate-5-phosphatase E(INPP5E)	INPP5E	613037	Regulator of transition zone function. INPP5E may regulate Hedgehog signaling via modulating the PI(4,5)P2-dependent recruitment of GPR161 to cilia. It is essential for maximal SMO accumulation at cilia after the Hh pathway (PMID: 27998989). Controls ciliary growth factor, PI3K signaling and stability (PMID: 19668215). Mutations cause Joubert Syndrome and promotes premature destabilization of cilia (PMID: 19668216).	PMID: 27998989, PMID: 19668215, PMID: 19668216.	Axoneme, adjacent to the base of the cilium	x		x		x
ENSG00000164066	inturned planar cell polarity protein(INTU)	INTU	610621	Governs apical actin assembly and thus controls the orientation, but not assembly, of ciliary microtubules (PMID: 16493421, PMID: 27158779). Cilia and planar polarity effector, essential for IFT-A complex assembly during ciliogenesis, forming a molecular module with Fuzzy (FUZ) and WDPCP. INTU functions upstream of Gli2 activation and is required for Smo-mediated Hh pathway activation (PMID: 28459465).	PMID: 16493421, PMID: 27158779, PMID: 28459465	Plasma Membrane		x	x		x

ENSG00000119509	inversin(INVS)	INVS	243305	Mutations cause infantile nephronophthisis type 2. Role in left-right axis specification (PMID: 12872123). Forms a stable complex with the cytoskeletal molecule tubulin and may be involved during early microtubule assembly in microtubule-organizing centers (PMID: 15213257).	PMID: 12872123, PMID: 15213257	Entire microtubule cytoskeleton. Primary apical cilia, Basal Body, Centrosome, nuclear compartments, and cell-cell contacts	x		x		x
ENSG00000173226	IQ motif containing B1(IQCB1)	IQCB1	609237	Also known as NPHP5. Required for extension of an axoneme after docking of the basal body to the cell membrane. Absence leads to mislocalization of BBS2, BBS5 and destroys BBSome integrity (PMID: 27328943). Interacts with CEP290 and is required for early cilia assembly, migration and/or anchoring of centrosomes to the cell cortex (PMID: 23446637). NPHP5-binding to the BBSome is crucial for BBSome integrity and ciliary trafficking (PMID: 25552655).	PMID: 27328943, PMID: 23446637, PMID: 25552655	Basal Body, distal region of centriolles, proximal outer segments	x		x		x
ENSG00000106012	IQ motif containing E(IQCE)	IQCE		Form a complex with EFCAB7. It anchors the EVC-EVC2 complex to positively regulate the Hh pathway (PMID: 24582806)	PMID: 24582806	Base of cilium, EvC zone	x				x
ENSG00000114473	IQ motif containing G(IQCG)	IQCG		Required for cilogenesis and spermiogenesis in mammals through a dynamic interaction with calmodulin in different intracellular calcium levels (PMID: 24849454).	PMID: 24849454	Cytoskeleton, sperm flagellum axoneme manchette.	x			x	x

ENSG00000169047	insulin receptor substrate 1(IRS1)	IRS1		IRS-1 is a docking protein that interacts with intracellular downstream signalling molecules involved in IGF-1 receptor signalling, interacts with Akt-1 at the transition neck region marked by Hsp90α (PMID: 25359884).	PMID: 25359884	Basal body, axoneme.	x					
ENSG00000077684	jade family PHD finger 1(JADE1)	JADE1		Ubiquitin ligase targeting β-catenin for proteasomal degradation. Component of the NPHP protein complex, interacting with NPHP4 which facilitates JADE-1 nuclear traslocation, leading to decreased canonical Wnt signaling (PMID: 22654112, PMID: 25100726).	PMID: 22654112, PMID: 25100726	Transition zone, basal body, nucleus, cytosol.	x		x			x
ENSG00000153814	JAZF zinc finger 1(JAZF1)	JAZF1		Regulator of multiciliated cell differentiation. It may function upstream of Foxj1 expression (PMID: 29661797).	PMID: 29661797	Nucleus		x				
ENSG00000109944	junctional cadherin complex regulator(JHY)	JHY		Required for ciliogenesis and proper differentiation of radial glia to functional ependymal cells. Mutations cause congenital hydrocephalus and loss of JHY results in the lack of the ciliary microtubule central pair (PMID: 29211732).	PMID: 29211732	Not reported						x

ENSG00000102781	katanin catalytic subunit A1 like 1(KATNAL1)	KATNAL1		Role in neuronal migration, neuronal morphology and ependymal ciliary function (PMID: 28373692). Regulates microtubule dynamics in Sertoli cells, a process that is essential for spermiogenesis and male fertility (PMID: 22654668)	PMID: 28373692, PMID: 22654668	Cytoskeleton, spindle pole, spindle	x				x
ENSG00000167216	katanin catalytic subunit A1 like 2(KATNAL2)	KATNAL2		Required for dynamic regulation of microtubules during axoneme formation and maintenance; it may have the ability to remodel microtubules at different cellular locations. Role during the regulation of Wnt/planar cell polarity (PCP)-dependent processes. Required for ciliogenesis, organogenesis and brain size control (PMID: 30096282). Interacts with Nubp1 and Nubp2 AAA proteins, which act as negative regulators of ciliogenesis (PMID: 26153462).	PMID: 30096282, PMID: 26153462	Basal body and ciliary axoneme in multiciliated cells, centrioles of non-ciliated cells and mitotic spindles.	x			x	x
ENSG00000162975	potassium voltage-gated channel modifier subfamily F member 1(KCNF1)	KCNF1		Ion channel gene that exhibits a significant negative effect on ciliogenesis but not on cell number. Potential new candidate in human ciliopathies (PMID: 26546361).	PMID: 26546361	Membrane subdomains at the base of primary cilia.	x				
ENSG00000143473	potassium voltage-gated channel subfamily H member 1(KCNH1)	KCNH1		Also known as Kv10.1. Promotes ciliary disassembly and cell progression through localised changes of membrane potential at the ciliary base (PMID: 28370099). Interacts with proteins involved in ciliary regulation, such as Rabaptin 5, cortactin, and HIF (PMID: 27113750).	PMID: 28370099, PMID: 27113750	Centrosome primary cilium.	x				

ENSG00000177807	potassium voltage-gated channel subfamily J member 10(KCNJ10)	KCNJ10		Ion channel gene that exhibits a significant negative effect on ciliogenesis but not on cell number (PMID: 26546361).	PMID: 26546361	Membrane subdomains at the base of primary cilia.	x				
ENSG00000115474	potassium voltage-gated channel subfamily J member 13(KCNJ13)	KCNJ13	603208	Mutations are associated with the human retinal dystrophies LCA and Snowflake Vitreoretinal Degeneration (PMID: 30009826).	PMID: 30009826	Photoreceptor connecting cilium	x				
ENSG00000053918	potassium voltage-gated channel subfamily Q member 1(KCNQ1)	KCNQ1		Ion channel gene that exhibits a significant negative effect on ciliogenesis but not on cell number. Genetic variants are known to cause Long QT syndrome. KCNQ1 could be implicated in renal ciliopathies (PMID: 26546361).	PMID: 26546361	Membrane subdomains at the base of primary cilia.	x				
ENSG00000110906	potassium channel tetramerization domain containing 10(KCTD10)	KCTD10		Component of cullin-3–RBX1–KCTD10 complex mediating CEP97 degradation and removal from the mother centriole, as well as removal of CP110 from the mother centriole to promote primary cilium formation. Binding partner of B9D1, mutations of which cause ciliopathies (PMID: 30404837).	PMID: 30404837	Mother centriole	x				

ENSG00000100379	potassium channel tetramerization domain containing 17(KCTD17)	KCTD17		Substrate-adaptor for Cul3-RING E3 ligases (CRL3s) that modulates polyubiquitylation of trichoplein and enables cells to assemble primary cilia by limiting Aurora-A activity. KCTD17 is required to initiate the axoneme extension during ciliogenesis (PMID: 25270598).	PMID: 25270598	Cytoplasm	x				x
ENSG00000115548	lysine demethylase 3A(KDM3A)	KDM3A		Negative regulator of ciliogenesis required for the controlled recruitment of IFT proteins into cilia through the modulation of actin dynamics. It binds to the actin cytoskeleton, creating a responsive actin gate involving ARP2/3 activity and regulating IFT within cilia and periciliary compartments. In the absence of KDM3A, access to cilia is unrestricted, so the assembly and growth of primary cilia is enhanced (PMID: 28246120).	PMID: 28246120	Nucleus, actin cytoskeleton, centrosome.	x				
ENSG00000122778	KIAA1549(KIAA1549)	KIAA1549	618613	Mutations are associated with Retinitis Pigmentosa (PMID: 30120214).	PMID: 30120214	Connecting cilium of the photoreceptor cells, synapses of the mouse retina	x				
ENSG00000047578	KIAA0556(KIAA0556)	KIAA0556	616784	Binds to microtubules and stabilise microtubule networks when overexpressed; it interacts with intraflagellar transport components and katanins associated with microtubules severing. Form a complex that regulates microtubule dynamics as well as ciliary functions. Negative regulator of ciliary length. Mutated in Joubert syndrome (PMID: 27245168, PMID: 26714646; PMID: 30982090; PMID: 32164589)	PMID: 27245168; PMID: 26714646; PMID: 30982090; PMID: 32164589	Cilium axoneme, basal body, centrosome	x				

				26714646).							
ENSG00000100578	KIAA0586(KIAA0586)	KIAA0586	616650	Required for ciliogenesis and sonic hedgehog/SHH signaling. Role in primary cilia formation and microtubule organisation (PMID: 19144723). Component of a CP110-containing protein complex required for centrosome and cilia function, Required for the centrosomal recruitment of RAB8A and for the targeting of centriole satellite proteins to centrosomes such as of PCM1. May play a role in early ciliogenesis in the disappearance of centriolar satellites that precedes ciliary vesicle formation. Essential for centrosomal migration prior to ciliogenesis (PMID: 24421332). Mutated in Joubert syndrome (PMID: 26096313) and in skeletal ciliopathies (PMID: 31816441; PMID: 32080096)	PMID: 1914472, PMID: 24421332, PMID: 26096313; PMID: 31816441; PMID: 32080096	Centrosome, distal end of centrioles, basal body	x				x
ENSG00000198920	KIAA0753(KIAA0753)	KIAA0753	617112	Mutated cause impaired ciliogenesis and result in Joubert Syndrome. Regulates centriole duplication positively by recruiting CDK2 to the centrosome, as well as WDR62 to the pericentriolar complex, which in turn recruits CEP63; interacts with CEP63, and regulate CEP63 level at the centrosome(PMID: 28220259, PMID: 24613305). Also cause skeletal ciliopathies (PMID: 31816441; PMID: 29138412).	PMID: 28220259, PMID: 24613305; PMID: 31816441; PMID: 29138412	Centriolar satellite, centrosome	x				x

ENSG00000197892	kinesin family member 13B(KIF13B)	KIF13B		Essential for establishment of a CAV1 membrane microdomain at the transition zone and promotes Sonic hedgehog signalling. Regulates transition zone membrane composition and ciliary SMO accumulation. Recruited to the ciliary base by NPHP4 (PMID: 28134340).	PMID: 28134340	Basal body, centrosome and primary cilia.	x				
ENSG00000118193	kinesin family member 14(KIF14)	KIF14	611279	Regulator of chromosome segregation, cytokinesis and midbody formation. Regulator of cilia formation, basal body biogenesis and Hedgehog (HH) signaling. Depletion of KIF14 specifically leads to defects in ciliogenesis and basal body biogenesis (PMID: 32348467). Mutations in humans and zebrafish cause ciliopathy-like kidney and microcephaly phenotypes (PMID: 30388224; PMID: 24128419)	PMID: 32348467; PMID: 30388224; PMID: 24128419	Primary cillium axoneme, spindle, nucleus, cytoplasm	x				
ENSG00000117245	kinesin family member 17(KIF17)	KIF17		Interacts with importin-β2 and Rab23 (PMID: 26136363). Motor protein that may be needed to position the ciliary localization signal CLS-containing protein at the ciliary base (PMID: 20526328). Interacts with the IFT46–IFT56 dimer in the IFT-B complex (PMID: 28077622). Required for photoreceptor OS formation and protein targeting; implicated in ciliary membrane protein trafficking (PMID: 18304522).	PMID: 26136363, PMID: 20526328, PMID: 28077622, PMID: 18304522	Basal body, axoneme, distal tip of the primary cilium, cytoplasm.	x		x		x

ENSG00000196169	kinesin family member 19(KIF19)	KIF19	Also known as KIF19A. Controls ciliary length through its microtubule depolymerising activity. It also has a role in microtubule-based active motility toward the plus-end (PMID: 27690357). Maintains the optimal length of motile cilia by depolymerising microtubules newly polymerised from axoneme at ciliary tips (PMID: 23168168).	PMID: 27690357, PMID: 23168168	Ciliary Tip	x		x		x
ENSG00000068796	kinesin family member 2A(KIF2A)	KIF2A	Promotes the primary cilium disassembly through its microtubule-depolymerising activity at the mother centriole, coupled with cell proliferation and blocking inappropriate ciliogenesis during the cycling phase. Controls the cytoplasmic tubulin concentration to regulate primary cilia disassembly (PMID: 25660017). Mutations cause primary microcephaly. Recruited by CEP170 (PMID: 31197141).	PMID: 25660017, PMID: 31197141	Basal body, subdistal appendages of the mother centrioles	x			x	x
ENSG00000186638	kinesin family member 24(KIF24)	KIF24	Physiological substrate of Nek2. It prevents the outgrowth of cilia in proliferating cells. Ensure the disassembly of a ciliary axoneme in the later stages of the cell cycle (PMID: 26290419). Interacts with CP110 and Cep97. Regulates ciliogenesis by dynamically controlling microtubule polymerisation (PMID: 21620453). Controls the formation of primary cilia by restricting the nucleation of cilia at centrioles, it regulates CP110 localisation at mother centrioles (PMID: 21663786).	PMID: 26290419, PMID: 21620453, PMID: 21663786	Centrioles, it preferentially localises to the distal end of mother centriole	x		x		x

ENSG00000165115	kinesin family member 27(KIF27)	KIF27	Essential for planarian ciliogenesis (PMID: 19933103). Kif27 expression is upregulated during mouse tracheal epithelial cells differentiation, consistent with their essential roles in motile ciliogenesis (PMID: 19305393). Interacts with Fu and both may regulate central pair assembly, probably through the recruitment or activation of central pair proteins such as Spag16 and Pcdp1. It could also interact with basal body components or accessory structures in controlling cilia orientation (PMID: 23907739).	PMID: 19933103, PMID: 19305393, PMID: 23907739	Ciliary base, between the transition zone and basal body	x		x		x
ENSG00000131437	kinesin family member 3A(KIF3A)	KIF3A	Required for cilia assembly, function and organisation of subdistal appendiges of centrioles. Recruits p150Glued to the subdistal appendages of mother centrioles, critical for centrosomes to function as microtubule-organising centres and to recruit other subdistal appendage components such as Ninein and Cep170. Loss of Kif3a abrogates microtubule anchoring and ciliary basal foot formation (PMID: 23386061).	PMID: 23386061	Distal appendages of mother centriole	x		x		x
ENSG00000101350	kinesin family member 3B(KIF3B)	KIF3B	Essential for ciliogenesis and the left-right asymmetry through intraciliary transportation of materials for ciliogenesis of motile cilia that produces the nodal flow (PMID: 9865700). Its phosphorylation inhibits IFT entry and flagellar lengthening restricting flagellar assembly (PMID: 30057303).	PMID: 30057303	Cilium	x		x		x

ENSG00000084731	kinesin family member 3C(KIF3C)	KIF3C		KIF3C dimerizes with KIF3A, redundant with KIF3B. Contributes to ciliogenesis, driving the differentiation of outer segments at later stages of cilia formation (PMID: 22308397, PMID: 23123805)	PMID: 22308397, PMID: 23123805	Cilium	x		x		x
ENSG00000170759	kinesin family member 5B(KIF5B)	KIF5B		Inhibitory role on cilia extension. Interacts with CCDC28B to regulate cilia length and the sub-cellular distribution of CCDC28B (PMID: 29445114). Required for intraflagellar transport and the formation of primary cilia (PMID: 31636894).	PMID: 29445114, PMID: 31636894	Pericentriolar region at the basal body	x				x
ENSG00000164627	kinesin family member 6(KIF6)	KIF6		Mediator of ciliogenesis and normal cilia beating in ependmal cells in vertebrates. Mutations are reported to be associated with neurological abnormalities in humans (PMID: 30475797).	PMID: 30475797	Axoneme and basal body of motile cilia	x				
ENSG00000166813	kinesin family member 7(KIF7)	KIF7	611254	Negative regulator of Hedgehog signalling at cilia in mammals by controlling the activity of Gli and Sufu. Controls the localization of Hh signaling complexes through its ability to regulate the dynamics of microtubule plus-ends. It inhibits microtubule growth to regulate the length of axonemal microtubules at the ciliary tip and control cilia structure (PMID: 24981634, PMID: 24952464, PMID: 19592253). A conserved non-motile kinesin, Kif7 defines a "cilium-tip compartment" by localizing to the distal ends of	PMID: 24981634, PMID: 24952464, PMID: 19592253; PMID: 31031197	Distal tip of primary cilia.	x		x		x

				axonemal microtubules; preferentially binds GTP-tubulin at microtubule ends over GDP-tubulin in the mature microtubule lattice (PMID: 31031197)								
ENSG00000237649	kinesin family member C1(KIFC1)	KIFC1		Required for ciliary membrane protein transport and serum starvation-induced ciliogenesis. Role in the Golgi transport of membrane proteins from the Golgi complex to the primary cilium and in the recruitment of ciliogenesis regulators and maintenance. Required for TTBK2 recruitment and CP110 removal from the distal end of the mother centriole to initiate cilium assembly (PMID: 29042452).	PMID: 29042452	Nucleus, spindle, golgi	x					
ENSG00000116014	KISS1 receptor(KISS1R)	KISS1R		Ciliary receptor which plays a role in reproductive function. Cilia are required for proper Kiss1r-mediated signaling on GnRH neurons	PMID: 24982149	Primary cilia membrane	x					
ENSG00000088970	kizuna centrosomal protein(KIZ)	KIZ	615757	Interacts with PLK1 in order to protect the centrosome from collapsing during spindle formation in mitosis. KIZ might have a critical role in this trafficking process. Mutations lead to autosomal-recessive Rod-Cone Dystrophy (PMID: 24680887).	PMID: 24680887	Basal body of monocilia, centrosome.	x					

ENSG00000126214	kinesin light chain 1(KLC1)	KLC1		Interacts with CCDC28B to regulate cilia length. Regulation the sub-cellular distribution of CCDC28B by loading CCDC28B onto KIF5B-based motors at the pericentriolar region (PMID: 29445114).	PMID: 29445114	Pericentriolar region at the basal body	x				
ENSG00000135338	LCA5, lebercilin(LCA5)	LCA5		Essential for the trafficking of proteins and vesicles to the photoreceptor outer segment. Mutations cause retinal degeneration, including cone dystrophy and autosomal recessive retinitis pigmentosa (PMID: 29673930, PMID: 21606596, PMID: 18826961). Interacts with a nearly complete IFT complex B particle, as well as with some IFT complex A proteins (PMID: 21606596).	PMID: 29673930, PMID: 21606596, PMID: 18826961	Basal Body, centrosome, primary Cilium, microtubule network.	x		x		x
ENSG00000050405	LIM domain and actin binding 1 (LIMA1)	LIMA1		Also known as EPLIN. Modulates actin stability to restrict primary cilia formation. Knockdown increases the levels of MyosinVa at the centrosome and primary cilia formation (PMID: 32496561)	PMID: 32496561	Actin filaments		x			
ENSG00000182541	LIM domain kinase 2(LIMK2)	LIMK2		Actin remodelling factor. Role in the negative regulation of ciliogenesis by upregulating the expression of YAP/TAZ target genes. Together with TESK1, LIMK2 plays a role in suppressing the directional trafficking of ciliary vesicles to the ciliary base (PMID: 25849865).	PMID: 25849865	Actin cytoskeleton		x			

ENSG00000198589	LPS responsive beige-like anchor protein(LRBA)	LRBA		Involved in trafficking events between the Golgi apparatus and the apical plasma membrane and/or the cilium base (PMID: 28814779)	PMID: 28814779	Photoreceptor, apical cytoplasm of somata, dendrites, and dendritic knobs.		x				x
ENSG00000010626	leucine rich repeat containing 23(LRRC23)	LRRC23		Radial spoke protein with function in ciliary motility and motile cilia formation. Mutations in Lrrc23 result in ciliary motility defects specifically in the cilia of the otic vesicle (PMID: 29475374).	PMID: 29475374	Axoneme of motile cilia	x				x	x
ENSG00000171757	leucine rich repeat containing 34(LRRC34)	LRRC34		Mutated in Joubert syndrome. Localized along the ciliary shaft in controls but only at ciliary base in LRRC34-/- patient cells, which displayed reduced ciliogenesis potential and muted response of SHH signaling to SAG compared with controls (PMID: 32055034)	PMID: 32055034	Along cilium	x				x	x
ENSG00000169683	leucine rich repeat containing 45(LRRC45)	LRRC45		Recruits the keratin-binding protein FBF1. Promotes cilia biogenesis in CP110-uncapped centrioles by organising centriolar satellites. It is required for centrosome cohesion at the proximal end of centrioles. Establishes the transition zone, recruitment of Rab8 and satellites to centrosomes, and promotes the docking of Rab8 GTPase-positive vesicles to the mother centriole (PMID: 30131441).	PMID: 30131441	Basal body: distal appendages.	x					

ENSG00000161328	leucine rich repeat containing 56(LRRC56)	LRRC56	618227	Interacts with the intraflagellar transport (IFT) protein IFT88. Required for the transport and addition of dynein arms to the distal end of the flagellum during cilia assembly. Mutations result in a disease entity within the group of mucociliary clearance disorders and loss of ciliary motility (PMID: 30388400).	PMID: 30388400	Motile cilium, distal flagellar portion of the axoneme, IFT trains	x				
ENSG00000129295	leucine rich repeat containing 6(LRRC6)	LRRC6	614930	Plays a role in cytoplasmic preassembly of dynein arms, essential for proper axonemal assembly (PMID: 23122589). Involved in the transport of the preassembled outer dynein arm complex to the axoneme and in transcriptional regulation of some dynein proteins (PMID: 27353389 PMID: 23527195). Mutations lead to primary ciliary dyskinesia with laterality defects (PMID: 23527195)	PMID: 23122589, PMID: 27353389 PMID: 23527195	Cilia, cytoplasm, proximal to basal body	x		x		x
ENSG00000188906	leucine rich repeat kinase 2(LRRK2)	LRRK2		Phosphorylates Rab8A, Rab10 and Rab12, mutations interferer with primary cilia formation by generating pRab10 that binds endogenous RILPL1 and blocks cilia formation. Regulates intracellular vesicular traffic during ciliogenesis (PMID: 30398148, PMID: 29125462).	PMID: 30398148, PMID: 29125462	Lysosome, ER, endosome, Golgi, mitochondrial outer membrane		x			
ENSG00000169641	leucine zipper protein 1(LUZP1)	LUZP1		Localizes to both actin filaments and the centrosome/basal body. Is an actin-stabilizing protein that regulates actin dynamics, at least in part, by mobilizing ARP2 to the centrosomes. Interacts with known ciliogenesis and cilia-length regulators. Depleting LUZP1 or its interacting protein, EPLIN, increases the levels of MyosinVa at the centrosome and primary cilia formation	PMID: 32496561	Actin filaments, basal body	x				

				(PMID: 32496561)							
ENSG00000163818	leucine zipper transcription factor like 1(LZTFL1)	LZTFL1	606568	Interacts with a BBS protein complex known as the BBosome to negatively regulate ciliary trafficking of this complex inhibiting BBosome ciliary entry. It also regulates ciliary trafficking of hedgehog signal transducer, Smoothened (PMID: 22072986). Coordinates the interactions between the BBosome and the IFT particle to regulate the removal of Ptch1 and Smo from cilia (PMID: 25446516). Depletion leads to a Bardet-Biedl syndrome phenotype (PMID: 22510444).	PMID: 22072986, PMID: 25446516, PMID: 22510444	Cytoplasm		x	x		x
ENSG00000127603	microtubule-actin crosslinking factor 1(MACF1)	MACF1		Role in coordination of actin and microtubules; critical in ciliogenesis, maintenance and regulation of apicobasal polarity. Potential interacting partner with basal body protein Mkks and TALPID3. It may be required for cilia-targeted vesicle exchange at this actin/microtubule interface. Loss of Macf1 abolished docking of the basal body and subsequent extension of the ciliary axoneme (PMID: 27783952).	PMID: 27783952	Distal end of microtubules, Cytoskeleton, cytoplasm, around the base of the connecting cilium.	x				

ENSG00000187391	membrane associated guanylate kinase, WW and PDZ domain containing 2(MAGI2)	MAGI2		SANS-Magi2 complex mediates endocytosis and regulates aspects of ciliogenesis and/or maintenance of primary cilia (PMID: 24608321).	PMID: 24608321	Endosome, cell membrane, cytoplasm of the apical inner segment. Periciliary membrane complex	x				x
ENSG00000111837	male germ cell associated kinase(MAK)	MAK	154235	Involved in the regulation of photoreceptor connecting cilium length. Mutations cause autosomal dominant Retinitis Pigmentosa (PMID: 22110072). Essential for the regulation of ciliary length, preventing excessive elongation of the cilia and is required for the long-term survival of photoreceptors (PMID: 21148103). Forms a complex with DYF-5 to regulate cilia morphology likely via modulation of microtubule stability (PMID: 30955935).	PMID: 22110072, PMID: 21148103, PMID: 30955935	Connecting cilia and outer segment axonemes	x		x		x
ENSG00000172005	mal, T-cell differentiation protein(MAL)	MAL		MAL overexpression is associated with shortened or absent cilia, abnormal cilium and cyst development. MAL may facilitate the organisation of apical cargo into microdomains to facilitate their targeting and trafficking (PMID: 22895261). Involved in the proper condensation of membranes at the ciliary base, required for efficient primary cilium extension (PMID: 25967552).	PMID: 22895261, PMID: 25967552	Punctuated distribution in the cilium, apical plasma membrane, ciliary base.	x		x		x

ENSG00000101460	microtubule associated protein 1 light chain 3 alpha(MAP1LC3A)	MAP1LC3A		Also known as LC3. Interacts with OFD1 and removes it from the centriolar satellite to initiate primary cilium biogenesis (PMID: 24343661)	PMID: 24343661	Autophagosome membrane, cytosol, late endosome, microtubule,	x				
ENSG00000101367	microtubule associated protein RP/EB family member 1(MAPRE1)	MAPRE1		Also known as EB1. Required for assembly of primary cilia, interacting with proteins implicated in microtubule minus-end anchoring at the centrosome to facilitate vesicular trafficking to the cilia base (PMID: 21768326, PMID: 17600711).	PMID: 21768326, PMID: 17600711	Proximal region of the basal body, centrosome: close to subdistal appendages of the mother centriole. Flagellar tip.	x		x		x
ENSG00000100030	mitogen-activated protein kinase 1(MAPK1)	MAPK1		ERK2. Downstream of TGFBR1 and 2, which phosphorylate ERK1/2 at base of cilium upon TGFB stimulation	PMID: 23746451	Ciliary base	x				
ENSG00000181085	mitogen-activated protein kinase 15(MAPK15)	MAPK15		Regulates motile and primary cilia formation, and the localisation of ciliary proteins involved in cilium structure, signaling and transport like BBS7 and ARL-13 (PMID: 29021280). Required for basal body migration and apical actin organisation in multiciliated cells, it is also important for left-right axis specification. Phosphorylates CapZIP, in cooperation with Dishevelled (PMID: 25823377).	PMID: 29021280, PMID: 25823377	Distal end of basal body	x			x	

ENSG00000102882	mitogen-activated protein kinase 3(MAPK3)	MAPK3		ERK1. Downstream of TGFBR1 and 2, which phosphorylate ERK1/2 at base of cilium upon TGFB stimulation	PMID: 23746451	Ciliary base	x				
ENSG00000119487	MAPK associated protein 1(MAPKAP1)	MAPKAP1		Also known as SIN1. Component of the mTOR complex 2. Interacts with CCDC28B to regulate cilia length and inhibits MEKK2, which activates different MAPKs that have been implicated in the regulation of IFT and ciliogenesis (PMID: 23727834, PMID: 26263164).	PMID: 23727834, PMID: 26263164	Plasma membrane, basal body	x				
ENSG00000136536	membrane associated ring-CH-type finger 7(MARCHF7)	MARCHF7		E3 ubiquitin ligase, works with USP9X to ubiquitinate NPHP5 and regulate ciliogenesis (PMID: 28498859).	PMID: 28498859	Nucleus, cytosol, plasma membrane, centrosome.	x		-		-
ENSG00000007047	microtubule affinity regulating kinase 4(MARK4)	MARK4		Required for initiation of axoneme extension during early ciliogenesis, after the docking of Rab8a-positive vesicles to the mother centriole. Regulates ODF2 localisation to the centriole (PMID: 23400999).	PMID: 23400999	Basal body, centrioles and axoneme	x				x

ENSG00000130368	MAS1 proto-oncogene, G protein-coupled receptor(MAS1)	MAS1		Neuronal ciliary GPCR. Galanin receptor.	PMID: 26167768	Base of primary cilia.	x				
ENSG00000234602	multiciliate differentiation and DNA synthesis associated cell cycle protein(MCIDAS)	MCIDAS		Promotes multiciliated cell formation during multiciliate cell differentiation. Mutations lead to severe axonemal defects, lack of ciliary motility-related proteins (PMID: 25048963). Promotes the transcriptional changes required for motile ciliogenesis and centriole assembly; it induces motile cilia outgrowth via FoxJ1, centriole assembly and basal body formation. Regulates the genes required for multiciliated cells differentiation (PMID: 22231168, PMID: 26933123).	PMID: 25048963, PMID: 22231168, PMID: 26933123	Nucleus		x			x
ENSG00000128285	melanin concentrating hormone receptor 1(MCHR1)	MCHR1		Involved in the primary cilia shortening through a unique signal from the ciliary membrane (PMID: 32240710). Candidate ciliary GPCR (PMID: 18256283).	PMID: 32240710, PMID: 18256283	Ciliary Membrane	x		x		x
ENSG00000073111	minichromosome maintenance complex component 2(MCM2)	MCM2		Function in transcriptional regulation/inhibition of genes involved in ciliogenesis and centrosome duplication. Helps to control left-right asymmetry development in zebrafish, facilitates formation and hence proper function of motile and primary cilia. Likely controls cilia length . Loss of MCM2 promotes transcription	PMID: 30329080	Nucleus		x			

				of a subset of genes, which cause cilia shortening and centriole overduplication (PMID: 30329080).							
ENSG00000187778	microspherule protein 1(MCRS1)	MCRS1		Required for dynein-dependent cargo transport to the centrosome. Interacts with the dynein intermediate chain DYNC111, and plays an essential role in both the establishment of centriolar satellites and the initiation of primary cilium assembly. MCRS1 also mediates centriolar recruitment of TTBK2 (PMID: 27263857).	PMID: 27263857	Centriolar satellites, nucleus and cytoplasm.	x			x	
ENSG00000111554	Mdm1 nuclear protein(MDM1)	MDM1		Upregulated during ciliogenesis (PMID: 23300604). Microtubule-binding protein. Negative regulator of centriole duplication through microtubule binding. Overexpression of MDM1 suppresses centriole duplication and results in stabilization of microtubules (PMID: 26337392). It might interact with members of the Hedgehog signaling pathway, as this pathway is tightly linked with cilia (PMID: 27658201)	PMID: 23300604, PMID: 26337392, PMID: 27658201	Mid-proximal region of centriole, in the lumen of the centriole, cillium	x		x		x
ENSG00000101752	mindbomb E3 ubiquitin protein ligase 1(MIB1)	MIB1		Interacts with and ubiquitylates AZI1 and PCM1, regulates the composition and function of centriolar satellites and suppresses primary ciliogenesis (PMID: 24121310). MIB1 destabilizes Talpid3 through poly-ubiquitylation and suppresses cilium assembly (PMID: 27146717). It redistributes to centrioles in response to	PMID: 24121310, PMID: 27146717, PMID: 25795303	Centriolar satellites, centrosomes.	x				

				conditions that induce centriole amplification, interacts with Plk4 and promotes its ubiquitylation, which is important for centriole homeostasis (PMID: 25795303).							
ENSG00000207990	microRNA 182(MIR182)	MIR182		Regulates the supply of molecular components to the apical membranes of photoreceptors in order to control the maintenance of outer segments, formation of inner segments, connecting cilia, and outer segments in a developing retina culture in vitro (PMID: 25002228).	PMID: 25002228	Not reported					
ENSG00000207691	microRNA 183(MIR183)	MIR183		Regulates the supply of molecular components to the apical membranes of photoreceptors in order to control the maintenance of outer segments, formation of inner segments, connecting cilia, and outer segments in a developing retina culture in vitro (PMID: 25002228). Depletion results in the upregulation of Rnf217 and misregulation of cilia-related genes (PMID: 32176259).	PMID: 25002228, PMID: 32176259	Not reported					
ENSG00000207811	microRNA 34b(MIR34B)	MIR34B		Required for the multiciliogenesis of multiciliated cells in the kidney and olfactory placode. It plays essential roles in the formation and organisation of cilia bundles. Serves as a guardian to maintain optimal levels of cmyb (PMID: 23698347). Essential for normal brain development, motile ciliogenesis, and spermatogenesis (PMID: 24982181).	PMID: 23698347, PMID: 24982181	Cytoplasm, nucleus		x			

ENSG00000207562	microRNA 34c(MIR34C)	MIR34C		Essential for normal brain development, motile ciliogenesis, and spermatogenesis (PMID: 24982181). Regulate motile ciliogenesis by repressing cp110 (PMID: 24899310).	PMID: 24982181; PMID: 24899310	Cytoplasm, nucleus	x				
ENSG00000125863	McKusick-Kaufman syndrome(MKKS)	MKKS	604896	Also known as BBS6. Associated with Bardet-Biedl and McKusick-Kaufman syndromes. Chaperonin-like BBS protein necessary for BBSome assembly in association with BBS10 and BBS12, and required for the stability of BBS2 and BBS7 (PMID: 20080638). Interacts with CEP290 in the ciliary transition zone to mediate ciliogenesis, maintenance of cilia integrity and function (PMID: 22446187).	PMID: 20080638, PMID: 22446187	Basal Body, pericentriolar material, Cytosol	x		x		x
ENSG00000011143	Meckel syndrome, type 1(MKS1)	MKS1	609883	Required for ciliogenesis of motile and non-motile cilia and post-migration of the centriole to the plasma membrane. Plays an essential role in modulating signaling in the non-canonical Wnt signaling pathway (PMID: 21045211). Modulates centrosome duplication to regular cilia length and number; mutants exhibit fewer and shorter cilia (PMID: 21045211, PMID: 19515853). Interacts with BBS4 to mediate trafficking of subset of membrane proteins to the primary cilium. Role in the structure of the cilia tip compartment (PMID: 28291807). Regulates ciliary INPP5E content (PMID: 26490104).	PMID: 21045211, PMID: 28291807, PMID: 26490104, PMID: 19515853	Mother centriole, basal body, transition zone	x		x		x

ENSG00000178053	myeloid leukemia factor 1(MLF1)	MLF1		Component of both primary and motile cilia. Upregulated during ciliogenesis (PMID: 23300604)	PMID: 23300604	Cilium.	x		x		x
ENSG00000154485	matrix metalloproteinase 21(MMP21)	MMP21	608416	Known to cause heterotaxy and primary ciliary dyskinesia (PMID: 30622330). Specialised role in the generation of left-right asymmetry (PMID: 26437028).	PMID: 30622330, PMID: 26437028	Secreted		x			
ENSG00000138587	meiosis specific nuclear structural 1(MNS1)	MNS1		Role in outer dynein arm (ODA) docking in the distal axonemes, stabilising structures in the motile cilium that helps the attachment of motor proteins to the microtubules to ensure a correct beating of the cilia; interacts with the ODA docking complex component CCDC114 and DNAH5. Deficiency causes laterality defects (PMID: 30148830). Essential for the assembly of sperm flagella (PMID: 22396656).	PMID: 30148830, PMID: 22396656	Axoneme	x		x		x
ENSG00000080823	MOK protein kinase(MOK)	MOK		Negative regulation of cilium length through the inhibition of the mTORC1 pathway and suppressing the synthesis of ciliary tubulin. It is part of, or transported by the IFT machinery (PMID: 25243405).	PMID: 25243405	Nucleus, ciliary base, cilium	x				x

ENSG00000185681	MORN repeat containing 5(MORN5)	MORN5		Associated with cytoskeletal structures in the sperm flagellum. Function in signaling during sperm development (PMID: 28742876).	PMID: 28742876	Centrosome, cytosol.	x			x	x
ENSG00000051825	M-phase phosphoprotein 9(MPHOSPH9)	MPHOSPH9		Forms a ring-like structure and regulates ciliogenesis by modulating CP110-CEP97 complex recruitment and localization at the mother centriole, by directly binding CEP97, which blocks cilia assembly in cycling cells able to form cilia. degradation of MPP9 facilitates the removal of the CP110-CEP97 complex from basal bodies and promotes cilia formation (PMID: 30375385).	PMID: 30375385	Distal end of mother centriole.	x				
ENSG00000198793	mechanistic target of rapamycin(MTOR)	MTOR		Mutations in MTOR Disrupt Neuronal Ciliogenesis and decreases cilia length, via induction of aberrant accumulation of OFD1 at centriolar satellites (PMID: 29937275). Downregulated by cilium bending, which in turn regulates cell size (PMID: 21145495). Inhibition of mTOR induces primary cilia formation; mTOR inhibition leads to the activation of autophagy, which suppresses the proteasome and prevents the degradation of ciliary proteins resulting in ciliogenesis (PMID: 32213867, PMID: 29180513, PMID: 25906314). Involved in the elongation and maintenance of cilium size by promoting the biosynthesis of proteins and lipids required for cilium elongation and maintenance (PMID: 29180513).	PMID: 29937275, PMID: 21145495, PMID: 32213867, PMID: 29180513, PMID: 25906314	Mitochondrion outer membrane, lysosome, ER, Golgi, cytoplasm	x				

ENSG00000118513	MYB proto-oncogene, transcription factor(MYB)	MYB		Activates endogenous Foxj1 expression, stimulates a cotransfected Foxj1 reporter in heterologous cells and regulates centriole amplification, which in turn activates centriole docking and the assembly of a motile cilium, so it drives a complete multiciliogenesis program (PMID: 24048590).	PMID: 24048590	Nucleus		x				x
ENSG00000133026	myosin heavy chain 10(MYH10)	MYH10		Regulators of the actin cytoskeleton. Required for centriole migration to the apical plasma membrane during the biogenesis of primary cilia. Regulates centrosomal recruitment of IFT88 required for ciliary tip structure (PMID: 25881509). Required for pericentrosomal recruitment of Cep290, PCM-1 and Rab11-positive recycling endosomes. It binds and antagonises Myh9 to increase actin dynamics and facilitate the assembly of the pericentrosomal preciliary complex that supplies materials for cilium growth and initiates ciliogenesis (PMID: 25494100).	PMID: 25881509, PMID: 25494100	Cytoskeleton		x				x
ENSG00000091536	myosin XVA(MYO15A)	MYO15A	602666	Essential for the graded elongation of stereocilia during their functional maturation and staircase formation (PMID: 14610277).	PMID: 14610277	Ciliary Tip of Stereocilia under the apical plasma membrane	x			x		x

ENSG00000071909	myosin IIIB(MYO3B)	MYO3B		Mediates movement along actin filaments in the cell (PMID: 26098909). Prevents photoreceptor loss of function caused by Myo3A defects, as it could provide functional redundancy for Myo3A (PMID: 19332056).	PMID: 19332056	Cytoskeleton. Photoreceptor inner segments, outer nuclear layer; outer segments of cones.	x				
ENSG00000197535	myosin VA(MYO5A)	MYO5A		Knockout results in reduction in ciliation, suggesting MYO5A is required for ciliogenesis. It might link microtubule cargo from Golgi or cilium (PMID: 28710093). It mediates the transportation of preciliary vesicles to the mother centriole, depletion inhibits the attachment of preciliary vesicles to the distal appendages of the mother centriole and decreases cilia assembly (PMID: 29335527). Interacts with the cilia-centrosomal protein RPRIP1L (PMID: 28266547).	PMID: 28710093, PMID: 29335527, PMID: 28266547	Centrosome. Preciliary vesicle, ciliary vesicle and ciliary sheath	x				x
ENSG00000137474	myosin VIIA(MYO7A)	MYO7A	276903	Responsible for Usher syndrome type 1B (PMID:7870171). required for normal hair cell development, as well as for the function of the mature hair cells (PMID: 9678669). It might be required for the transport of newly synthesized outer segment proteins to the site of disk membrane assembly (PMID: 9227854). Myosin VIIa participates in the transport of opsin through the connecting cilium (PMID: 10414956).	PMID: 7870171, PMID: 9678669, PMID: 9227854, PMID: 10414956	Connecting cilium, photoreceptor axoneme, periphery of the transition zone	x		x		x

ENSG00000072864	nudE neurodevelopment protein 1(NDE1)	NDE1		Negative regulator of ciliary length via interaction with LC8. Its depletion causes longer cilia and a delay in cell cycle re-entry (PMID: 26880200, PMID: 21394081).	PMID: 26880200, PMID: 21394081	Subdistal appendage of the mother centriole	x				x
ENSG00000111859	neural precursor cell expressed, developmentally down-regulated 9(NEDD9)	NEDD9		Transducer of integrin attachment, migration, and anti-apoptotic signals at focal adhesions. Promotes activation and stabilisation of AurA, which phosphorylates ciliary HDAC6 to induce disassembly of the primary cilium. (PMID: 17604723).	PMID: 17604723	Basal body, centriole.	x				
ENSG00000137601	NIMA related kinase 1(NEK1)	NEK1		Role in centrosome integrity, affecting both ciliogenesis and centrosome stability; it may suppress cilia by phosphorylating VHL. NEK1 overexpression leads to cilia resorption in cultured cells. Mutations lead to defective cilia (PMID: 23255108). NEK1 binds to the microtubule-dependent motor protein, Kif3A, probably influencing microtubule-dependent processes, including stabilization of axonemal microtubules (PMID: 18387364). Compites with CP110 for binding to the Cep104 ZNF domain. Mutations cause ciliopathies (PMID: 28017521),	PMID: 23255108, PMID: 1838736, PMID: 28017521	Basal body, pericentriolar satellite region of the centrosome	x		x		x

ENSG00000163491	NIMA related kinase 10(NEK10)	NEK10	NEK10 is a ciliated-cell specific kinase that regulates the motile ciliary proteome to promote ciliary length, and its deficiency produces short motile cilia . Required for effective airway mucociliary clearance (PMID: 31959991). NEK1 localisation is mediated by its interaction with PCM1 (PMID: 29581457).Mutations are associated with primary ciliary dyskinesia (PMID: 32414360). Mutation of NEKL-4/NEK10 and TTLL genes suppress neuronal ciliary degeneration caused by loss of CCPP-1 deglutamylase function (PMID: 33064774)	PMID: 31959991, PMID: 29581457, PMID: 32414360; PMID: 33064774	Centriolar satellites	x				
ENSG00000117650	NIMA related kinase 2(NEK2)	NEK2	Phosphorylates the centrosomal linker proteins C-Nap1, Cep68 and Rootletin to promote centriole separation, and is involved in the development of left-right asymmetry by regulating balancing ciliogenesis and resorption (PMID: 26493400). Phosphorylates Kif24, a depolymerising kinesin (PMID: 26290419). Influences centrosome-anchored microtubules organisation and promotes cilium disassembly at the onset of mitosis (PMID: 22613497). NEK2 removes distal appendage components from the mother centriole, preventing cilia maintenance during mitosis (PMID: 32211891).	PMID: 26493400,PMID: 26290419, PMID: 22613497, PMID: 32211891	Proximal end of centrosomes, distal end of mother centriole, basal body.	x		x		x

ENSG00000136098	NIMA related kinase 3(NEK3)	NEK3		Biallelic loss of function NEK3 mutations were found in 2 families with abnormal cardiac left-right patterning. NEK3 depletion was associated with deacetylation of α -tubulin and downregulation of NUP205, NUP188, and NUP155 and defective ciliary ultrastructure (PMID: 33230144)	PMID: 33230144	No data					
ENSG00000114904	NIMA related kinase 4(NEK4)	NEK4		Interacts with RPGRIP1 and RPGRIP1L to regulate cilium integrity and/or assembly. It associates with ciliary rootlets to regulate cilia stability	PMID: 21685204	Basal Body, Ciliary Rootlet	x		x		x
ENSG00000160602	NIMA related kinase 8(NEK8)	NEK8	609799	Mutations cause nephronophthisis (NPHP9); it can be transported into cilia by Inv (PMID: 20169535). Controls or modulates the ciliary localization of PC1 and PC2, as well as ciliary length (PMID: 18235101). It is speculated that Nek8 activity may be required to initiate ciliogenesis, and degradation may be required to prevent it (PMID: 22106379). Component of the PC2 channel complex required to mediate nodal calcium flux, indispensable for the establishment of left-right asymmetry (PMID: 23274954).	PMID: 20169535, PMID: 18235101, PMID: 22106379, PMID: 23274954	Centrosomes, proximal segment of cilia (Inv compartments), cytoplasm, nucleus.	x		x		x

ENSG00000215041	neuralized E3 ubiquitin protein ligase 4(NEURL4)	NEURL4		Controls CP110 levels at mother centriole. It transiently translocates to mother centriole to remove CP110 and facilitate proximity between both centrioles and and promote axoneme formation (PMID: 28385950). Promotes ubiquitylation of CP110 to maintain centrosome homeostasis and prevent the formation of ectopic microtubular organizing centres (PMID: 22441691).	PMID: 28385950, PMID: 22441691	Daughter centriole	x					
ENSG00000116044	nuclear factor, erythroid 2 like 2(NFE2L2)	NFE2L2		Also called NRF2. Primary Cilium-Autophagy-Nrf2 (PAN) Axis Activation Commits Human Embryonic Stem Cells to a Neuroectoderm Fate (PMID: 27020754). Negative regulator of primary ciliogenesis (via p62) and the Hh signaling pathway (through PTCH1), which blocks ciliary translocation of BBS4 and SMO (PMID: 32053600). Contradictory paper says that it promotes ciliogenesis (PMID: 31554934)	PMID: 27020754, PMID: 32053600, PMID: 31554934	Nucleus		x				
ENSG00000064300	nerve growth factor receptor(NGFR)	NGFR		Likely to provide signals to drive granule cell progenitor proliferation (PMID: 22995307). It may be involved in neurogenesis and memory-encoding functions in dentate gyral granule cells (PMID: 20875398).	PMID: 22995307, PMID: 20875398	Ciliary Membrane, granule cell cilia	x		x			x

ENSG00000100503	ninein(NIN)	NIN		Also called ninein. Interacts with γ -tubulin-containing complexes. It plays roles in promoting microtubule-nucleation and -anchoring processes at the centrosome (PMID: 15784680).	PMID: 15784680	Subdistal appendages of mother centriole	x		x		x
ENSG00000101004	ninein like(NINL)	NINL		Also called NLP. It is involved in microtubule nucleation, anchoring and outgrowth (PMID: 18826961). NLP interacts with ciliopathy-associated proteins lebercilin, USH2A and CC2D2A. Interacts with DZANK1 in vesicle transport towards the photoreceptor outer segments and its development/maintenance, in photoreceptor cells (PMID: 26485514).	PMID: 18826961, PMID: 26485514	Basal Body, Centrosome	x		x		x
ENSG00000103024	NME/NM23 nucleoside diphosphate kinase 3(NME3)	NME3		Facilitates DNA-repair mechanisms. Binds to several NPHs, including NEK8, CEP164, and ankyrin repeat and sterile α motif domain-containing 6 (ANKS6). Required for determining left-right asymmetry and ciliogenesis. Potential nephronophthisis disease gene (PMID: 30111592).	PMID: 30111592	Basal body	x				
ENSG00000112981	NME/NM23 family member 5(NME5)	NME5		Hydrocephalus and infertility due to dysfunctional motile cilia in KO mouse (PMID: 21746835). Role in ciliary beating (PMID: 24424412). Mutations cause of human PCD (PMID: 32185794).	PMID: 21746835, PMID: 24424412, PMID: 32185794	Axoneme, radial spokes.	x		x		x

ENSG00000143156	NME/NM23 family member 7(NME7)	NME7		It may regulate vesicular transport from the Golgi to the plasma membrane; its depletion inhibits Hh-stimulated accumulation of SMO in cilia (PMID: 21289087). Plays a role in human laterality patterning (PMID: 27060491). NME7 is a γ -tubulin ring complex component that regulates microtubule-nucleating activity in centrosomes. It is also required for ciliary signaling and protein trafficking to primary cilia (PMID: 24807905).	PMID: 21289087, PMID: 27060491, PMID: 24807905	Centrosome	x		x		x
ENSG00000086288	NME/NM23 family member 8(NME8)	NME8	607421	Also called TXNDC3. Candidate gene for PCD. Encodes a thioredoxine-nucleoside diphosphate kinase expressed in testis and respiratory epithelial cells. It is shown to have binding properties to microtubules (PMID: 17360648). Human ortholog of the sea urchin IC1 gene that encodes a component of sperm outer dynein arms (PMID: 19410201).	PMID: 17360648, PMID: 19410201	Sperm flagellum	x		x		x
ENSG00000148400	notch receptor 1(NOTCH1)	NOTCH1		It is glycosylated by GALNT11 to establish an essential balance between motile and immotile cilia at the left-right organiser, controlling laterality (PMID: 24226769).	PMID: 24226769	Plasma Membrane, Nucleus		x			x

ENSG00000214513	notochord homeobox(NOTO)	NOTO		Essential for ciliogenesis in the posterior notochord, and the establishment of left-right asymmetry (PMID: 17884984). Transcription factor regulating expression of cilia genes through Foxj1 and Rfx3. Mutants have disorganised or incomplete axonemal microtubules (PMID: 22357932).	PMID: 17884984, PMID: 22357932	Nucleus		x		x		x
ENSG00000144061	nephrocystin 1(NPHP1)	NPHP1	607100	Involved in centrosome and cilia function. Causative gene in the majority of familial cases of nephronophthisis and recently associated with Joubert syndrome (PMID: 15138899, PMID: 22982934).	PMID: 15138899, PMID: 22982934	Transition zone, connecting cilium	x			x		x
ENSG00000113971	nephrocystin 3(NPHP3)	NPHP3	608002	Mutations cause nephronophthisis type 3 (PMID: 22328406). Interaction with Inversin is essential for NPHP3 localisation (PMID: 20169535). Cooperates with Inversin and promotes ANKS6 phosphorylation by NEK8 in renal cilia. It may be involved in the regulation of noncanonical Wnt-pathway (PMID: 29395339).	PMID: 22328406, PMID: 20169535, PMID: 29395339	Basal body, Inv compartment.	x			x		x
ENSG00000131697	nephrocystin 4(NPHP4)	NPHP4	607215	Involved in familial juvenile nephronophthisis and retinitis pigmentosa. Component of the transition zone barrier that controls the movement of proteins between the flagellar and cytoplasmic compartments. Potentially interacts with MKS/B9 proteins to prevent ciliary membrane proteins from diffusing out of the cilium (PMID: 25150219). Probably involved in actin and microtubule organisation in cell-cell and cell-matrix adhesion signaling (PMID:	PMID: 25150219, PMID: 15661758	Basal body, transition zone. axoneme.	x			x		x

				15661758).								
ENSG00000185149	neuropeptide Y receptor Y2(NPY2R)	NPY2R		Encodes a G protein-coupled receptors involved in energy homeostasis, and could be a major driver of hyperphagia-induced obesity in BBS (PMID: 24316073).	PMID: 24316073	Cilium membrane	x					x
ENSG00000164129	neuropeptide Y receptor Y5(NPY5R)	NPY5R		Encodes a G protein-coupled receptors involved in energy homeostasis (PMID: 24316073).	PMID: 24316073	Cilium membrane	x					
ENSG00000012504	nuclear receptor subfamily 1 group H member 4(NR1H4)	NR1H4		Regulates autophagy by controlling transcription of genes involved in autophagic pathways. It negatively regulates ciliogenesis by inhibiting autophagy (PMID: 29771182)..	PMID: 29771182	Nucleus		x				

ENSG00000090273	nuclear distribution C, dynein complex regulator(NUDC)	NUDC	613524	Regulates actin dynamics and organisation, as well as ciliogenesis, by stabilising cofilin 1. Suppresses ciliogenesis in mammals (PMID: 26704451).	PMID: 26704451	Cytoplasm		x		x	x
ENSG00000168101	nudix hydrolase 16 like 1(NDUT16L1)	NUDT16L1		Also called SDOS. Controls cilium biogenesis by regulating the translation of cilia mRNAs involved in ciliogenesis. Controls synthesis of constituent proteins of the primary cilium.	PMID: 30260431	Nucleus and cytoplasm (endoplasmic reticulum, associated translating ribosomes)		x			
ENSG00000133961	NUMB endocytic adaptor protein(NUMB)	NUMB		Potential role in regulation of protein trafficking and polarised delivery of Cng channels to photoreceptor cilia, through recycling endosomes for delivery to OS plasma membrane (PMID: 25319694).	PMID: 25319694	Cytoplasm, inner segment of photoreceptor cilia.		x			
ENSG00000155561	nucleoporin 205(NUP205)	NUP205		Encodes a protein of the inner ring of the nuclear pore complex. Implicated as causes of steroid resistant nephrotic syndrome. When knocked down in Xenopus, Nup205 depletion results in loss of cilia and abnormal cardiac morphology, and mispositioning of intracellular ciliary structures such as basal bodies and rootlets (PMID: 33065118). Downregulated in NEK3-/- patients who are predisposed to cilia-related abnormal cardiac left-right patterning	PMID: 33065118	Nuclear pore		x			

ENSG00000126883	nucleoporin 214(NUP214)	NUP214		Component of ciliary pore complex that regulates protein transport into cilia (PMID: 22388888)	PMID: 22388888	Transition zone	x		x		x
ENSG00000163002	nucleoporin 35(NUP35)	NUP35		Inner ring nucleoporin, component of ciliary pore complex that regulates protein transport into cilia (PMID: 22388888)	PMID: 22388888	Transition zone, inner ring	x		x		x
ENSG00000075188	nucleoporin 37(NUP37)	NUP37		Outer ring nucleoporin, component of ciliary pore complex that regulates protein transport into cilia (PMID: 22388888)	PMID: 22388888	Transition zone, outer ring	x		x		x
ENSG00000213024	nucleoporin 62(NUP62)	NUP62		Central chanel nucleoporin, component of ciliary pore complex that regulates protein transport into cilia (PMID: 22388888). Involved in active transport of cytosolic proteins into both ciliary and nuclear compartments (PMID: 25264252).	PMID: 22388888, PMID: 25264252	Transition zone	x		x		x

ENSG00000102900	nucleoporin 93(NUP93)	NUP93		Inner ring nucleoporin, component of ciliary pore complex that regulates protein transport into cilia (PMID: 22388888). Involved in proper left-right patterning in developing embryos (PMID: 33065118).	PMID: 22388888, PMID: 33065118	Transition zone	x		x		x
ENSG00000110713	nucleoporin 98(NUP98)	NUP98		Interact with Nek2 to regulate cilium integrity, as removal of Nup98 from the ciliary nucleoporin complex is mandatory for ciliary resorption (PMID: 26493400). Component of a diffusion barrier that limits diffusion of molecules >70kDa between cytoplasm and ciliary matrix. Knockdown of NUP98 stops cilium elongations, suggesting it is required for cilia length regulation (PMID: 29731308).	PMID: 26493400, PMID: 29731308	Transition zone	x			x	
ENSG00000122126	OCRL, inositol polyphosphate-5-phosphatase(OCRL)	OCRL	300535	Interacts with GTPases and is involved in intracellular trafficking and cilia maintenance; mutation are associated with Lowe syndrome (PMID: 22543976). Required for cilia assembly by contributing to Rab8/PIP27 dependent protein trafficking to the primary cilia (PMID: 22228094) and modulation of cilia length (PMID: 22680056). Required for primary cilia to respond to pressure stimulation through its interaction with vanilloid 4 (PMID: 25143588).	PMID: 22543976, PMID: 22228094, PMID: 22680056, PMID: 25143588	Trans-Golgi network, basal body, transition zone and axoneme.	x		x		x

ENSG00000136811	outer dense fiber of sperm tails 2(ODF2)	ODF2		Required for localization of ninein to the appendages of the mother centriole, for the formation of the appendages necessary for primary cilia formation, as it appears to originate from the Odf2-positive end (PMID: 15852003). It associates with Cep128 in the construction of subdistal appendages (PMID: 30623524). Critical in the formation of basal feet, important for the polarization of basal bodies and coordinated beating of cilia (PMID: 22265411).	PMID: 15852003, PMID: 30623524, PMID: 22265411	Distal and subdistal appendages of the mother centriole.	x		x			x
ENSG00000046651	OFD1, centriole and centriolar satellite protein(OFD1)	OFD1	300170	Associated with oral-facial-digital type I syndrome. Absence of the Ofd1 protein leads to basal body dysfunction, which in turn prevents the assembly of the primary cilium, and causes failure of left-right axis specification (PMID: 16311594). It associates with the distal ends of centriolar microtubules to build distal appendages, recruit Ift88, and stabilise centriole length and regulate ciliogenesis (PMID: 20230748). Its presence at the centriolar satellite suppresses primary ciliogenesis and removal of OFD1 by autophagy promotes it (PMID: 24089205).	PMID: 16311594, PMID: 20230748, PMID: 24089205	Distal end of centrioles, centriolar satellites, nucleus	x		x			x
ENSG00000169856	one cut homeobox 1(ONECUT1)	ONECUT1		Also called HNF6. Transcription factor that controls the expression of a network of genes involved in ciliogenesis, cilium function and in hereditary polycystic diseases (PMID: 16472605).	PMID: 16472605	Nucleus		x				x

ENSG00000102076	opsin 1 (cone pigments), long-wave-sensitive(OPN1LW)	OPN1LW		Encodes photoreceptor opsins and is associated with blue cone monochromacy (PMID: 30614359). Requires IFT20 for localisation (PMID: 21307337).	PMID: 30614359, PMID: 21307337	Cone photoreceptor outer segment	x					x
ENSG00000268221	opsin 1 (cone pigments), medium-wave-sensitive(OPN1MW)	OPN1MW		Encodes photoreceptor opsins and is associated with blue cone monochromacy (PMID: 30614359). Requires IFT20 for localisation (PMID: 21307337).	PMID: 30614359, PMID: 21307337	Cone photoreceptor outer segment	x					x
ENSG00000166160	opsin 1 (cone pigments), medium-wave-sensitive 2(OPN1MW2)	OPN1MW2		Encodes photoreceptor opsins and is associated with blue cone monochromacy (PMID: 30614359). Requires IFT20 for localisation (PMID: 21307337).	PMID: 30614359, PMID: 21307337	Cone photoreceptor outer segment	x					
ENSG00000128617	opsin 1 (cone pigments), short-wave-sensitive(OPN1SW)	OPN1SW		Encodes photoreceptor opsins and is associated with blue cone monochromacy (PMID: 30614359). Requires IFT20 for localisation (PMID: 21307337).	PMID: 30614359, PMID: 21307337	Cone photoreceptor outer segment	x					x

ENSG00000125510	opioid related nociceptin receptor 1(OPRL1)	OPRL1		Neuronal ciliary GPCR. As a GPCR, it may play sensory or signalling roles in the ciliary vesicle at early stages of ciliogenesis	PMID: 26167768	Base of cillium and ciliary region of photoreceptors.	x				
ENSG00000085840	origin recognition complex subunit 1(ORC1)	ORC1	601902	Mutations are associated with Seckel syndrome and/or Meier-Gorlin syndrome (PMID: 23516378). Knockdown affects ciliogenesis and Hh signaling. It prevents reduplication events that could lead to supernumerary centrioles, which has been associated with cilia defects (PMID: 30696958).	PMID: 23516378, PMID: 30696958	Centrosome	x		x		x
ENSG00000174944	purinergic receptor P2Y14(P2RY14)	P2RY14		Neuronal ciliary GPCR	PMID: 26167768	Neuronal cilia membrane, base of primary cilia	x				
ENSG00000112530	PARK2 coregulated(PACRG)	PACRG		Interacts with Rib72, FAP252, and FAP20 to regulate ciliary motility (PMID: 27770595). Required for proper activation of dynein-driven microtubule sliding and for the assembly of regulatory components. Mutations lead to reduced assembly of inner-arm dynein IDA b, the beak-MIP structures and cause severe motility defects (PMID: 31116684).	PMID: 27770595, PMID: 31116684	Axoneme	x		x		x

ENSG00000124507	protein kinase C and casein kinase substrate in neurons 1(PACSIN1)	PACSIN1		Promotes mother centriole to basal body transition at early stages in ciliogenesis. Required for the establishment of an extracellular membrane channel, through interaction with EHD1 to form the membrane tubules that connect the developing cilium to the cell surface. Regulates actin and microtubule cytoskeleton assembly (PMID: 30683896).	PMID: 30683896	Preciliary membrane during ciliogenesis, Ciliary pocket membrane in mature cilia	x					
ENSG00000100266	protein kinase C and casein kinase substrate in neurons 2(PACSIN2)	PACSIN2		Promotes mother centriole to basal body transition at early stages in ciliogenesis. Required for the establishment of an extracellular membrane channel, through interaction with EHD1 to form the membrane tubules that connect the developing cilium to the cell surface. Regulates actin and microtubule cytoskeleton assembly (PMID: 30683896). Modulates primary cilium length in kidney epithelial cells and contributes to normal kidney tubular development/structure and repair (PMID: 23235565).	PMID: 30683896, PMID: 23235565	Preciliary membrane during ciliogenesis, Ciliary pocket membrane in mature cilia	x					
ENSG00000007168	platelet activating factor acetylhydrolase 1b regulatory subunit 1(PAFAH1B1)	PAFAH1B1		Also known as LIS1. It may regulate outer dynein arm activity and ciliary beating in the trachea. Essential role in cytoplasmic dynein regulation, mitosis and neuronal migration. Interacts with NudC(PMID: 17314247). Regulator of cytoplasmic dynein and alters its microtubule-binding affinity (PMID: 22855525).	PMID: 17314247, PMID: 22855525	Motile cilium axoneme, basal body and punctate structures throughout the cytoplasmic microtubule network	x		x			x

ENSG00000148498	par-3 family cell polarity regulator(PARD3)	PARD3		Also called PAR3. Interacts with Par6 and aPKC to form a polarity complex that regulates the stability of the ciliary axoneme (PMID: 15324661). It is required for the elongation of primary cilia, possibly by facilitating anterograde transport of cargo. Interacts with Kif3a, anterograde motor for IFT (PMID: 18070914, PMID: 25144710). PAR3 is also required for apical localisation of centrosome and may also be involved in the proper orientation of the axoneme (PMID: 20138861).	PMID: 15324661, PMID: 18070914, PMID: 25144710, PMID: 20138861	Along the length of cilium	x		x		x
ENSG00000102981	par-6 family cell polarity regulator alpha(PARD6A)	PARD6A		Interacts with Par3 and aPKC to form a polarity complex that regulates the stability of the ciliary axoneme (PMID: 15324661)	PMID: 15324661	Along the length of cilium	x		x		x
ENSG00000179270	photoreceptor cilium actin regulator(PCARE)	PCARE	613425	Also called C2ORF71. Predominantly expressed in the retina. Important for photoreceptor outer segment morphogenesis and visual function in zebrafish (PMID: 29946172). Interacts with WASF3 and recruits it to the primary cilium; coexpression results in expansion of the ciliary tip (PMID: 32312818). Mutations are associated with retinitis pigmentosa (PMID: 20398886)	PMID: 32312818; PMID: 29946172; PMID: 20398886	Cilium, Basal body, actin	x		x		x

ENSG00000150275	protocadherin related 15(PCDH15)	PCDH15	605514	Also called USH1F. Mutations cause retinitis pigmentosa in Usher syndrome type 1F and non-syndromic hearing loss (PMID: 14570705). It may be involved in the development or maintenance of the stereocilia bundles on the apical surface of hair cells (PMID: 11138007). It is also speculated to play a role in regulation of planar polarity in neurosensory cells (PMID: 11487575), probably through the activation of Rhoa (PMID: 28883094).	PMID: 14570705, PMID: 11138007, PMID: 11487575, PMID: 28883094	Connecting Cilium, Stereocilium, ciliary membrane.	x		x		x
ENSG00000078674	pericentriolar material 1(PCM1)	PCM1		Centriolar satellite protein/component that serves as a scaffold to target proteins to centrosome during centrosome maturation. When phosphorylated, it recruits Plk1 to the pericentriolar matrix before mitotic entry (PMID: 23345402). Involved in the formation of primary cilia by associating with CEP290 and BBSome during the ciliogenesis: required for ciliary localisation of Rab8. Role in the recruitment of major centrosomal proteins required for the formation of the non-motile primary cilium (PMID: 18772192, PMID: 31023719). Depletion reduces ciliogenesis (PMID: 21985783). Its localisation is dependent on BBS4, mislocalisation leads to deanchoring of centrosomal microtubules (PMID: 15107855). Required for focal ciliary integrity (PMID: 33214552)	PMID: 23345402, PMID: 18772192, PMID: 31023719, PMID: 21985783, PMID: 15107855; PMID: 33214552	Centriolar Satellites	x		x		x

ENSG00000160299	pericentrin(PCNT)	PCNT		Mediates microtubule organization (nucleation and anchoring) (PMID: 22031837). Form a complex with intraflagellar transport proteins and polycystin-2, required for primary cilia assembly and motile cilia. Recruits protein complexes involved in cilia assembly and calcium signaling to centriole (PMID: 15337773). Interacts with SYNE-2 to form a complex important for ciliogenesis and outer segment formation during retinal development (PMID: 30054381)	PMID: 22031837, PMID: 15337773, PMID: 30054381	Centrioles and basal body. Basal Body Complex of the photoreceptor connecting cilium	x					x
ENSG00000105650	phosphodiesterase 4C(PDE4C)	PDE4C		Catabolizes cAMP, located in renal primary cilia. Component of the AKAP150 complex, it promotes the hydrolysis of cAMP to negatively regulate cAMP signaling (PMID: 21670265).	PMID: 21670265	Cilium	x					x
ENSG00000132915	phosphodiesterase 6A(PDE6A)	PDE6A	180071	Also known as RP43. Mutations have been associated with autosomal recessive retinitis pigmentosa (PMID: 26321862, PMID: 29118501), and increase intracellular cGMP triggering cell death most likely through a calcium/sodium imbalance (PMID: 30998820).	PMID: 26321862, PMID: 29118501, PMID: 30998820	Rod photoreceptor outer segment peripheral membrane	x					x
ENSG00000133256	phosphodiesterase 6B(PDE6B)	PDE6B	180072	Also known as RP40. Mutations have been associated with autosomal recessive rod-cone dystrophy and increase intracellular cGMP triggering cell death most likely through a calcium/sodium imbalance (PMID: 30998820, PMID: 27588261).	PMID: 30998820, PMID: 27588261	Rod photoreceptor outer segment peripheral membrane	x					x

ENSG00000156973	phosphodiesterase 6D(PDE6D)	PDE6D	602676	Interacts with ARL13B, INPP5E and CEP164 in a functional network involved in nephronophthisis and Joubert syndrome. Required for ciliary targeting of full-length prenylated INPP5E (PMID: 23150559). Joubert Syndrome disease gene (PMID: 24166846; PMID: 32055034).	PMID: 23150559, PMID: 24166846; PMID: 32055034	Cytoplasm, basal body.	x		x		x
ENSG00000185527	phosphodiesterase 6G(PDE6G)	PDE6G	180073	Localised to rod outer segment by ARL13B (PMID: 30573647). Encodes for the inhibitory γ subunit of rod photoreceptor cyclic GMP-phosphodiesterase. Mutations are associated with retinitis pigmentosa (PMID: 20655036).	PMID: 30573647, PMID: 20655036	Rod outer segment	x				x
ENSG00000186862	PDZ domain containing 7(PDZD7)	PDZD7	612971	Retinal disease modifier in patients with digenic Usher syndrome. Homolog of whirlin (USH2D) and harmonin (USH1C). Part of the Usher protein complex (PMID: 20440071). It positions and stabilises GPR98 (PMID: 23055499).	PMID: 20440071, PMID: 23055499	Basal body and perinuclear region; ankle-link region of stereocilia.	x		x		x
ENSG00000083535	progesterone immunomodulatory binding factor 1(PIBF1)	PIBF1	300933	Also known as CEP90. Centriolar satellite protein required for centrosome protein recruitment, centrosomal accumulation of PCM-1 and primary cilia formation, as well as BBS4 recruitment and localisation to the cilium (PMID: 23110211). Mutations have been identified as cause of the ciliopathies Joubert and Jeune syndromes (PMID: 26167768).	PMID: 23110211, PMID: 26167768	Centriolar satellites	x	x	x		x

ENSG00000173947	primary cilia formation(PIFO)	PIFO	607532	Controls cilia retraction as well as the liberation and duplication of the basal body/centrosome during primary cilia disassembly. Enhances AurA Activation and regulates left-right asymmetry. Mutations cause cilia retraction, basal body detachment and overreplication defects (PMID: 20643351). Critical component of the Smo ciliary targeting complex and Hh pathway activation (PMID: 26901434).	PMID: 20643351, PMID: 26901434	Trans-Golgi network (TGN), cytoplasm, vesicles of TGN. Centrioles and basal body during cilia assembly and disassembly	x					x
ENSG00000155629	phosphoinositide-3-kinase adaptor protein 1(PIK3AP)	PIK3AP		PIK3AP1, also known as BCAP. Centriolar satellite protein that plays a role as negative regulator of ciliogenesis, specifically initiation (PMID: 28775150).	PMID: 28775150	Basal body, centriolar satellite	x					
ENSG00000011405	phosphatidylinositol-4-phosphate 3-kinase catalytic subunit type 2 alpha	PIK3C2A		Mutations cause syndromic short stature, skeletal abnormalities, and cataracts associated with ciliary dysfunction (PMID: 31034465). Critical for the formation of cilia and for receptor mediated endocytosis. Promotes the activation and pericentriolar localisation of Rab11+ vesicles, triggers the Rab8-dependent pathway and translocation of Smo to the primary cilium. Controls the production of PtdIns3P at the endocytic recycling compartment located at the base of the primary cilium (PMID: 24697898). PI3KC2α-dependent and VPS34-independent generation of PI3P controls primary cilium-mediated autophagy in response to shear stress	PMID: 31034465, PMID: 24697898, PMID: 31941925	Perinuclear area, Pericentriolar recycling compartment around the ciliary base	x					

				(PMID: 31941925).								
ENSG00000196455	phosphoinositide-3-kinase regulatory subunit 4(PIK3R4)	PIK3R4		Also known as VPS15. Interacts with the golgin GM130 to form a complex and ensure efficient loading of IFT20 in vesicles addressed from the cis-Golgi and transported to the basal body of the cilium (PMID: 27882921).	PMID: 27882921	Golgi		x				x
ENSG00000072062	protein kinase cAMP-activated catalytic subunit alpha(PRKACA)	PRKACA		Catalytic subunit of PKA. At centriolar satellites. Part of a multimeric complex nucleated by PCM1. Phosphorylates NEK10 for CHIP-mediated ubiquitination and proteolysis resulting in cilia resorption. Patched1-ArhGAP36-PKA-Inversin axis determines the ciliary translocation of Smoothened for Sonic Hedgehog pathway activation. PMID: 30598432	PMID: 29581457; PMID: 30598432	Centriolar satellites	x					x
ENSG00000165059	protein kinase cAMP-activated catalytic subunit gamma(PRKACG)	PRKACG		Catalytic subunit of PKA. At centriolar satellites. Part of a multimeric complex nucleated by PCM1. Phosphorylates NEK10 for CHIP-mediated ubiquitination and proteolysis resulting in cilia resorption. Patched1-ArhGAP36-PKA-Inversin axis determines the ciliary translocation of Smoothened for Sonic Hedgehog pathway activation. PMID: 30598432	PMID: 29581457; PMID: 30598432	Centriolar satellites	x					x

ENSG00000008710	polycystin 1, transient receptor potential channel interacting(PKD1)	PKD1	601313	Also known as PC2. Mutation causes autosomal dominant polycystic kidney disease (PKD). Dimerizes with PKD2 to form a channel in the ciliary membrane (PMID: 20392267) and mediate transduction of an extracellular mechanical stimulus into a Ca ²⁺ signaling response involved in the differentiation of kidney tubular epithelial cells (PMID: 12514735, PMID: 26788466). PC1 inhibits calcium release by reducing PC2-IP3R interaction (PMID: 26788466). Required for the asymmetric localization of Vangl2 and involved in the establishment of the asymmetric positioning of planar cell polarity proteins (PMID: 26245976).	PMID: 20392267, PMID: 12514735, PMID: 26788466, PMID: 26245976	Cillium, basal body	x		x		x
ENSG00000158683	polycystin 1 like 1, transient receptor potential channel interacting(PKD1L1)	PKD1L1	609721	Interacts with Pkd2 to form a complex that functions as the nodal flow sensor in the motile cilia. This interaction is required for left-right patterning (PMID: 21307098, PMID: 21307093)	PMID: 21307098, PMID: 21307093	Cilium	x		x		x
ENSG00000118762	polycystin 2, transient receptor potential cation channel(PKD2)	PKD2	173910	Also known as PC2. Mutation causes autosomal dominant PKD. Dimerizes with PKD1 to form a channel in the ciliary membrane (PMID: 20392267) and mediate transduction of an extracellular mechanical stimulus into a Ca ²⁺ signaling response involved in the differentiation of kidney tubular epithelial cells (PMID: 12514735, PMID: 26788466). PC2 enhances calcium release from the ER by stimulating the activity of the IP3 receptor (PMID: 26788466). Required for the asymmetric localization of Vangl2 and involved in the	PMID: 8650545, PMID: 20392267, PMID: 12514735, PMID: 26788466, PMID: 26245976	Cillium	x		x		x

				establishment of the asymmetric positioning of planar cell polarity proteins (PMID: 26245976).							
ENSG00000107593	polycystin 2 like 1, transient receptor potential cation channel(PKD2L1)	PKD2L1		PKD1L1-PKD2L1 acts as a ciliary calcium channel controlling ciliary calcium concentration and thereby modifying SMO-activated GLI2 translocation and GLI1 expression (PMID: 24336288). Plays a role in localization with β 2-adrenergic receptor (β 2AR) on the neuronal primary cilia (PMID: 29230552).	PMID: 24336288, PMID: 29230552	Cilium membrane	x				x
ENSG00000170927	polycystic kidney and hepatic disease 1 (autosomal recessive)(PKHD1)	PKHD1	606702	Also known as FPC. Functional and/or structural component of primary cilia in kidney tubular cells (PMID: 14978161). Mutations are involved in utosomal recessive polycystic kidney disease (PMID: 15458427, PMID: 11898128, PMID: 14983006). PKHD1 may associate with a common pathway that alters ciliary function and cation channel activity and may act as an intraflagellar motor-binding protein. It also interacts with polycystins, which suggest it could be directly or indirectly involved in a common cystogenic pathway (PMID: 14983006).	PMID: 14978161, PMID: 15458427, PMID: 11898128, PMID: 14983006	Basal Body, Cilium, cytoplasm.	x		x		x

ENSG00000100078	phospholipase A2 group III(PLA2G3)	PLA2G3		Negative ciliogenesis regulator. Depletion of PLA2G3 facilitates cilium extension and induces ciliogenesis independent of serum starvation (PMID: 20393563). Mediator of SREBP1c-induced cilium repression by distorting vesicular trafficking (PMID: 25904332).	PMID: 20393563, PMID: 25904332	Pericentriolar region.	x					x
ENSG00000166851	polo like kinase 1(PLK1)	PLK1		Regulates of centrosome maturation. Induces the phosphorylation of the transition zone protein nephrocystin-1 and modulates trafficking required for ciliary maintenance or disassembly (PMID: 22701722). Interacts with and activates histone deacetylase 6 (HDAC6) to promote ciliary deacetylation and resorption. Plk1 is recruited to the pericentriolar region by PCM1 (PMID: 23345402).	PMID: 22701722, PMID: 23345402	Transition Zone	x			x		x
ENSG00000142731	polo like kinase 4(PLK4)	PLK4	605031	Role in pericentrosomal localisation and integrity of centriolar satellite and ciliogenesis through PCM1 phosphorylation (PMID: 26755742). Interacts with CEP78. Over-expression induces centrosome amplification, which prevents the formation of basal bodies leading to loss of primary cilia (PMID: 26701933, PMID: 27246242).	PMID: 26755742, PMID: 26701933, PMID: 27246242	Centrioles	x					x
ENSG00000164087	POC1 centriolar protein A(POC1A)	POC1A	614783	Stable component of centrioles that and plays a role in their formation and maintenance. Mutations cause ciliogenesis defects and impaired mitosis, as well as primordial dwarfism (PMID: 26336158, PMID: 23015594, PMID: 22840364).	PMID: 26336158, PMID: 23015594, PMID: 22840364	Inner luminal walls and proximal ends of centrioles in the basal body.	x			x		x

ENSG00000139323	POC1 centriolar protein B(POC1B)	POC1B	614784	Stable component of centrioles that and plays a role in their formation and maintenance; required for ciliogenesis (PMID: 26336158, PMID: 23015594, PMID: 25044745). Disease gene for recessive retinal degeneration and severe syndromic ciliopathy (PMID: 25018096).	PMID: 26336158, PMID: 23015594, PMID: 25044745, PMID: 25018096	Basal body	x				x
ENSG00000152359	POC5 centriolar protein(POC5)	POC5		Essential for assembling the distal half of the centriole and the elongation of the centrioles. Mutations of POC5 alter its subcellular localization and induce ciliary retraction (PMID: 19349582), in addition, impair cell cycle, cilia length and centrosome protein interactions (PMID: 30845169).	PMID: 19349582, PMID: 30845169	Distal portion of centrioles.	x			x	
ENSG00000185900	protein-O-mannose kinase(POMK)	POMK		Hydrocephalus and infertility due to dysfunctional motile cilia, cerebellar dysplasia in KO mouse PMID: 21746835	PMID:21746835	ER		x	x		x
ENSG00000186951	peroxisome proliferator activated receptor alpha(PPARA)	PPARA		Ciliogenesis is reciprocally regulated by PPARA and NR1H4/FXR through controlling autophagy in vitro and in vivo. Regulates autophagy by controlling transcription of genes involved in autophagic pathways, and activation of PPARA facilitated ciliogenesis independently of cellular nutritional states by activating autophagy (PMID: 29771182).	PMID: 29771182	Nucleus		x			

ENSG00000178125	protein phosphatase 1 regulatory subunit 42(PPP1R42)	PPP1R42		Regulates PP1 activity at the centrosome. It activates PP1 to negatively regulate NEK2, suppressing centrosome separation (PMID: 23718219). PPP1R42 overexpression induces an increase in cilia number and length (PMID: 29141582).	PMID: 23718219, PMID: 29141582	Centrosome, Basal Body	x				
ENSG00000011485	protein phosphatase 5 catalytic subunit(PPP5C)	PPP5C		PP5 interacts with and directly dephosphorylates Dvl2. Overexpression enhances ciliogenesis (PMID: 29426949).	PMID: 29426949	Centrosome, Basal Body	x				x
ENSG00000102103	polyglutamine binding protein 1(PQBP1)	PQBP1		Interacts with and inhibits Dynamin 2 to induce neuronal ciliogenesis. Forms a complex with the GTPase Dynamin 2 and thereby inhibits its GTPase activity. Mutations of the PQBP1 WW domain cause recessive X-linked intellectual disability (PMID: 23994472).	PMID: 23994472	Base of neuronal cilium, nucleus.	x				
ENSG00000117450	peroxiredoxin 1(PRDX1)	PRDX1		Antioxidant protein, involved in the regulation of cilium disassembly (PMID: 24480440) and decreases the activity of Aurora A, through the regulation of the critical NEDD9-Aurora A-HDAC6 signal axis (PMID: 32357862).	PMID: 24480440, PMID: 32357862	Not reported in human cells. Flagella and basal bodies of <i>Dunaliella salina</i>	x				

ENSG00000139174	prickle planar cell polarity protein 1(PRICKLE1)	PRICKLE1		Also known as PK1. Regulates cell polarity and directional cell migration. It may also be involved in ciliogenesis. Pk1 mutants exhibited primary and motile cilia defects (PMID: 26883626).	PMID: 26883626	Cytosol, nucleus membrane		x			
ENSG00000163637	prickle planar cell polarity protein 2(PRICKLE2)	PRICKLE2		Also known as PK2. Planar cell polarity core component involved in the development and function of motile cilia (PMID: 24708399). Role in anterograde transport. Required for basal body and cilia orientation (PMID: 24938409).	PMID: 24708399, PMID: 24938409	Apical membrane		x			
ENSG00000012211	prickle planar cell polarity protein 3(PRICKLE3)	PRICKLE3		Also known as PK3. Required for γ-tubulin and Nedd1 recruitment to the basal body, necessary for basal body organization and cilia growth. Pk3 knockdown inhibites ciliogenesis (PMID: 27062996).	PMID: 27062996	Centrosome/Basal Body, cytoplasm.	x				
ENSG00000007062	prominin 1(PROM1)	PROM1	607300	Controls recruitment of ciliary membrane components, histone deacetylase, and transcription factors in stem cells. Mutations cause various human retinal disorders by disrupting the cilium-derived photoreceptor outer segment (PMID: 30523147).	PMID: 30523147	Nucleus, rods and cones outer segments		x		x	x

ENSG00000105618	pre-mRNA processing factor 31(PRPF31)	PRPF31		Mutated in autosomal dominant RP type 11. Splicing factor implicated in the correct splicing of transcripts encoding proteins important for ciliogenesis (PMID: 26167768). PRPF31 loss negatively regulates ciliogenesis, causing defects in cilia incidence and structural organisation (PMID: 30315276).	PMID: 26167768, PMID: 30315276	Basal body, connecting cilium, nuclear speckles	x				
ENSG00000101161	pre-mRNA processing factor 6(PRPF6)	PRPF6	613979	Mutated in autosomal dominant RP type 60. Splicing factor implicated in the correct splicing of transcripts encoding proteins important for ciliogenesis (PMID: 26167768).	PMID: 26167768	Basal body, apical inner segment, apical connecting cilium of photoreceptor cells, nuclear speckles	x				
ENSG00000174231	pre-mRNA processing factor 8(PRPF8)	PRPF8	606419	Mutated in autosomal dominant RP type 13. Splicing factor implicated in the correct splicing of transcripts encoding proteins important for ciliogenesis. It has been implicated in the process of centriolar under-duplication (PMID: 26167768).	PMID: 26167768	Basal body, apical inner segment, apical connecting cilium of photoreceptor cells, nuclear speckles	x				
ENSG00000112619	peripherin 2(PRPH2)	PRPH2	179605	Photoreceptor-specific tetraspanin protein, essential for the morphogenesis of the outer segment (PMID: 30819798). Role in generating and/or sensing the curvature associated with creation and maintenance of OS membrane architecture (PMID: 23886945). Mutations are associated with RP (PMID: 25805137).	PMID: 30819798, PMID: 23886945, PMID: 25805137	Outer segment disk membranes, Luminal side of the late endosome	x				x

ENSG00000185920	patched 1(PTCH1)	PTCH1		Also known as PTC1. Transcriptional target of Hh signaling that inhibits Smo by preventing its accumulation within cilia and activation. Shh binds and inactivates Ptc1, allowing activation of Smo (PMID: 17641202, PMID: 32435793).	PMID: 17641202, PMID: 32435793	Cilium, base of cilium	x		x		x
ENSG00000171522	prostaglandin E receptor 4(PTGER4)	PTGER4		Also known as EP4. Activates cAMP-mediated signaling cascade. Together with COX-2, is required for cilia formation and elongation. Increases the intracellular cAMP and activates protein kinase A (PKA) signaling (PMID: 25173977).	PMID: 25173977	Cilium, base of cilium	x				
ENSG00000169398	protein tyrosine kinase 2(PTK2)	PTK2		Interacts with both basal bodies and actin; its downregulation leads to defects in ciliogenesis (PMID: 24434137). It is a regulator of the actin cytoskeleton, cell adhesion, migration and plays a scaffolding role in multiciliated cells (PMID: 27895123).	PMID: 24434137, PMID: 27895123	Cytoskeleton, basal bodies.	x				
ENSG00000158079	protein tyrosine phosphatase domain containing 1(PTPDC1)	PTPDC1		Involved in centrosomal duplication, cytokinesis, and plays a role in ciliary length determination (PMID: 19284988, PMID: 21289087). Depletion may lengthen cilia and accumulate higher levels of Gli3 via increased anterograde transport (PMID: 21289087).	PMID: 19284988, PMID: 21289087	Centrosome, cytoplasm, nucleus.	x		x		x

ENSG00000179295	protein tyrosine phosphatase, non-receptor type 11(PTPN11)	PTPN11		Also called SHP2. Responsible for craniofacial skeletal formation by regulation of primary cilia formation in chondrocytes, especially in the craniofacial region. Positively regulates ciliogenesis through IFT regulation in condyle cartilage. Deficiency shortens length and reduces number of primary cilia (PMID: 25919282).	PMID: 25919282	Nucleus, cytoplasm	x				
ENSG00000186867	pyroglutamylated RFamide peptide receptor(QRFPR)	QRFPR		Ciliary GPCR	PMID: 24316073	Cilium membrane	x				
ENSG00000084733	RAB10, member RAS oncogene family(RAB10)	RAB10		Negative regulator of ciliogenesis. Rab10 phosphorylation enhances binding to RILPL1, which in turn strengthens its ability to block ciliogenesis (PMID: 30398148). Mediates membrane protein trafficking (PMID: 21856246). Interacts in a protein complex with Sec8 at the basal body (PMID: 20576682).	PMID: 30398148, PMID: 21856246, PMID: 20576682	Centrosome, pericentriolar membranes. Basal body of nascent cilia.	x				x
ENSG00000103769	RAB11A, member RAS oncogene family(RAB11A)	RAB11A		Regulator of the endocytic recycling compartment (ERC), binds to Rabin8 and transport it to the centrosome where it activates Rab8 to assemble the ciliary membrane from the ciliary vesicle (PMID: 20308558, PMID: 31204173, PMID: 22718902). Rabin8-Rab11a interaction and preciliary trafficking are promoted by PI3K and Akt inhibitors (PMID: 31204173).	PMID: 20308558, PMID: 31204173, PMID: 22718902	Centrosome, Basal Body	x		x		x

ENSG00000090565	RAB11 family interacting protein 3(RAB11FIP3)	RAB11FIP3	Also known as FIP3. Rab11 effector, required for regulation of preciliary trafficking and ciliogenesis. FIP3 increases Rabin8 affinity for Rab11 and forms a complex with them (PMID: 31204173). Interact with Arf4, ASAP1 and Rab11 to form a complex that regulates budding of post-Trans-Golgi Network carriers targeted to the primary cilia (PMID: 19153612), which means that FIP3 is an essential regulator of ciliary membrane targeting, expansion and compartmentalisation of signaling receptors (PMID: 25673879)	PMID: 31204173, PMID: 19153612, PMID: 25673879	Trans-Golgi Network, Basal body.	x		x		x
ENSG00000124839	RAB17, member RAS oncogene family(RAB17)	RAB17	Involved in membrane trafficking, necessary for ciliogenesis (PMID: 17646400). Involved in the function of apical sorting endosomes in polarized epithelial cells (PMID: 9490718)	PMID: 17646400, PMID: 9490718	Plasma Membrane		x	x		x
ENSG00000112210	RAB23, member RAS oncogene family(RAB23)	RAB23	Negative regulator of the Hedgehog signaling pathway, required for retrograde transport away from primary cilia (PMID: 17646400). It mediates the ciliary recycle of Smo (PMID: 20375059). Facilitates binding of cytoplasmic Kif17 with ciliary import carrier importin β 2, essential for its targeting to the primary cilium (PMID: 26136363).	PMID: 17646400, PMID: 20375059, PMID: 26136363	Cell membrane, endosome, cytoplasm, cilioplasm.	x		x		x

ENSG00000157869	RAB28, member RAS oncogene family(RAB28)	RAB28	612994	Associated with autosomal recessive cone-rod dystrophy. IFT-associated cargo with BBSome-dependent cell autonomous and non-autonomous functions at the ciliary base, maintaining the structural and functional integrity of cilia (PMID: 27930654). Ciliary Rab28 and the BBSome negatively regulate extracellular vesicle shedding (PMID: 32101165)	PMID: 27930654; PMID: 32101165	Basal body, cilium and periciliary membrane.	x					x
ENSG00000117280	RAB29, member RAS oncogene family(RAB29)	RAB29		Colocalizes and interacts with Rab8, Rab11 and IFT20. Participates in ciliogenesis by controlling cilium growth, by promoting the transport of endosome-associated ciliary cargo to the base of the cilium. Required for Smo trafficking to the ciliary membrane.	PMID: 26021297	Cis-golgi, trans-Golgi network, base of cilium	x					
ENSG00000127328	RAB3A interacting protein(RAB3IP)	RAB3IP		Also known as Rabin8. GTP loading by Rabin8 promotes Rab8 association with post-Golgi vesicles and targets them to polarised areas of the plasma membrane. BBS1 subunit of the BBSome associates with the C terminus of Rabin8 at the basal body. Activates Rab8 by loading with GTP, which is necessary for Rab8 to enter the primary cilium, promoting ciliogenesis and apical membrane formation (PMID: 17574030). Phosphorylation of Rabin8 might modulate its interactions with ASAP1 and Rab11a. It binds to the catalytically active ASAP1 in complex with Rab11a and FIP3 (PMID: 25673879). Interacts with Chibby to promote recruitment of Rab8 and efficient assembly of ciliary vesicles. Interaction with Rab11 stimulates its GEF activity (PMID: 25313408)	PMID: 17574030, PMID: 25673879, PMID: 25313408, PMID: 21273506	Basal Body, Centrosome, membrane.	x		x			x

				and is required for Rabin8 preciliary membrane trafficking to the centrosome and for ciliogenesis (PMID: 21273506).						
ENSG00000109113	RAB34, member RAS oncogene family(RAB34)	RAB34		Rab34 regulates the early steps in serum starvation-induced ciliogenesis in hTERT-RPE1 cells, through a unique N-terminal region, by regulating the ciliary vesicle fusion step and/or the subsequent expansion. Required for ciliogenesis in only certain cell types (PMID: 32669361). Rab34 is required for the successive fusion of small preciliary vesicles in order to generate ciliary vesicles, it is also required for the migration of the mother centriole from perinuclear region to cell membrane. Required for an early step of ciliary vesicle formation and Hh signaling in vivo. Mutation results in significant decrease of ciliogenesis in both cultured cells and mice (PMID: 30301781). Potential transcriptional target of Gli transcription factors (PMID: 17442700).	PMID: 32669361, PMID: 30301781, PMID: 17442700	Golgi, cytoplasm, nucleus, cilium	x			
ENSG00000167461	RAB8A, member RAS oncogene family(RAB8A)	RAB8A		Rab8 targets vesicles to the cilium and promotes ciliary membrane elongation. It is activated by Rabin8 (PMID: 17574030, PMID: 22433857). Inactivating Rab8 (GDP-bound state) blocks ciliogenesis and induces BBS phenotypes in zebrafish (PMID: 17574030). Rab8a, Rab8b and Rab10 work synergistically for ciliogenesis (PMID: 24213529). Rab8a	PMID: 17574030, PMID: 22433857, PMID: 24213529, PMID: 17646400, PMID: 20631154; PMID: 32669361	Ciliary Membrane, Cilium	x		x	x

			specifically interacts with cenexin/ODF2, a basal body and microtubule binding protein required for cilium biogenesis (PMID: 17646400). Regulates the trafficking of rhodopsin in photoreceptors. Interacts with RPGR which regulates its activity and localisation. Involved in polarised trafficking of vesicles from the trans-Golgi network (PMID: 20631154). Involved in serum-starvation-induced ciliogenesis (PMID: 32669361)							
ENSG00000177548	rabaptin, RAB GTPase binding effector protein 2(RABEP2)	RABEP2	RABEP2 localization at the centrosome is regulated by SDCCAG8. As a component of cilium, it may be a component of the intraciliary transport mechanism. siRNA mediated knockdown in hTERT-RPE1 cells abolished cilia formation, indicating a critical role in ciliogenesis (PMID: 27224062).	PMID: 27224062	Basal body, centrosome, ciliary compartment	x				
ENSG00000144134	RAB, member of RAS oncogene family-like 2A(RABL2A)	RABL2A	RABL2 paralog. Functionally redundant with RABL2B as they can rescue ciliogenesis when both genes have been knocked out (PMID: 28625565). Associates with ciliary GPCR in mammalian primary cilia and regulates theirs localisation (PMID: 30578315). Two variants in RABL2A were identified as cause of male infertility and ciliopathy (PMID: 33075816). RABL2 might play a role in trafficking of ciliary proteins as well as ciliogenesis. RABL2A/B knock out lead to decrease in number of primary cilia in RPE1 cells (PMID: 30578315).	PMID: 28625565, PMID: 30578315, PMID: 33075816	Base of cilium	x			x	x

ENSG00000079974	RAB, member of RAS oncogene family-like 2B(RABL2B)	RABL2B	RABL2 paralog. Functionally redundant with RABL2A as they can rescue ciliogenesis when both genes have been knocked out (PMID: 28625565). Associates with ciliary GPCR in mammalian primary cilia and regulates their localisation (PMID: 30578315). It is recruited by CEP19 to the ciliary base where it mediates release of IFT-B from pre-docked complexes to initiate IFT ciliary entry; CEP19-RABL2B complex is then a critical regulator that initiates intraflagellar transport at the ciliary base (PMID: 28625565). Variant in RABL2B gene could be a risk factor in Iranian infertile men with oligoasthenoteratozoospermia (PMID: 28138870).	PMID: 28625565, PMID: 30578315, PMID: 28138870	Base of cilium	x			x	x
ENSG00000132341	RAN, member RAS oncogene family(RAN)	RAN	Ran GTP regulates ciliary protein transport and ciliogenesis (PMID: 21998203). RanGTP gradient regulates trafficking and entry of kinesin motor KIF17 into cilium. It has been proposed that Ran plays a role in regulating cellular compartmentalisation by controlling the shuttling of cytoplasmic proteins into ciliary compartments (PMID: 20526328). Interacts with Importin β 2, which interacts directly with cargoes, releasing them upon binding Ran GTP and regulating localisation of RP2 (PMID: 21285245).	PMID: 21998203, PMID: 20526328, PMID: 21285245	Cilium, Basal body, Cytosol.	x		x		x

ENSG00000099901	RAN binding protein 1(RANBP1)	RANBP1		Negative regulator of Ran GTP, as it stimulates Ran GTP to GDP conversion. RanBP1 Knockdown results in an increased concentration of Ran GTP at basal bodies, leading to ciliogenesis. Overexpression antagonizes primary cilia formation.	PMID: 21998203	Cilium, Basal Body, Cytosol.	x		x		x
ENSG00000172348	regulator of calcineurin 2(RCAN2)	RCAN2		Negative regulator of calcineurin and modulates Ca2+ dependent signalling. Required to maintain normal cilia length. Depletion in WT RPE-1 cells results in shorter cilia, whilst in giantin KO cells induces longer cilia (PMID: 29643119).	PMID: 29643119	Basal Body, Centriole	x				
ENSG00000198771	RCSD domain containing 1(RCSD1)	RCSD1		Also known as CAPZIP. Ortholog duboraya (dub), has been reported to be essential for actin organization in the cells lining Kupffer's vesicle, regulates ciliogenesis in a phosphorylation-dependent manner as well as left-right patterning in zebrafish (PMID: 17013396). Phosphorylated by ERK7. Ciliogenesis regulations through CapZIP, requires Dvl binding to both ERK7 and CapZIP (PMID: 25823377).	PMID: 17013396, PMID: 25823377	Actin cytoskeleton		x			
ENSG00000087903	regulatory factor X2(RFX2)	RFX2		Involved in regulation of ALMS1 in vivo, by binding its promoter (PMID: 20381594). Coordinates multiple, distinct gene expression programs in MCCs, regulating genes encoding axonemal dynein subunits, components of the transition zone and BBSome, centriolar genes and planar cell polarity (PCP) effector fritz/wdpcp. It also regulates the extracellular matrix component dag1, which is a	PMID: 20381594, PMID: 24424412, PMID: 22227339	Nucleus		x			x

				regulator of MCC insertion (PMID: 24424412). Critical regulator of vertebrate ciliogenesis. Knockdown results in phenotypes associated with defective cilia, such as disruption of neural tube closure and left–right asymmetry (PMID: 22227339).							
ENSG00000080298	regulatory factor X3(RFX3)	RFX3		Transcription factor regulating expression of ciliary assembly (like Dync2li1, Foxj1 and Bbs4) and axonemal dyneins involved in ciliary motility (Dnahc11, Dnahc9 and Dnahc5), by binding directly to the promoters of their genes. Regulates the orthologs of genes involved in human PCD. It may also cooperate with FOXJ1 to regulate the expression of cilia genes (PMID: 19671664). Regulates the expression of D2lic in mouse, a gene implicated in intraflagellar transport, a proces required for the assembly and maintenance of cilia. RFX3 is essential for the differentiation of nodal monocilia and for LR patterning (PMID: 15121860). Together with FOXJ1, plays a role in modulating differentiation in multiciliated cells (PMID: 23822649).	PMID: 19671664, PMID: 15121860, PMID: 23822649	Nucleus		x	x		x
ENSG00000111783	regulatory factor X4(RFX4)	RFX4		Modulates Shh signaling and Gli3 processing by regional control of ciliogenesis and function of primary cilia. Rfx4 nuclear translocation and/or transcriptional activity, may be regulated by phosphorylation. Required for ciliogenesis in the neural tube (PMID: 19887680). It appears to regulate different genes than RFX3, as Rfx4 KO embryos exhibite different	PMID: 19887680, PMID: 29298325, PMID: 24530844	Nucleus		x			x

				phenotypes from those with Rfx3 and Foxj1 mutations (PMID: 29298325). RFX4 expression is controlled positively by RFX7 or negatively by Foxj1 during neural tube closure (PMID: 24530844).							
ENSG00000181827	regulatory factor X7(RFX7)	RFX7		Highly expressed in Brain. RFX7 controls positively the formation of cilia by regulating the expression of RFX4 gene. Involved in complete neural tube closure (PMID: 24530844).	PMID: 24530844	Nucleus		x			
ENSG00000163914	rhodopsin(RHO)	RHO	180380	Cilia-specific GPCR essential for vision. Multiple roles in trafficking, interacts with Arf4, TcTex-1 and SARA (PMID: 23966685). Mutations cause protein misfolding and retention in the endoplasmic reticulum, leading to cellular stress and cell death. Mutations are associated with retinitis pigmentosa (PMID: 32343782).	PMID: 23966685, PMID: 32343782	Rod photoreceptor outer segment	x				x
ENSG00000188026	Rab interacting lysosomal protein like 1(RILPL1)	RILPL1		Rab effector-related protein, regulates protein concentration by removal of membrane proteins from the cilium. Depletion results in accumulation of signaling proteins in the ciliary membrane, including Smo. Interacts with activated Rab34, Rab36, Rab12 and Rab40B (PMID: 23264467). Interacts with the LRRK2-phosphorylated forms of Rab8A and Rab10 (PMID: 29125462). Enhances pRab10 clustering on pericentriolar membranes to	PMID: 23264467, PMID: 29125462, PMID: 30398148	Cilium, Basal Body (distal end of mother centriole)	x		x		x

				suppress ciliogenesis (PMID: 30398148).							
ENSG00000150977	Rab interacting lysosomal protein like 2(RILPL2)	RILPL2		Rab effector-related protein that exhibit dynamic localisation; regulates protein concentration by removal of membrane proteins from the cilium.. Depletion results in accumulation of signaling proteins in the ciliary membrane, including Smo. Interacts with activated Rab34 and Rab36 (PMID: 23264467). Binds to phosphorylated Rab8A, Rab10, and Rab12. As it binds Rab8a much stronger than RILPL1, RILPL2 might be a key effector specific for pRab8A (PMID: 29125462).	PMID: 23264467, PMID: 29125462	Cilium, Centrosome	x		x		x
ENSG00000111913	RHO family interacting cell polarization regulator 2(RIPOR2)	RIPOR2	611410	Mutations cause deafness. Interacts with Myh9, which is related to deafness. Regulates the abundance and post translational modifications of proteins highly expressed within the kinocilium , which suggests a role in hair bundle morphogenesis (PMID: 30280293).	PMID: 30280293	Basal end of stereocilium, kinocilium	x				x
ENSG00000149489	retinal outer segment membrane protein 1(ROM1)	ROM1		Outer segment protein. Its localisation depends on AHI1 (PMID: 20592197). Non-covalently linked PRPH2/ROM1 complex can initiate OS formation, whilst covalent oligomerisation is required for proper OS maturation (PMID: 32716032, PMID: 29961824). Creates and maintains the rim region	PMID: 20592197, PMID: 32716032, PMID: 29961824, PMID: 30630813	Rod photoreceptor outer segment	x				x

				of rod discs and cone lamellae and help to regulate disc size and alignment (PMID: 29961824). Homozygous frameshift mutation has been recently associated with pattern dystrophy (PMID: 30630813).							
ENSG00000145491	rhophilin associated tail protein 1 like(ROPN1L)	ROPN1L		Functions as a key molecule in the maintenance of basal ciliary beat frequency. Its role in ciliary motility could be played via regulation of axonemal signaling (PMID: 22021175)	PMID: 22021175	Cilium, Cytosol	x		x		x
ENSG00000104237	retinitis pigmentosa 1 (autosomal dominant)(RP1)	RP1	603937	Nonsense mutation associated with retinitis pigmentosa (PMID: 10401003). Microtubule-associated protein containing the doublecortin (DCX) domain. Disruption of Rp1 in mice causes misorientation of outer segment discs, suggesting a role in outer segment organisation. Stimulates microtubule polymerization and stabilise existing microtubules in vitro and in vivo, therefore, it controls the length and stability of the photoreceptor axoneme (PMID: 15269252). Due to Rp1 localisation, it may be involved in the formation of outer segments (PMID: 11773008). Plays synergistic role with Rp1L1 in affecting photosensitivity and OS morphogenesis of rod photoreceptors (PMID: 19657028).	PMID: 10401003, PMID: 15269252, PMID: 11773008, PMID: 19657028	Connecting Cilium and outer segment portion of photoreceptor axoneme.	x		x		x

ENSG00000183638	retinitis pigmentosa 1 like 1(RP1L1)	RP1L1	608581	Shares similar DCX tandem repeats and a 34 aa domain (RP1D) with Rp1. Plays essential and synergistic roles with Rp1 in affecting photosensitivity and OS morphogenesis of rod photoreceptors. Mutations in RP1L1 could underlie retinopathy or modify RP1 disease expression in humans (PMID: 19657028). Mutations in RP1L1 have been associated with occult macular dystrophy and RP (PMID: 20826268)	PMID: 19657028, PMID: 20826268	Connecting Cilium and outer segment portion of photoreceptor axoneme.	x				
ENSG00000102218	retinitis pigmentosa 2 (X-linked recessive)(RP2)	RP2	312600	Interacts with Arl3 and mediates the localization of GFP-Kif17 to the cilia tip and interacts with the regulator of Hedgehog (Hh) signaling Kif7. This suggests RP2 is implicated primary cilia protein trafficking via the regulation of kinesin motor proteins at the ciliary tip (PMID: 28444310). Loss of RP2 leads to fragmentation of the Golgi network and dispersal of vesicles cycling cargo from the Golgi complex to the cilium, including IFT20. It indicates a role for RP2 in regulating vesicle traffic and docking from the Golgi to the base of the cilium (PMID: 20106869). Interacts with polycystin 2 and regulates its trafficking within the cilium (PMID: 20729296). RP2 entry to the ciliary compartment is regulated by Importin β 2 (PMID: 21285245). Mutations are associated with with X-linked retinitis pigmentosa (PMID: 9697692).	PMID: 28444310, PMID: 20106869, PMID: 20729296, PMID: 21285245, PMID: 9697692	Golgi, Basal Body and along the length of the ciliary compartment	x		x		x

ENSG00000156313	retinitis pigmentosa GTPase regulator(RPGR)	RPGR	312610	Role in regulation of nascent disc formation in photoreceptor cilium, by regulating actin-mediated membrane extension. Absence of RPGR impaired ciliogenesis and cell attachment (PMID: 21933838). Mutations cause type 3 retinis pigmentosa (PMID: 8673101). RPGR is suggested to regulate cilia function and facilitate the trafficking of proteins along the photoreceptor cilium. Interacts with ciliary proteins, such as RPGRIP1, IQCB1, CEP290 and RPGRIP1L, as well as Rab8a (PMID: 20631154).	PMID: 8673101, PMID: 21933838, PMID: 20631154	Basal Body, Centriole, cilium.	x		x		x
ENSG00000092200	retinitis pigmentosa GTPase regulator interacting protein 1(RPGRIP1)	RPGRIP1	605446	Known ciliopathy gene CORD13. Mutations are known causes of RP and Leber congenital amaurosis. Potential structural component of the ciliary axoneme, which anchors RPGR within the cilium (PMID: 11104772, PMID: 15772089). Required for disk morphogenesis and RPGR retention in the connecting cilium (PMID: 12651948). Essential for rod OS formation, probably by regulating the transport of proteins from the IS to the OS (PMID: 19679561). Required for ciliary targeting of NPHP4 and SDCCAG8, Rab8 localisation in photoreceptors and is involved in rhodopsin ciliary vesicle transport in zebrafish (PMID: 29203866).	PMID: 11104772, PMID: 12651948, PMID: 19679561, PMID: 29203866, PMID: 15772089	Basal Body/Centriole, Cilium, Connecting Cilium	x		x		x

ENSG00000103494	RPGRIP1 like(RPGRIP1L)	RPGRIP1L	610937	Mutations cause Joubert (PMID: 17558407) and Meckel syndrome (PMID: 17558409). It functions as cilium-specific scaffold that recruits a Nek4 signaling network to regulate cilium assembly and stability. Interacts with RPGR (PMID: 21685204). Loss of RPGRIP1L in human primary cell cultures and mice increases ciliary axonemal length (PMID: 30728336). Interacts with MKS and NPHP proteins, and could potentially act as part of the gate for cytoplasmic proteins (PMID: 30237221). Interacts with Psmd2, regulating the activity of the ciliary proteasome at the transition zone to control ciliary signaling. This proteosomal function could be involved in the control of ciliary length (PMID: 26150391). Rpgrip1l positively controls dishevelled levels. In the mouse and zebrafish, Rpgrip1l is required for positioning the basal body along the planar polarity axis; morphants display planar cell polarity defects (PMID: 22927466).	PMID: 17558407, PMID: 17558409, PMID: 21685204, PMID: 30728336, PMID: 30237221, PMID: 26150391, PMID: 22927466	Basal Body, transition zone.	x		x		x
ENSG00000189306	ribosomal RNA processing 7 homolog A(RRP7A)	RRP7A		Localizes to centrosomes, cilia and nucleoli. Highly expressed in neural stem cells in developing human forebrain, and a missense mutation is associated with microcephaly. Targeted mutation of Rrp7a leads to defects in neurogenesis and proliferation in a mouse stem cell model and in zebrafish causes reduced brain size, impaired neurogenesis and cell proliferation, and defective ribosomal RNA processing. Patient-derived fibroblasts display defects in	PMID: 33199730	Centrosomes, cilia and nucleoli	x				

				ribosomal RNA processing, primary cilia resorption, and cell cycle progression (PMID: 33199730)							
ENSG00000160188	radial spoke head 1 homolog(RSPH1)	RSPH1	609314	Mutations have been shown to cause PCD (PMID: 25473808, PMID: 24568568). Integral component of the ciliary axoneme expressed during ciliated cell differentiation (PMID: 24568568). Mainly expressed in tissues with motile cilia or flagella; absence of RSPH1 leads to abnormal axonemal configuration with central-complex defects, suggesting a role for RSPH1 in proper building of central-complexes and radial spokes (PMID: 23993197).	PMID: 25473808, PMID: 24568568, PMID: 23993197	Cilium, axoneme, radial spoke region of the sperm flagellum	x		x		x
ENSG00000130363	radial spoke 3 homolog(RSPH3)	RSPH3	615876	Protein kinase A-anchoring protein that scaffolds the cAMP-dependent protein kinase holoenzyme (PMID: 19684019). Mutations are associated with PCD, with CC/RS defects like abnormal axoneme configuration and absence of RS. Required for proper building of RSs and the CC in humans (PMID: 26073779). Mutations in RSPH3 in Chlamydomonas reinhardtii lead to paralyzed flagella due to loss of the entire radial spoke structure (PMID: 29475374).	PMID: 19684019, PMID: 26073779, PMID: 29475374	Cilium, Axoneme	x		x		x

ENSG00000111834	radial spoke head 4 homolog A(RSPH4A)	RSPH4A	612647	Component of spoke heads. Plays a central role in the assembly of all the three-types of RS. Contributes to the generation of the planar beating of motile cilia by building the distal architecture of radial spokes, regulating the motion pattern of the mouse motile cilia. Mutations lead to most severe phenotype of PCD with central complex defects and near absence of radial spokes (PMID: 26073779; PMID: 25789548) Essential for the triplet radial spoke head assembly of the mouse motile cilia (PMID: 32203505).	PMID: 32203505; PMID: 26073779; PMID: 25789548	Cilium, Axoneme	x		x		x
ENSG00000104941	radial spoke head 6 homolog A(RSPH6A)	RSPH6A		RSPH6A is required for sperm flagellar assembly and male fertility in mice. Interacts with RSPH1, RSPH4A, RSPH9 and RSPH10B, where nteraction of RSPH6A and RSPH9 is important to incorporate RSPH9 into the flagella. Absence of RSPH6A leads to immotile spermatozoa in mice (PMID: 30185526). Expressed exclusively in testicular germ cells during meiosis. Rsph6a is phosphorylated by PKA during sperm capacitation (PMID: 30239614).	PMID: 30185526, PMID: 30239614	Radial spoke of motile cilium, flagellar axoneme.	x				
ENSG00000172426	radial spoke head 9 homolog(RSPH9)	RSPH9	612648	Component of the axonemal radial spoke head. Mutations cause PCD with defects of the central microtubular pair, where altered function of the radial spoke heads causes 9+2 cilia motility to resemble the rotary 9+0 cilia movement (PMID: 19200523). Essential for ciliary structure and motility in mouse ependymal cilia; deletion causes hydrocephalus. Dysfunction of RSPH9 changes the motion pattern of motile cilia. (PMID: 32709945). Required for RS	PMID: 19200523, PMID: 32709945, PMID: 30383886	Radial spoke of motile cilium	x		x		x

				head assembly and central pair maintenance in mammalian ependymal cilia (PMID: 30383886).							
ENSG00000176225	rotatin(RTTN)	RTTN	610436	Regulates early embryonic axial rotation, anteroposterior and dorsoventral patterning in mice and is required for maintenance of a normal ciliary structure (PMID: 22939636). Required for assembly of full-length centrioles through its STIL-RTTN interaction; it also serves as an upstream effector of CEP295 (PMID: 28811500). Centrosomal protein possibly involved in centriolar elongation and ciliogenesis. Mutations cause polymicrogyria (PMID: 30879067) and missense or splicing variants have been found in patients with microcephaly (PMID: 29967526).	PMID: 22939636, PMID: 28811500, PMID: 30879067, PMID: 29967526	Basal Body, axoneme.	x		x		x
ENSG00000175792	RuvB like AAA ATPase 1(RUVBL1)	RUVBL1		Role in the pre-assembly and targeting of ciliary protein complexes. Ruvbl1 may be a major organiser of dynein arm complex assembly (PMID: 29959317). Essential for cilia motility in both zebrafish and mouse, cooperating with Ruvbl2 through the assembly of ciliary ODAs and IDAs; requires Ruvbl2 for stability. Essential for the stabilization of axonemal dynein intermediate chain 1 (DNAI1) and DNAI2 in mouse, and facilitates dynein arm assembly in cytosolic foci enriched with R2TP-like complexes (PMID: 29959317, PMID: 29113992, PMID: 28574207)	PMID: 29959317, PMID: 29113992, PMID: 28574207	Cytoplasm, nucleus, basal bodies, cilium.	x			x	x

				29113992). Depletion of HsRuvBL1 in RPE1 cells leads to the loss of primary cilia (PMID: 28574207).							
ENSG00000183207	RuvB like AAA ATPase 2(RUVBL2)	RUVBL2		Essential for cilia motility in both zebrafish and mouse, through the assembly of ciliary ODAs and IDAs and cooperating with Ruvbl1. Interacts with Hsp90 to regulate the stability of both DNAI1 and DNAI2. Requires Ruvbl1 for stability (PMID: 29113992). Involved in epigenetic and transcriptional regulation of cilia motility genes; interacts with LRRC6 in a complex involved in dynein arm formation, essential for cilia motility in zebrafish. Candidate gene for PCD (PMID: 23858445).	PMID: 29113992, PMID: 23858445	Nucleus, cytoplasm		x		x	x
ENSG00000103449	spalt like transcription factor 1(SALL1)	SALL1	602218	Zinc-finger transcription factor that mediates transcriptional repression via its interaction with the nucleosome-remodeling deacetylase (NuRD) complex. Truncated SALL1 interacts with or sequester factors related to cilia function, including the negative regulators of ciliogenesis CCP110 and CEP97. SALL1 mutations can disrupt cilia formation and function (PMID: 29395072). Truncated SALL1 increases the ubiquitin proteasome-mediated degradation of LUZP1, contributing to Townes-Brocks Syndrome (PMID: 32553112).	PMID: 29395072, PMID: 32553112	Nucleus, cytoplasm		x			

ENSG00000156876	SAS-6 centriolar assembly protein(SASS6)	SASS6	609321	Required for daughter centriole formation and primary cilia assembly in mammalian cells. Recruited to centrioles by STIL (PMID: 22349705). Interacts with SAS-5 and both are mutually dependent for their centriolar distribution (PMID: 15665853). Cooperates with PLK4 and STIL to initiate centriole duplication (PMID: 27911707). Depletion has been shown to reduce the microtubule numbers in Chlamydomonas and Drosophila centrioles, suggesting that SASS6 has a direct role in the regulation of centriolar microtubule formation (PMID: 26422590).	PMID: 22349705, PMID: 15665853, PMID: 27911707, PMID: 26422590	Centrioles, cytoplasmic foci	x		x		x
ENSG00000155875	stabilizer of axonemal microtubules 1(SAXO1)	SAXO1		FAM154A. A microtubule-stabilizing protein specific to cilia and related structures. Regulates cilium axonemal length and promotes microtubule assembly in Hela , RPE1 cells and Trypanosome Brucei (PMID: 33200789). It binds specifically to centriole and cilium microtubules and is responsible for their targeting and stabilisation. SAXO1 is also reported to be present across the length of human sperm flagellum where it colocalizes with α -tubulin, where it could play a role in its biogenesis, stability and length (PMID: 25673876)	PMID: 33200789, PMID: 25673876	Centrioles/basal body, cilium, flagellar axoneme.	x				x
ENSG00000140386	S-phase cyclin A associated protein in the ER(SCAPER)	SCAPER	611611	Varians have been associated with intellectual disability and autosomal recessive RP. Mutations affect cilia length, suggesting a rol for SCAPER in restricting cilia elongation or enhancing cilia resorption (PMID: 30723319). Although it is ubiquitously expressed in human tissues, deleterious mutations appear to affect	PMID: 30723319, PMID: 28794130	Ciliary axoneme, basal body	x				

				only retina and brain. In the retina, SCAPER appears to be important for photoreceptor function and/or maintenance rather than for development (PMID: 28794130).							
ENSG00000151466	sodium channel and clathrin linker 1(SCLT1)	SCLT1		Recruited to centrioles via CEP83. Subsequently, it recruits FBF1 and CEP164. Depletion of SCLT1 in RPE1 cells disrupts CEP164 localisation and blocks cilia formation (PMID: 23348840). Variants have been identified in patients with a severe ciliopathy phenotype. Candidate gene of autosomal recessive Oro-Facio-Digital (OFD) syndrome (PMID: 24285566). Recruits LRRC45 to the distal appendages of mother centriole (PMID: 30131441).	PMID: 23348840, PMID: 24285566, PMID: 30131441	Centrioles.	x		x		x
ENSG00000054282	serologically defined colon cancer antigen 8(SDCCAG8)	SDCCAG8		Mutations cause nephronophthisis type 10. Interacts with ODF1 and FAM161A (PMID: 20835237, PMID: 24722439) and FA161A; plays a role in intracellular trafficking and is essential for the survival and maintenance of both rods and cones (PMID: 24722439). Mutations have also been reported in patients with Bardet-Biedl syndrome (PMID: 22190896). Required for the formation of outer segments (PMID: 22825473).	PMID: 20835237, PMID: 24722439, PMID: 22190896, PMID: 22825473	Basal Body, Centrioles, Transition Zone, connecting cilium	x		x		x

ENSG00000161956	SUMO specific peptidase 3(SENP3)	SENP3		SUMO specific peptidase 3 (SENP3). Dominant form of de-sumoylation enzymes in human retina (PMID: 30621560)	PMID: 27746045, PMID: 30621560	Cytoplasm, nucleus		x				
ENSG00000168385	septin 2(SEPT2)	SEPT2		Component of a diffusion barrier at the ciliary base, essential for retaining receptor-signaling pathways in the primary cilium, and is required for cilium-dependent Shh signal transduction. Depletion inhibits ciliogenesis (PMID: 20558667). Ciliary localisation of SEPT2, suggest that it is involved in organization of the axonemal microtubular structure, controlling ciliogenesis and ciliary length (PMID: 24317785). Forms a complex with SEPT7 and SEPT9 to positively control ciliogenesis and the length of cilia (PMID: 23572511).	PMID: 20558667, PMID: 24317785, PMID: 23572511	Cytoplasm, ciliary base, axoneme	x		x			x
ENSG00000125354	septin 6(SEPT6)	SEPT6		Associates with acetylated α -tubulin and controls ciliogenesis. Knockdown results in aberrant cilium formation. Required for left-right patterning of visceral organs in zebrafish by controlling nodal signaling and through KV ciliogenesis. Sept6 is also required for ciliogenesis in the neural tube (PMID: 24469395).	PMID: 24469395	Cytoplasm, ciliary base, axoneme	x					x

ENSG00000122545	septin 7(SEPT7)	SEPT7		Forms a complex with SEPT2 and SEPT9 to positively control ciliogenesis and the length of cilia (PMID: 23572511). Component of a diffusion barrier at the ciliary base, essential for retaining receptor-signaling pathways in the primary cilium (PMID: 20558667). Knockdown affects ciliogenesis and disrupts expression of Hedgehog target genes (PMID: 20671153).	PMID: 23572511, PMID: 20558667, PMID: 20671153	Cytoplasm, ciliary base, axoneme	x		x			x
ENSG00000164690	sonic hedgehog(SHH)	SHH		Binds to Ptc1 and inactivates it, leading to accumulation of Smo and activation of signaling. Primary cilia regulates its activity (PMID: 17641202). Shh controls patterning and proliferation through the primary cilium (PMID: 31825820).	PMID: 17641202, PMID: 31825820	Ciliary Membrane	x		x			x
ENSG00000181788	siah E3 ubiquitin protein ligase 2(SIAH2)	SIAH2		Shh signaling maintains Siah2 expression (which supports ciliogenesis in a feed-forward fashion by restraining cilium disassembly) in cerebellar granule neuron (CGN) progenitors (GNPs) in a Ras/Mapk-dependent manner. This maintains GNPs to control CGN occupancy in the germinal zone (GZ) (PMID: 33082319)	PMID: 33082319	Cytoplasm		x				
ENSG00000068903	sirtuin 2(SIRT2)	SIRT2		Regulates cell cycle progression at mitosis, replication and G1/S transition. SIRT2 acts as a sensor that links growth factor-dependent signal transduction and primary cilia formation by regulating the cell cycle under nutrient-present conditions. Suppression leads to a cell cycle arrest and promotes formation of primary cilia, by inhibiting mTOR signaling mainly via mTORC1 (PMID:	PMID: 32213867, PMID: 24203696	Cytoplasm, centrosome, nucleus, membrane	x					

				32213867). Functions as a tubulin deacetylase. Regulates ciliogenesis and centrosome amplification. Inhibition of SIRT2 induces hyperacetylation of α -tubulin and blocks cilia disassembly. SIRT2 could also be responsible for aberrant centrosome amplification and polyploidy in polycystic kidney disease (PMID: 24203696).							
ENSG00000180638	solute carrier family 47 member 2(SLC47A2)	SLC47A2		Also known as MATE2. It is a multidrug and toxin extrusion transporter as well as transportes of other positively charged small molecules, and a brain-specific central gene in the ciliary module (PMID: 22558177).	PMID: 22558177	Membrane, primary cilium compartment	x		x		x
ENSG00000109062	SLC9A3 regulator 1	SLC9A3R1		Also known as EBP50 or NHERF1. Regulates Wnt and PCP signaling PCP signaling and the development of functional motile cilia via binding the planar cell polarity genes Frizzled (Fzd) and Vangl. Knockdown in zebrafish embryos causes severe hydrocephalus. Ablation causes reduced number and altered orientation of the cilia but does not change the structural organization of the cilium (PMID: 27055101).	PMID: 27055101	Apical cell membrane, cytoplasm		x		x	x
ENSG00000175387	SMAD family member 2(SMAD2)	SMAD2		Mediator in the activation of TGF- β signaling pathway and controls cilia length (PMID: 25959824). Downstream of TGFBR1 and 2, which phosphorylate SMAD2/3 at base of cilium upon TGF- β stimulation (PMID: 23746451).	PMID: 25959824, PMID: 23746451	Basal body, nucleus	x				

ENSG00000166949	SMAD family member 3(SMAD3)	SMAD3		Downstream of TGFBR1 and 2, which phosphorylate SMAD2/3 at base of cilium upon TGF- β stimulation (PMID: 23746451). Sox2 can interact with Smad3 and inhibit TGF- β 1/Smad3-mediated transcriptional activity (PMID: 20011520).	PMID: 23746451, PMID: 20011520	Basal body, nucleus	x				
ENSG00000128602	smoothened, frizzled class receptor(SMO)	SMO		Translocates to the primary cilium membrane and activates downstream signaling. It is enriched in cilia upon stimulation with Shh (PMID: 17641202). Smo traslocation to the cilium results in Glis activation (PMID: 28344083). Hh signalling depends upon a Smo ciliary localization (PMID: 16136078). Defect in Smo localisation results from defective retrograde transport (PMID: 16229832). For instance, defects in intraflagellar transport components lft27 and the BBSome, cause ciliary accumulation of Smo without pathway activation (PMID: 32435793).	PMID: 17641202, PMID: 28344083, PMID: 16136078, PMID: 16229832, PMID: 32435793	Cilium, ciliary Membrane, cell membrane	x		x		x
ENSG00000132639	synaptosome associated protein 25(SNAP25)	SNAP25		Regulate rhodopsin delivery; controls ciliogenesis through the expansion of the ciliary membrane and rod outer segments biogenesis (PMID: 19454479). Potentially involved in membrane trafficking into the outer segment (PMID: 9452464).	PMID: 19454479, PMID: 9452464	Cilium, ciliary membrane, base of cilium.	x		x		x

ENSG00000099940	synaptosome associated protein 29(SNAP29)	SNAP29		SNARE membrane fusion regulator that binds to EHD1 and EHD3. This interaction is required for assembly of ciliary vesicle from small distal appendage vesicles (PMID: 25686250). Role in Golgi trafficking and vesicle fusion. Mutations cause CEDNIK syndrome (PMID: 15968592, PMID: 21073448).	PMID: 25686250, PMID: 15968592, PMID: 21073448	Ciliary pocket membrane, Golgi	x					x
ENSG00000086300	sorting nexin 10(SNX10)	SNX10		Required for the formation of primary cilia and essential for left-right patterning of visceral organs. Binds to V-ATPase to regulate its centrosomal trafficking and Rab8a vesicle trafficking to the ciliary membrane. Those events are crucial for ciliogenesis (PMID: 21844891).	PMID: 21844891	Centrosome/basal body, early endosome	x		x			x
ENSG00000115234	sorting nexin 17(SNX17)	SNX17		SNX17 functions in early stages of ciliogenesis. Regulates the homeostasis of centriolar satellite proteins including PCM1, CEP131, and OFD1 during serum starvation induced ciliogenesis. Recruits USP9X to antagonise MIB1-induced degradation of PCM1, maintaining the integrity of centriolar satellites. SNX17 deficiency leads degradation of USP9X and PCM1, and disrupts ciliogenesis upon serum starvation (PMID: 31671755).	PMID: 31671755	Golgi, early endosome, cytoplasm		x				
ENSG00000064199	sperm autoantigenic protein 17(SPA17)	SPA17		Regulatory role in a PKA-independent AKAP complex in both germinal cells and somatic ciliated epithelia. It may also play a role in cell motility (PMID: 19604394, PMID: 15034007). Binds the RII binding domain of AKAP3 along the length of the flagella (PMID: 15257753).	PMID: 19604394, PMID: 15034007, PMID: 15257753	Cilium, flagellum, cytoplasm.	x		x			x

ENSG00000104450	sperm associated antigen 1(SPAG1)	SPAG1	603395	Possible role in the cytoplasmic assembly and/or trafficking of the ciliary dynein arms. Mutations have been reported in patients with PCD with ODA and IDA defects, severe ciliary dysmotility and laterality defects (PMID: 24055112).	PMID: 24055112	Cytoplasm		x				x
ENSG00000144451	sperm associated antigen 16(SPAG16)	SPAG16		Orthologue of Chlamydomonas reinhardtii PF20, known to be essential to the structure and function of the motile axoneme. Encodes two proteins, SPAG16L and SPAG16S. SPAG16L is essential for sperm flagellar motility. Interacts with SPAG6, which in turn interacts with SPAG17. Mutation results in instability of interacting proteins in the central pair of the sperm axoneme. (PMID: 16382026, PMID: 17699735). SPAG16S, identified only in male germ cells, plays a role in the gene expression machinery of male germ cells required for spermatogenesis, and it also modulates Spag16L mRNA expression (PMID: 21655194). Transcription of SPAG16 is regulated through S-SOX5 binding to the promoter (PMID: 28137312).	PMID: 16382026, PMID: 17699735, PMID: 21655194, PMID: 28137312	Cytoplasm, central apparatus of the axoneme (SPAG16L), nuclear speckles (SPAG16S).	x		x			x
ENSG00000155761	sperm associated antigen 17(SPAG17)	SPAG17		Encodes a central pair protein present motile cilia or flagella with a '9+2' axoneme structure and plays both a structural and functional role in the ciliary axoneme. Interacts with SPAG6 and SPAG16. KO mice have immotile nasal and tracheal cilia, respiratory problems and hydrocephalus (PMID: 23418344). Essential for proper development of the sperm flagellum and	PMID: 23418344, PMID: 32988999, PMID: 29690537	Central pair of motile cilia	x		x			x

				development or stability of the C1 microtubule structure within the central pair of the respiratory motile cilia (PMID: 32988999). Disruption of SPAG17 affects the recruitment of Pcdp1 and IFT20 to the manchette, which indicated a role in the transport of cargo proteins (PMID: 29690537).							
ENSG00000077327	sperm associated antigen 6(SPAG6)	SPAG6		Orthologue of Chlamydomonas PF16. Axoneme central apparatus protein, essential for sperm flagellar motility and motility of ependymal cilia, as well as maintenance of the structural integrity of the sperm flagella by serving as a scaffold protein (PMID: 12167721). Role in ciliogenesis, axoneme orientation, and cell polarity. Spag6-deficient epithelial cells in the brain and trachea lose rotational polarity and Spag6-efficient mice suffer hydrocephalus and are sterile because of sperm motility defects (PMID: 25333478). Mutated in PCD with male infertility with severe asthenoteratospermia (PMID: 32124190)	PMID: 12167721, PMID: 25333478; PMID: 32124190	Axoneme, center of the transition zone at the site of central pair assembly, cytoplasm.	x		x		x
ENSG00000042317	spermatogenesis associated 7(SPATA7)	SPATA7	609868	Expressed in spermatocytes and in the mature retina. Mutations have been associated with Leber Congenital Amaurosis and Retinitis Pigmentosa (PMID: 19268277). Component of a protein complex which interacts with CETN2 and CETN3 to stabilise the photoreceptor distal connecting cilium and proximal OS axoneme (PMID: 30647131). Interacts with photoreceptor-specific ciliary proteins RPGR and RPGRIP1 to maintain distal connecting cilium (DCC) structure; it also	PMID: 19268277, PMID: 30647131, PMID: 29899041	Cytoplasm of inner segment of photoreceptors		x	x		x

				regulates protein localization into the DCC (PMID: 29899041).							
ENSG00000101222	sperm flagellar 1(SPEF1)	SPEF1		Also known as CLAMP. Regulator of PCP signaling, via localisation of asymmetric PCP components and regulates asymmetric microtubule accumulation. Required for proper PCP-dependent establishment of cilia polarity (PMID: 29514918). Microtubule-bundling protein, and central pair MT regulator highly expressed in 9 + 2 mammalian cilia. Essential for the assembly and functions of central pair MTs, central pair apparatus, and for ciliary retentions of other central pair proteins such as Spag6 and Hydin (PMID: 30535028).	PMID: 29514918, PMID: 30535028	Axoneme, along central pair, ciliary rootlet	x			x	x
ENSG00000152582	sperm flagellar 2(SPEF2)	SPEF2		Essential for correct sperm tail development and male fertility. Interacts with intraflagellar transport protein IFT20. Mutation in SPEF2 was identified as a cause for immotile short tail sperm in pigs (PMID: 19889948). Required for ciliary motility and spermatogenesis. In mouse, loss of SPEF2 function results in PCD with severe spermatogenic defects (PMID: 21715716). Functional defects also lead to PCD due to a defect of the central pair complex. Interacts with HYDIN, which regulates assembly of SPEF2 to the ciliary axoneme, as it cannot	PMID: 19889948, PMID: 21715716, PMID: 31545650, PMID: 31942643	Golgi, cytoplasm basal body, midpiece of the sperm tail, sperm manchette, ciliary axoneme	x		x		x

				be detected in the axonemes of HYDIN-mutant cells (PMID: 31545650). Required for the flagellar assembly of central pair complex and peri-axonemal structure in humans (PMID: 31942643).							
ENSG00000072310	sterol regulatory element binding transcription factor 1(SREBF1)	SREBF1		Also known as SREBP1. Suppresses primary ciliogenesis. Induces large levels of lysophosphatidylcholine and lysophosphatidylethanolamine , which are typically generated at the outer membrane leaflet. It increases the positive membrane curvature in the plasma membrane and vesicle membranes, causing Rab11-positive vesicles to deviate away from the pericentrosomal preciliary compartment at the ciliary base, suppressing ciliogenesis (PMID: 25904332).	PMID: 25904332	Nucleus, Golgi.		x			
ENSG00000196220	SLIT-ROBO Rho GTPase activating protein 3(SRGAP3)	SRGAP3		Regulates the actin cytoskeleton, forming a signalling complex with WAVE-1 to regulate its activation. 10% of SrGAP3 knockout mice die from hydrocephalus and display ependymal cells with reduced density of cilia but without alterations in the length (PMID: 26104135).	PMID: 26104135	Cytoplasm		x			x

ENSG00000176101	SS nuclear autoantigen 1(SSNA1)	SSNA1	Required for ciliary transport and signalling. It may regulate postciliary endocytosis of constitutive GPCRs and facilitate endocytosis from the cell surface following transit through cilia, as knockdown inhibits Gli3 ciliary accumulation but retain normal length cilia (PMID: 21289087).	PMID: 21289087	Nucleus, centrosome	x		x		x
ENSG00000278195	somatostatin receptor 3(SSTR3)	SSTR3	Ciliary GPCR. Localized to the plasma membrane of neuronal cilia (PMID: 10199624, PMID: 23392690). Ciliary trafficking of SSTR3 into neural cilia is promoted by TULP3. Overexpression induces premature growth of primary cilia and increases expression levels of Kif3A, IFT88 and cytoplasmatic dynein (PMID: 23392690). Possible influence on cell survival and/or apoptotic processes, as well as cell cycle (PMID: 18992731).	PMID: 10199624, PMID: 23392690, PMID: 18992731	Ciliary Membrane	x				
ENSG00000117155	SSX family member 2 interacting protein(SSX2IP)	SSX2IP	Centriolar satellite and microtubule anchoring protein. Interacts with PCM-1. Essential for the assembly of primary cilia and the recruitment of Cep290 to both satellites and the basal body; it also promotes the ciliary entry of the entire BBSome. Knockdown limits targeting of somatostatin receptor 3, and significantly reduces axoneme length. Possible novel cooperator of Rab8 (PMID: 24356449). Binds the microtubule-nucleator γ -tubulin complex. Knockdown of its orthologue in zebrafish, results in ciliary defects and disturbs left-right asymmetry. Microtubule anchoring role to the basal body is required for primary cilia formation (PMID: 24397932).	PMID: 24356449, PMID: 24397932	Centriolar satellites, basal body.	x				x

ENSG00000123473	SCL/TAL1 interrupting locus(STIL)	STIL	Centrosomal protein necessary for SAS6 recruitment to centrioles, which suggests an essential role in daughter centriole formation. Required for mouse embryonic left-right specification, For PLK4-induced centriole duplication and primary cilia formation. Mutated in autosomal recessive primary microcephaly (PMID: 22349705). Phosphorylation of STIL promotes its binding to CPAP (PMID: 31115335). Key component of de novo centriole biogenesis in mammalian. Null mutants embryos show lack of centrioles and primary cilia, with absence of Hedgehog signaling (PMID: 25486474).	PMID: 22349705, PMID: 31115335, PMID: 25486474	Centrioles	x		x		x
ENSG00000118046	serine/threonine kinase 11(STK11)	STK11	Downregulated in diseases associated with ciliopathies, including BBS, PKD and PHP. Regulates multiciliogenesis and differentiation of ciliated cells by phosphorylating MARK3 which suppresses ERK1/2 (PMID: 31636950). Required for CCL2 regulation within cilia, via interactions with NPHP1, ANKS3 and NEK7 (PMID: 29925518). Bending of cilium activates Lkb1 pathway, which inhibits mTORC1 activity and regulation cell size (PMID: 20972424). Duct-specific deletion causes duct dilations and shorter cilia with swollen tips, indicative of perturbed intraciliary transport (PMID: 28159992).	PMID: 31636950, PMID: 29925518, PMID: 20972424, PMID: 28159992	Cilium, basal body.	x				

ENSG00000163482	serine/threonine kinase 36(STK36)	STK36	Also known as FU. Participates in Hh signaling, 9+2 cilia biogenesis and function, as well as assembly or maintenance of the central pair apparatus. Interacts with Kif27 (PMID: 19305393), Spag16 and Pcdp1 (PMID: 23907739). Required for cilia orientation and the integrity of the whole radial spokes head. Essential for recruiting radial spokes head proteins to the axoneme. Mutations are associated with a novel PCD variant (PMID: 28543983). Stk36 null show craniofacial abnormalities and defects in organs where hedgehog signaling is required (PMID: 21746835).	PMID: 19305393, PMID: 23907739, PMID: 28543983, PMID: 21746835	Between radial spokes and central pair in motile cilia, along axonemes	x		x		x
ENSG00000211455	serine/threonine kinase 38 like(STK38L)	STK38L	Phosphorylates Rabin8 which is crucial role in ciliogenesis. Regulates the process of vesicular trafficking and tethering to the pericentrosome, promoting local activation of Rab8 and ciliary membrane formation. Defects in NDR2-mediated Rabin8 phosphorylation impair preciliary membrane assembly and ciliogenesis, leading to accumulation of Rabin8-/Rab11-containing vesicles at the pericentrosome. Knockdown in RPE1 cells supresses ciliogenesis (PMID: 23435566, PMID: 28122914). Mutated in canine early retinal degeneration, suggesting a role in photoreceptor cell function (PMID: 20887780).	PMID: 23435566, PMID: 28122914, PMID: 20887780	Cytosol, peroxisome		x	x		x

ENSG00000133115	stomatin like 3(STOML3)	STOML3		Localizes to Basal Body, mechanosensing channel. BBS4 and BBS8 knock-out impede its function. (PMID: 17959775, PMID: 21646512). Localised to olfactory cilia of mature olfactory sensory neurons. Involved in olfactory signaling (PMID: 12122055).	PMID: 17959775, PMID: 21646512, PMID: 12122055	Basal Body, proximal portion of olfactory cilia.	x		x		x
ENSG00000103266	STIP1 homology and U-box containing protein 1(STUB1)	STUB1	607207	Also known as CHIP. E3 ubiquitin ligase. Ubiquitylates and degrades NEK10. Novel regulator of protein turnover at ciliary sites, it couples GPCR signaling to cilia dynamics. Downregulation of CHIP prevents cilia resorption whilst increased expression of CHIP is linked to reduced NEK10 levels and ciliogenesis; role in control of cilium stability. Mutations have been reported in patients with autosomal recessive spinocerebellar ataxia-16 SCAR16 (PMID: 29581457).	PMID: 29581457	Centriolar satellites	x				
ENSG00000166900	syntaxin 3(STX3)	STX3		Major partner for SNAP25 in photoreceptor cells and regulates rhodopsin delivery by mediating docking and fusion of rhodopsin transport carriers, expansion of the ciliary membrane and rod outer segments biogenesis (PMID: 19454479). Mislocalises to the outer segment in Bardet-Biedl syndrome (PMID: 31694913, PMID: 26216965).	PMID: 19454479, PMID: 31694913, PMID: 26216965	Cilium, base of cilium, plasma membrane of inner segment in photoreceptors.	x		x		x

ENSG00000107882	SUFU negative regulator of hedgehog signaling(SUFU)	SUFU	Negative regulator of Gli, essential for repressing Hh signaling. SuFu prevents nuclear translocation of Gli proteins. PKA inhibits SuFu-Gli complexes and blocks their ciliary localisation (PMID: 20956384, PMID: 21209912). Mutations have been associated with Joubert syndrome, cranio-facial dysmorphisms and congenital ocular motor apraxia (PMID: 33024317, PMID: 28965847).	PMID: 20956384, PMID: 21209912, PMID: 33024317, PMID: 28965847	Ciliary Tip, cytoplasm.	x		x		x
ENSG00000131018	spectrin repeat containing nuclear envelope protein 1(SYNE1)	SYNE1	Also known as Nesprin1. Integral components of ciliary rootlets of multiciliated cells. Interacts with rootletin (PMID: 28625779). Essential for the relocalisation of centrosome proteins and components of microtubule motor complexes to the nuclear envelope, and interacts with PCM-1 (PMID: 27802164). Mutations are associated with cerebellar and non-cerebellar pathologies of unknown molecular pathogenesis (PMID: 29630990). Mediates centrosome migration and determination of centrosome/basal body positioning during early cell polarization, an essential step for ciliogenesis (PMID: 19596800).	PMID: 28625779, PMID: 27802164, PMID: 29630990, PMID: 19596800	Ciliary rootlets, nuclear envelope of rod photoreceptors.	x			x	x
ENSG00000054654	spectrin repeat containing nuclear envelope protein 2(SYNE2)	SYNE2	Also known as Nesprin2. Colocalises with meckelin at filopodia prior to the establishment of cell polarity and ciliogenesis. Mediates centrosome migration and determination of centrosome/basal body positioning during early cell polarization, an essential step for ciliogenesis. It may form a linking network between the centrosome and the actin cytoskeleton (PMID: 19596800). Role in nuclear	PMID: 19596800, PMID: 30054381	Actin cytoskeleton		x	x		x

				migration and positioning during retinal development. Interaction partner of Pcmt in photoreceptors. Pcmt-Syne-2 complex is important for ciliogenesis and outer segment formation during retinal development and for the attachment of the centrosome to the nucleus in cycling cells (PMID: 30054381).							
ENSG00000169762	transmembrane anterior posterior transformation 1(TAPT1)	TAPT1	616897	Regulates intracellular protein trafficking and organisation of cellular organelles; it could also play a role in ciliary assembly and signaling. Defective TAPT1 mislocalises to the cytoplasm, disrupts Golgi morphology, trafficking and normal ciliogenesis. Knockdown of tapt1b in zebrafish induces severe malformations of the craniofacial skeleton (PMID: 26365339).	PMID: 26365339	Centrosome and/or basal body	x				x
ENSG00000102125	tafazzin(TAZ)	TAZ		Cytoplasmic retention of YAP/TAZ correlates with active ciliogenesis either in spatially confined cells or in cells treated with an actin filament destabilizer. Moreover, knockdown of YAP/TAZ is sufficient to induce ciliogenesis, whereas YAP/TAZ hyperactivation suppresses serum starvation-mediated ciliogenesis.	PMID: 25849865	Cytoplasm		x			
ENSG00000111490	TBC1 domain family member 30(TBC1D30)	TBC1D30		PMID:17646400 involved in membrane trafficking, necessary for ciliogenesis (PMID: PMID: 17646400). GAP protein that regulates activity of specific Rab GTPases including RAB3A22 and RAB8A (PMID: 23263489).	PMID: 17646400, (PMID: 23263489)	Basal Body	x		x		x

ENSG00000146350	TBC1 domain family member 32(TBC1D32)	TBC1D32	615867	Mutations have been found in patients with a severe ciliopathy phenotype, including autosomal recessive OFD (PMID: 24285566, PMID: 32573025). Interacts with Dzip1l to regulate neural tube patterning and ciliogenesis before the docking of ciliary vesicle to the basal body (PMID: 29487109). Ciliary gene implicated in SHH signaling, controls ciliary morphology and Gli2 localisation within the cilium. Required for proper association between the ciliary membrane and axoneme and axonemal shape (PMID: 20159594).	PMID: 24285566, PMID: 32573025, PMID: 29487109, PMID: 20159594	Cilium	x					x
ENSG00000145979	TBC1 domain family member 7(TBC1D7)	TBC1D7		GTPase-activating protein that controls specific Rabs involved in primary cilium formation. TBC1D7 specifically acts on Rab17.	PMID:17646400	Basal Body	x		x			x
ENSG00000113838	TBCC domain containing 1(TBCCD1)	TBCCD1		Required for centrosome and Golgi apparatus perinuclear positioning. Depletion reduces primary cilia assembly efficiency/ability and affects cell shape and migration. TBCCD1-silenced cells showed disorganised Golgi apparatus (PMID: 20168327). Candidate for primary cause of centrosome amplification due to gain of function (PMID: 32681070).	PMID: 20168327, PMID: 32681070	Centrosome, basal body of primary and motile cilia.		x				

ENSG00000139437	trichoplein keratin filament binding(TCHP)	TCHP		Negative regulator of ciliogenesis. Polyubiquitinated at the mother centriole by CRL3-KCTD17. Depletion induces aberrant ciliogenesis, whereas overexpression blocks ciliogenesis (PMID: 25270598). Interacts with Odf2 and ninein. Controls Aurora A (AurA) activation at the centrioles in G1 phase. Trichoplein–AurA pathway is required for cell cycle progression through the suppression of primary cilia assembly (PMID: 22529102). TCHP is deubiquitinated and stabilized by USP8, critical to restrict ciliogenesis in dividing cells (PMID: 29472535).	PMID: 25270598, PMID: 22529102, PMID: 29472535	Subdistal/medial zone of centrioles, cytoplasm		x				x
ENSG00000213123	Tctex1 domain containing 2(TCTEX1D2)	TCTEX1D2		Dynein-2 light chain, required for cilia function. Associates with WDR34 (PMID: 25205765) and WDR60, which is required for ciliary protein trafficking, as KO cells show defects in retrograde ciliary protein trafficking (PMID: 29742051). Causative gene for short-rib polydactyly syndromes. Associates with Dync2h1 and other subunits of the Dync2 complex to deliver specific cargo necessary for ciliogenesis to the base of the cilium (PMID: 25830415). Mutations cause Jeune asphyxiating thoracic dystrophy (PMID: 26044572).	PMID: 25205765, PMID: 29742051, PMID: 25830415, PMID: 26044572.	Ciliary base		x			x	x
ENSG00000204852	tectonic family member 1(TCTN1)	TCTN1	609863	Required for cilium-dependent Hh signal transduction, for transition zone localisation of interacting proteins. Form a complex with Tctn3, Mks1, Tmem216, Tmem67, Cep290, Cc2d2a, B9d1 and Tctn2, which regulates ciliogenesis and ciliary membrane composition. Mutations have been identified to cause JBTS (PMID: 21725307). Mutations disrupts ciliogenesis and Hh	PMID: 21725307, PMID: 28800946, PMID: 29725084.	Transition Zone		x		x		x

				signaling (PMID: 28800946). Role in selective trafficking of ciliary proteins to cilia, required for ciliogenesis (PMID: 29725084).							
ENSG00000168778	tectonic family member 2(TCTN2)	TCTN2	613846	Form a complex with Tctn3, Mks1, Tmem216, Tmem67, Cep290, Cc2d2a, B9d1 and Tctn1 to regulate ciliogenesis and ciliary membrane composition (PMID: 21725307). Mutations disrupts ciliogenesis, Hh signaling (PMID: 28800946) and neural tube patterning (PMID: 29725084). Regulator of Shh pathway. Role in selective trafficking of ciliary proteins to cilia, required for ciliogenesis (PMID: 29725084). Indispensable for ciliogenesis in the neural tube (PMID: 21565611). Mutations have also been identified as cause of Meckel–Gruber syndrome (PMID: 32655147).	PMID: 21725307, PMID: 28800946, PMID: 29725084, PMID: 21565611, PMID: 32655147	Transition Zone	x		x		x
ENSG00000119977	tectonic family member 3(TCTN3)	TCTN3	613847	Form a complex with Tctn2, Mks1, Tmem216, Tmem67, Cep290, Cc2d2a, B9d1 and Tctn1 to regulate ciliogenesis and ciliary membrane composition (PMID: 21725307). Mutations disrupts ciliogenesis and Hh signaling (PMID: 28800946) and are also the causes of Joubert syndrome (PMID: 30286481). Regulator of Shh pathway. Role in selective trafficking of ciliary proteins to cilia, required for ciliogenesis (PMID: 29725084). Role for GLI3 processing and function. Mutations cause severe orofacioidigital syndrome, Mohr-Majewski syndrome	PMID: 21725307, PMID: 28800946, PMID: 30286481, PMID: 22883145	Transition Zone	x		x		x

				(PMID: 22883145).								
ENSG00000167858	tektin 1(TEKT1)	TEKT1		Positive regulator of cilium length. Its expression is controlled by the transcription factor FOXJ1. In zebrafish, it is required for fully efficient motility of cilia in various organs, and for ciliogenesis or maintenance of motile cilia in the KV. Mutations of which may cause ciliary dyskinesia (PMID: 29121203).	PMID: 29121203	Centrosome, basal bodies of both primary and motile cilia, axoneme of motile cilia.	x			x		x
ENSG00000092850	tektin 2(TEKT2)	TEKT2		Also known as Tektin-t. Forms filaments in flagellar, ciliary, and axonemal microtubules. Required for normal flagellar structure and function. Participates in dynein inner arm assembly or attachment to microtubules. Loss of Tektin-t results in impaired motility of cilia and flagella due to deficiency in flagellar construction. Candidate gene for PCD (PMID: 15340058).	PMID: 15340058	Cilium, periphery of the outer dense fiber (ODF) in the sperm flagella	x		x			x
ENSG00000163060	tektin 4(TEKT4)	TEKT4		Axonemal protein required for flagella motility in mouse sperm. Expression reduced in inasthenozoospermic men (PMID: 22774605, PMID: 16596631)	PMID: 22774605, PMID: 16596631	Cilium, cortex of outer dense fiber	x		x			x

ENSG00000153060	tektin 5(TEKT5)	TEKT5		Role in flagella assembly during spermiogenesis and is implicated in sperm motility (PMID: 20378928).	PMID: 20378928	Cilium, midpiece and principal piece of the sperm flagellum	x		x		x
ENSG00000107140	testis-specific kinase 1(TESK1)	TESK1		Role in the actin dynamics and actin cytoskeleton remodelling, via inactivation of the actin-severing protein CFL. Suppresses ciliogenesis by upregulating the expression of YAP/TAZ target genes, as well as modulating directional trafficking of ciliary vesicles to the ciliary base. TESK1 overexpression interferes with serum starvation-mediated ciliogenesis (PMID: 25849865).	PMID: 25849865	Cytoplasm		x			
ENSG00000105329	transforming growth factor beta 1(TGFB1)	TGFB1		Involved in defective differentiation of ciliated cells (PMID: 31784664). Regulates the transformation of RPE into fibrous cells (PMID: 15965172). Increases receptor localisation and activation of SMAD2/3 and ERK1/2 at the ciliary base. TGF-B signaling is regulated by the primary cilium. Involved in clathrin-dependent endocytosis (CDE) (PMID: 23746451).	PMID: 31784664, PMID: 15965172, PMID: 23746451	Ciliary tip (membrane), endocytic vesicles at the ciliary base.		x			
ENSG00000106799	transforming growth factor beta receptor 1(TGFB1)	TGFB1		Phosphorylates and activates downstream effectors MEK1/2-ERK1/2 and receptor SMADs, SMAD2/3 (PMID: 23746451). Involved in CDE (PMID: 27364476).	PMID: 23746451, PMID: 27364476	Base of primary cilium; ciliary tip (membrane).	x				

ENSG00000163513	transforming growth factor beta receptor 2(TGFBFR2)	TGFBFR2		Phosphorylates and activates downstream effectors MEK1/2-ERK1/2 and receptor SMADs. SMAD2/3 (PMID: 23746451). Involved in CDE (PMID: 27364476).	PMID: 23746451, PMID: 27364476	Base of primary cilium; ciliary tip (membrane).	x				
ENSG00000179029	transmembrane protein 107(TMEM107)	TMEM107	616183	Critical regulator of cilia formation and morphology (PMID: 26518474). Controls ciliary protein composition. Functions redundantly with NPHP4 to regulate cilium integrity, transition zone docking and assembly of membrane to microtubule Y-link connectors. Recruits ciliopathy proteins MKS1, TMEM231 and TMEM237. Mutated in OFD and JBTS patients (PMID: 26595381). In mice, mutations decrease cilia number and disrupt Hh signaling. Essential for craniofacial development (PMID: 28954202). Acts synergistically with Gli2 as a positive mediator of Shh and ciliogenesis (PMID: 22698544). Loss of Tmem107 results in disruption in early L-R patterning events at the node (PMID: 31887266).	PMID: 26518474, PMID: 26595381, PMID: 28954202, PMID: 22698544, PMID: 31887266	Transition Zone	x				x
ENSG00000149483	transmembrane protein 138(TMEM138)	TMEM138	614459	Mutations cause JBTS. Required for ciliogenesis. Mutations lead to shorter cilia, and knockdown cells show defects of ciliogenesis. It might mark vesicles en route to the base of cilium, which is known to be crucial for ciliary assembly (PMID: 22282472).	PMID: 22282472	Basal Body, Cilium	x		x		x

ENSG00000186889	transmembrane protein 17(TMEM17)	TMEM17		CEP-290-dependent transition zone protein that facilitates cilium formation, being necessary for ciliogenesis; mutated in OFD6 (PMID: 26982032).	PMID: 26982032	Transition Zone	x			x	x
ENSG00000187049	transmembrane protein 216(TMEM216)	TMEM216	613277	Required for ciliogenesis. It might mark vesicles en route to the base of cilium, which is known to be crucial for ciliary assembly. Mutations cause JBTS (PMID: 22282472). Mutated in OFD6 (PMID: 26982032) and in Meckel syndrome MKS (PMID: 21493627). Mutations or knockdown result in impaired ciliogenesis and centrosome docking. Modulates hyper-responsiveness of signaling pathways mediated by Dvl and RhoA, suggesting they might be part of a complex to mediate cellular polarization and centrosomal apical docking (PMID: 20512146).	PMID: 22282472, PMID: 26982032, PMID: 21493627, PMID: 20512146	Basal Body, transition zone, Golgi.	x		x		x
ENSG00000205084	transmembrane protein 231(TMEM231)	TMEM231	614949	Interacts with several JBTS proteins as part of the B9 complex, involved in the formation of the diffusion barrier between the cilia and plasma membrane, maintaining the integrity of the cilia. Knockdown results in a reduction of cilia formation, and perturbs localisation of components of the B9 complex at the transition zone and ciliary localisation of signalling receptors. Mutations have been associated with JBTS (PMID: 23012439). Mutated in MKS (PMID: 23349226). Critical for organizing the MKS complex and controlling ciliary composition. Mutated in	PMID: 23012439, PMID: 23349226, PMID: 25869670	Transition Zone	x		x		x

				OFD3. In mice, mutations disrupt the localisation of proteins Arl13b and Inpp5e to cilia (PMID: 25869670).							
ENSG00000155755	transmembrane protein 237(TMEM237)	TMEM237	614423	Also known as JBTS-14 . Interacts functionally with NPHP4, MKS2, MKSR1, MKSR2, and MKS5 at the transition zone. Its interaction with NPHP4 promotes basal body-TZ attachment to the membrane and cilium formation. Loss of TMEM237 results in defective ciliogenesis and deregulation of Wnt signaling. Mutated in Joubert syndrome related disorders (PMID: 22152675).	PMID: 22152675	Transition Zone	x		x		x
ENSG00000164953	transmembrane protein 67(TMEM67)	TMEM67	609884	Also known as Meckelin or MKS3. Component of the MKS-JBTS module at the transition zone. Essential for phosphorylation of the non-canonical Wnt receptor ROR2. Negative regulator of Shh signalling. Required for centriolar migration to the apical membrane during cell polarisation, regulation of actin cytoskeleton remodelling and RhoA activity (PMID: 26035863). Required for the elongation and maintenance of the outer segments in retina. It may play a role in photoreceptor intraciliary transport (PMID: 22247471). Mutations are a major cause of MKS, JBTS and NPHP (PMID: 23393159). MKS function is required for ciliary structure and function; plays a role in regulating cilia length and number through modulating centrosome	PMID: 26035863, PMID: 22247471, PMID: 2339315, PMID: 19515853. PMID: 21422230	Basal Body, transition zone, ciliary membrane	x		x		x

				duplication (PMID: 19515853). In complex with other TZ proteins, it contributes to an early ciliogenic event and establish basal body/TZ membrane attachments (PMID: 21422230).							
ENSG00000083312	transportin 1(TNPO1)	TNPO1		Also known as Importin β 2. Regulated RP2 trafficking into the ciliary compartment (PMID: 21285245). Interacts with KIF17; this interaction is required for the ciliary entry of KIF17 (PMID: 20526328). TNPO1 for a complex with Rab8 and CTSs (ciliary target signals) to mediate selective entry and retention of cargos within cilia (PMID: 27633000).	PMID: 21285245, PMID: 20526328, PMID: 27633000	Axoneme, nucleus, basal body	x		x		x
ENSG00000198718	TOG array regulator of axonemal microtubules 1(TOGARAM1)	TOGARAM1	619185	Mutated in JBTS37. TOGARAM1 localized along the ciliary axoneme in hTERT-RPE1 cells and was associated with longer cilia when knocked down. Binds directly to ARMC9, forming a protein module that localizes to microtubule-like structures at the centriole and proximal portion of the cilium. The interaction with ARMC9 mapped to the N-terminal protein of TOGARAM1 (PMID: 32453716; PMID: 32747439).	PMID: 32453716; PMID: 32747439	Ciliary axoneme	x			x	x
ENSG00000197579	TOP1 binding arginine/serine rich protein(TOPORS)	TOPORS	609507	Component of the photoreceptor sensory cilium. Potential role in the regulation of primary cilia-dependent photoreceptor development and function. Knockdown leads to increased cell death in the retina of zebrafish (PMID: 21159800). Mutations have been associated with autosomal dominant retinitis pigmentosa (PMID: 17924349, PMID: 18509552).	PMID: 21159800, PMID: 17924349, PMID: 18509552	Basal Body of connecting cilium, Centrosome, nucleus.	x		x		x

ENSG00000078900	tumor protein p73(TP73)	TP73		Also called P73. Might be critical for the initiation of cilia assembly by indirectly promoting the multiplication of centrosomes (PMID: 25201258). Transcription factor critical to multiciliogenesis. Required for multiciliated cells differentiation, function and for the direct regulation of transcriptional modulators of multiciliogenesis, including Foxj1, Rfx2 and Rfx3 (PMID: 31028178, PMID: 27257214, PMID: 26947080). Required for basal body docking, axonemal extension and motility (PMID: 27257214).	PMID: 25201258, PMID: 31028178, PMID: 26947080, PMID: 27257214	Golgi, mitochondria, cytoplasm, nucleus		x			
ENSG00000179636	tubulin polymerization promoting protein family member 2(TPPP2)	TPPP2		IFT transport and signalling defects after knockdown, ciliary frequency not affected. Candidate cilia gene.	PMID: 21289087	Cytoplasm, microtubules		x	x		x
ENSG00000163870	transmembrane protein adipocyte associated 1(TPRA1)	TPRA1		Also known as GPR175. Positively regulates Hh signaling via inhibiting PKA activity by lowering cAMP levels. Its ciliary localisation and activity is Hh-dependent (PMID: 26451044).	PMID: 26451044	Cilium	x				

ENSG00000204104	TRAF3 interacting protein 1 (TRAF3IP1)	TRAF3IP1	607380	Also known as IFT54. Evolutionarily conserved role in ciliogenesis, interacts with tubulin and actin. Mutations lead to altered cilia mediated hedgehog signaling (PMID: 21945076). Plays a critical role in IFT, as well as building and maintaining functional cilia (PMID: 18369462). IFT-B subunit which regulates ciliary composition. Inactivation is embryonic lethal and causes characteristic ciliopathy phenotypes (PMID: 26487268). Interacts with and stabilises IFT20 and Rabaptin5, which in turn interacts with Rab8, suggesting a potential role for IFT in regulating the assembly of ciliary membranes. It may also participate in tubulin transport for ciliogenesis. Loss of IFT54 impairs cilia assembly as well as cytoplasmic MT dynamics (PMID: 28417161). Interacts with both kinesin-II and IFT dynein and regulates anterograde IFT (PMID: 33368450).	PMID: 21945076, PMID: 26487268, PMID: 18369462, PMID: 28417161, PMID: 33368450	Basal Body, Cilium, transition zone, Cytoplasm, Cytoskeleton	x		x		x
ENSG00000160218	trafficking protein particle complex 10 (TRAPPC10)	TRAPPC10		Component of a mammalian TRAPPII complex. Required for Rabin8 centrosomal targeting prior to ciliogenesis. Knockdown impairs cilia formation by inhibiting Rabin8 recruitment to the centrosome (PMID: 21273506).	PMID: 21273506	Basal Body, Centrosome	x		x		x

ENSG00000054116	trafficking protein particle complex 3(TRAPPC3)	TRAPPC3		Component of a mammalian TRAPPII complex. Required for Rabin8 centrosomal targeting prior to ciliogenesis. Knockdown impares cilia formation by inhibiting Rabin8 recruitment to the centrosome (PMID: 21273506). Candidate gene for BBS (PMID: 27894351).	PMID: 21273506, PMID: 27894351	Basal Body, Centrosome	x		x		x
ENSG00000167632	trafficking protein particle complex 9(TRAPPC9)	TRAPPC9	611966	Component of a mammalian TRAPPII complex. Required for Rabin8 centrosomal targeting prior to ciliogenesis. Knockdown impares cilia formation by inhibiting Rabin8 recruitment to the centrosome (PMID: 21273506).	PMID: 21273506	Basal Body, Centrosome	x		x		x
ENSG00000119401	tripartite motif containing 32(TRIM32)	TRIM32	602290	Inactivating homozygous mutations cause ciliopathy BBS11 (PMID: 16606853). Ubiquitinates NPHP5 which triggers protein delocalization and loss of cilia. Loss of BBS11 causes an accumulation of NPHP5 at the centrosome (PMID: 28498859).	PMID: 16606853, PMID: 28498859	Centrosome	x		x		x
ENSG00000100815	thyroid hormone receptor interactor 11(TRIP11)	TRIP11		Anchors IFT20 to the Golgi complex. Interacts with IFT20 and to regulate sorting and/or trafficking of proteins from Golgi to the ciliary membrane. Required for efficient targeting of polycystin-2 to the cilium. Cilia in mutant cells are shorter (PMID: 19112494)	PMID: 19112494	Golgi complex		x	x		x

ENSG00000111199	transient receptor potential cation channel subfamily V member 4(TRPV4)	TRPV4		Mediates Ca ²⁺ influx in response to mechanical and osmotic stimuli in ciliated epithelia, which in turns regulates the ciliary beat frequency (PMID: 18719094). Mediates oscillatory fluid shear mechanotransduction in various cells via the primary cilium (PMID: 29491434).	PMID: 18719094, PMID: 29491434	Cilium, base of cilium	x				x
ENSG00000196367	transformation/transcription domain associated protein(TRRAP)	TRRAP		Central regulator of multiciliated cells formation. Acts downstream of Notch 2 and upstream Multicilin in regulation of ciliated cell formation; binds to the promoters and regulates the expression of genes involved in multiciliated cell differentiation and function, including several genes associated with human ciliopathies like PCD. Required for FOXJ1 expression. Knockdown of TRRAP decreased the number of ciliated cells, and knockout mice are not viable (PMID: 29588376).	PMID: 29588376	Golgi, nucleus		x			
ENSG00000165699	tuberous sclerosis 1(TSC1)	TSC1		Regulates the length of cilia. Form a complex with TSC2 that negatively regulates the nTOR pathway and, in turn, cell growth, proliferation and survival (PMID: 19008302). Mutations have been associated with Tuberous sclerosis complex and polycystic kidney disease. Loss of Tsc1 or Tsc2 enhances ciliary formation and elongation (PMID: 18845692). Required for activation of canonical TGF- β signaling. Regulates Smoothened (SMO)-dependent HH signaling in cilia (PMID: 29396625). Role in maintaining cell polarity (PMID: 19321600).	PMID: 19008302, PMID: 18845692, PMID: 29396625, PMID: 19321600	Golgi, basal body.	x			x	

ENSG00000103197	tuberous sclerosis 2(TSC2)	TSC2		Form a complex with TSC1 that negatively regulates the TOR pathway and, in turn, cell growth, proliferation and survival (PMID: 19008302).Mutations have been associated with Tuberous sclerosis complex and polycystic kidney disease. Loss of Tsc1 or Tsc2 enhances ciliary formation and elongation (PMID: 18845692). Regulates Smoothened (SMO)-dependent HH signaling in cilia (PMID: 29396625). Role in maintaining cell polarity. Associates with the GSK3/axin complex to inhibit canonical Wnt-signalling (PMID: 19321600). Required for the normal trafficking of PC-1 from the Golgi to the membrane (PMID: 20169078).	PMID: 19008302, PMID: 18845692, PMID: 29396625, PMID: 19321600, PMID: 20169078	Golgi, membrane, nucleus	x				
ENSG00000135951	testis specific 10(TSGA10)	TSGA10	607166	Role in spermatogenesis. Possible role in the structure of primary cilium, as it was found to colocalise with γ -tubulin in COS-1 cells (PMID: 16643851).	PMID: 16643851	Perinuclear region, Centriole, cytoplasm, sperm tail	x				x
ENSG00000128881	tau tubulin kinase 2(TTBK2)	TTBK2	611695	Regulator of the initiation of ciliogenesis in vivo. Ttbk2 mutants lack cilia. Promotes the removal of CP110 from the mother centriole, the recruitment and retention of IFT proteins at the distal end of basal body, to initiate the assembly of the ciliary axoneme (PMID: 23141541). Mutations have been associated with dominant human neurodegenerative disorder Spinocerebellar Ataxia type 11 (PMID: 18037885). Interacts in a	PMID: 23141541, PMID: 18037885, PMID: 24982133, PMID: 25297623, PMID: 31455668, PMID: 30532139	Distal end of mother centriole, transition zone	x		x		x

				complex with CP164, which is required for its ciliary localisation. Promotes assembly of distal appendages at both centrioles and Cep83 localisation (PMID: 24982133). Phosphorylates Cep164, Cep97 and inhibits the interaction between Cep164 and Dvl3 (PMID: 25297623). Phosphorylated CEP83, which is required for ciliary vesicle docking and CP110 removal (PMID: 31455668). Remains at the transition zone of the cilium following ciliogenesis, to regulate cilium stability (PMID: 30532139).							
ENSG00000149292	tetratricopeptide repeat domain 12(TTC12)	TTC12		Centrosome component (PMID: 23300604). Involved in the assembly and/or transport of axonemal dynein arms. Loss of function affects both ODA and IDA. In paramecia, knockdown led to the absence of dynein arms (PMID: 31978331).	PMID: 23300604, PMID: 31978331	Centrosome, cytoplasm.	x		x		x
ENSG00000052841	tetratricopeptide repeat domain 17(TTC17)	TTC17		Interacts with C2orf62 and regulates ciliogenesis and actin polymerisation in ciliated cells. Depletion reduces the number of primary cilia but does not alter their length (PMID: 24475127).	PMID: 24475127	Actin cytoskeleton, cytosol and plasma membrane.		x			

ENSG00000123607	tetratricopeptide repeat domain 21B(TTC21B)	TTC21B	612014	Encodes a retrograde intraflagellar transport-A complex, associates with the motor protein dynein2 to regulate retrograde trafficking in the primary cilium. Knockdown induces microtubule anchoring defects (PMID: 24876116). Mutated in nephronophthisis, syndromic Jeune Asphyxiating Thoracic Dystrophy (PMID: 21258341), Focal Segmental glomerulosclerosis (PMID: 24876116) and Joubert Syndrome (PMID: 29615573).	PMID: 21258341, PMID: 24876116, PMID: 24876116, PMID: 29615573	Axoneme, base of primary cilium.	x		x		x
ENSG00000103852	tetratricopeptide repeat domain 23(TTC23)	TTC23		Mutated in OFD. TTC23 localizes to the shaft of the cilia in controls but is restricted mainly to the base of the cilia in patient fibroblasts. Patient fibroblasts show normal ciliogenesis but increased length of cilia compared with control (PMID: 32055034). Affinity purification and mass spectrometry show TTC23 to strongly associate with IQCE and EFCAB7, components of the Ellis-van Creveld (EVC) zone (PMID: 29459677)	PMID: 32055034; PMID: 29459677	Along cilium	x				
ENSG00000204815	tetratricopeptide repeat domain 25(TTC25)	TTC25	617095	Member of the ODA-docking complex in humans and mice. Plays an essential role in the correct function of motile cilia and is therefore important for the development of left-right patterning. Mutations result in PCD and randomisation of left-right body asymmetry. TTC25 deficiency in mice leads to absence of ODA (PMID: 27486780). May interact with CFAP53 to facilitate ODA proteins transport into the axoneme and regulate its stable docking with the axonemal microtubules (PMID: 33347437). TTC25 morphant cells show fewer and	PMID: 27486780, PMID: 33347437, PMID: 17961536	Base of cilium, axoneme.	x			x	

				extremely shorter cilia. (PMID: 17961536).							
ENSG00000105948	tetratricopeptide repeat domain 26(TTC26)	TTC26		Also known as IFT56. Regulates normal levels and distribution of IFTB proteins in the cilia. Required for the accumulation of Gli2 and Gli3 at the distal tips of cilia; essential for normal microtubule architecture within the ciliary axoneme and trafficking of Shh signaling proteins. IFT56 associates with motility factors, function results in impaired motility (PMID: 28264835). Required for normal development and differentiation in retina and pronephros (PMID: 22718903). Role in transporting ciliary cargoes related to ciliary motility and for assembling cilia of full length (PMID: 24596149). Forms a dimer with IFT46, responsible for the interaction of IFT-B with ARL13B (PMID: 27927754).	PMID: 28264835, PMID: 22718903, PMID: 24596149, PMID: 27927754	Basal Body, Cilia/flagella, transition zone of primary and photoreceptor cilia.	x		x		x
ENSG00000137473	tetratricopeptide repeat domain 29(TTC29)	TTC29		Also known as NYD-SP14. Strongly expressed in tissues with motile cilia and flagella. Functions in cilia and flagella, most likely as an inner-arm dynein subunit. Chlamydomonas homologue encodes 44kDa subunit of axonemal dynein (PMID: 17981992). Involved in functional interaction with the IFT machinery, participating in anterograde IFT. Mutations have been associated with multiple morphological	PMID: 17981992, PMID: 31735294	Axoneme	x		x		x

				abnormalities of the flagella (PMID: 31735294).								
ENSG00000197557	tetratricopeptide repeat domain 30A(TTC30A)	TTC30A		Also known as IFT70A. Component of the IFT-B complex. Crucial for interaction with IFT52–IFT88 dimer in the IFT-B complex, which is required for ciliogenesis. Knockout cells do not form cilia. (PMID: 296541164). Required for flagella assembly. Depletion results in the formation of short flagella (PMID: 20534810).	PMID: 29654116, PMID: 2053481	Cilium, flagellum, IFT	x		x			x
ENSG00000196659	tetratricopeptide repeat domain 30B(TTC30B)	TTC30B		Also known as IFT70B. Component of the IFT-B complex. Redundant roles with IFT70A. Interacts with the IFT52–IFT88 dimer in the IFT-B1 subcomplex, required for ciliogenesis. Knockout cells do not form cilia. (PMID: 296541164). Required for flagella assembly. Depletion results in the formation of short flagella (PMID: 20534810).	PMID: 29654116, PMID: 2053481	Cilium, flagellum, IFT	x		x			x
ENSG00000139865	tetratricopeptide repeat domain 6(TTC6)	TTC6		Mutated in MKS. TTC6 localizes to the centrosome and ciliary shaft in controls but only to the centrosome in patient fibroblasts (PMID: 32055034)	PMID: 32055034	Along cilium	x					

ENSG00000165533	tetratricopeptide repeat domain 8(TTC8)	TTC8	608132	Also known as BBS8. Component of the BBSome complex, interacts with PCM1. Mutations leads to Bardet-Biedl syndrome with randomization of left-right body axis symmetry (PMID: 14520415). Role in intraciliary/flagellar transport, coordinating the velocity of the kinesin molecular motors during anterograde transport. Interacts with Vangl2 to establish and maintain of vertebrate left-right asymmetry. It may also be responsible for maintaining normal cilium length in the KV but not cilia number (PMID: 20643117). Required for the development, maintenance and survival of photoreceptors and outer segments in both rods and cones (PMID: 29126234). Knockdown reduces ciliary length and increases levels of HDAC6 (PMID: 30446775).	PMID: 14520415, PMID: 20643117, PMID: 29126234, PMID: 30446775	Centrosome, basal body, connecting cilium	x		x		x
ENSG00000112742	TTK protein kinase(TTK)	TTK		Also known as MPS1. Negative regulator of ciliogenesis. Interacts with VDAC3 at the centrosome to promote ciliary disassembly during cell cycle entry and proliferation. Essential for the recruitment of Centrin 2 to the procentrioles (PMID: 23388454).	PMID: 23388454	Centrosome	x		x		x
ENSG00000100271	tubulin tyrosine ligase like 1(TTLL1)	TTLL1		Involved in the post-translational polyglutamylation of tubulin in axonemal microtubules within cilia and flagella; this glutamylation is required to generate a curved cilia waveform. Essential for normal motility of respiratory ciliary and for the biogenesis and integrity of sperm flagella, as PCD phenotype is seen in KO mice (PMID: 20442420).	PMID: 20442420	Cytoplasm, microtubules		x			x

ENSG00000214021	tubulin tyrosine ligase like 3(TTLL3)	TTLL3		Functions in posttranslational glycylation of microtubules. Required for either elongation or stability of axonemes (PMID: 19531357). Depletion leads to shortening and loss of axonemes (PMID: 21262966). Required for primary cilia formation and control of cell proliferation. Only glycyase expressed in colon (PMID: 25180231).	PMID: 19531357, PMID: 21262966, PMID: 25180231	Cilium	x		x		x
ENSG00000119685	tubulin tyrosine ligase like 5(TTLL5)	TTLL5	612268	Recruited by ARL13B and FIP5 (PMID: 30120249). Higher activity initiating glutamate chains than elongating them. Required for RPGRORF15 function as it seems to be highly selective (PMID: 27162334). Essential for the correct function of sperm flagella. It might also be responsible for the tubulin polyglutamination in the microtubule triplets of the centrioles, increasing the centriole stability. Mutations have been identified as cause of recessive retinal dystrophy (PMID: 24791901), and reduced sperm motility and infertility in humans (PMID: 28173158).	PMID: 30120249, PMID: 24791901, PMID: 27162334, PMID: 28173158	Centrosome, base of connecting cilium.	x				x
ENSG00000170703	tubulin tyrosine ligase like 6(TTLL6)	TTLL6		Main axonemal tubulin glutamylase of ciliated cells. Knockdown caused loss of axonemal glutamylation but not basal body glutamylation (PMID: 21262966). Ciliary entry and distribution of TTLL6 is regulated by CEP41 (PMID: 22246503). Critical in the regulation of axonemal motility and ciliary beating in ependymal cells, as it cannot be replaced by any of the other glutamylases (PMID: 23897886).	PMID: 21262966, PMID: 22246503, PMID: 23897886	Cilium, basal body	x		x		x

ENSG00000138892	tubulin tyrosine ligase like 8(TTLL8)	TTLL8		It may function in initiating glycylation modification by attaching the first glycine to tubulin. Interacts with TTLL10 and is required for TTLL10 efficient tubulin glycylation function (PMID: 19427864). Redundant function with TTLL3, as both are unique monoglycylation enzymes in mammals. Required for stability and maintenance of motile ependymal cilia. Absence of TTLL3 leads to absence of glycylation on axonemes, destabilising motile ependymal cilia and resulting in ciliary disassembly (PMID: 23897886, PMID: 25180231).	PMID: 19427864, PMID: 23897886, PMID: 25180231	Cilium	x					x
ENSG00000131044	tubulin tyrosine ligase like 9(TTLL9)	TTLL9		α -tubulin glutamylase required for the elongation of the polyglutamate side chains, and for the normal assembly and function of cilia (PMID: 18586949, PMID: 20188560).	PMID: 18586949, PMID: 20188560	Cilium	x		x			x
ENSG00000166402	tubby bipartite transcription factor(TUB)	TUB	601197	Mutations have been associated with retinal and cochlear degeneration, as well as obesity (PMID: 24375934). Function in ciliary GPCR trafficking in a similar way to TULP3 (PMID: 28154160).	PMID: 24375934; PMID: 28154160	Base of the photoreceptor cilium, ciliary rootlet, cilium	x					x

ENSG00000167552	tubulin alpha 1a(TUBA1A)	TUBA1A		Alfa tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000167553	tubulin alpha 1c(TUBA1C)	TUBA1C		Alfa tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000127824	tubulin alpha 4a(TUBA4A)	TUBA4A		Alfa tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000137267	tubulin beta 2A class IIa(TUBB2A)	TUBB2A		Beta tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x		x		x

ENSG00000137285	tubulin beta 2B class IIb(TUBB2B)	TUBB2B		Beta tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000258947	tubulin beta 3 class III(TUBB3)	TUBB3		Beta tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule.	PMID: 11516662	Basal Body, Centriole, Cytosol	x				
ENSG00000188229	tubulin beta 4B class IVb(TUBB4B)	TUBB4B	602660	Beta tubulin isoform. Component of the cytoskeleton, centrosomes and protofilaments, structures that associate laterally to form a microtubule. Mutations in TUBB4B Cause a Distinctive Sensorineural Disease of LCA with hearing loss	PMID: 11516662, PMID: 29198720	Microtubules		x		x	x
ENSG00000074935	tubulin epsilon 1(TUBE1)	TUBE1		Interacts with delta-tubulin to form the triplet microtubules or to stabilize them against depolymerization (PMID: 28906251). Function in the centriole lumen and on the cytoplasmic side is essential for the centriole-to-centrosome conversion. Depletion of TUBE1 affects microtubules biogenesis and reduces CEP44 recruitment to centrioles (PMID: 11148142).	PMID: 28906251, PMID: 11148142	Centriole	x		x		x

ENSG00000130640	tubulin gamma complex associated protein 2(TUBGCP2)	TUBGCP2	Also known as GCP2. Gamma tubulin complex protein. These type of complexes of γ -tubulin are used as templates for the longitudinal association with α/β -tubulin dimers during the nucleation process (PMID: 29514869). Component of γ -tubulin small complexes that then associate with associate with GCP4, GCP5 and GCP6 to form ring complexes (PMID: 21725292).	PMID: 29514869, PMID: 21725292	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000126216	tubulin gamma complex associated protein 3(TUBGCP3)	TUBGCP3	Also known as GCP3. Gamma tubulin complex protein. These type of complexes of γ -tubulin are used as templates for the longitudinal association with α/β -tubulin dimers during the nucleation process (PMID: 29514869). Component of γ -tubulin small complexes that then associate with associate with GCP4, GCP5 and GCP6 to form ring complexes (PMID: 21725292).	PMID: 29514869, PMID: 21725292	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000137822	tubulin gamma complex associated protein 4(TUBGCP4)	TUBGCP4	Also known as GCP4. Required for the initial assembly of γ -tubulin ring complexes and for their stabilization after assembly. These complexes are used as templates for the longitudinal association with α/β -tubulin dimers during the nucleation process (PMID: 29514869, PMID: 32317396). Associates with γ -tubulin small complexes, GCP5 and GCP6 (PMID: 21725292). Mutations have been identified in patients with autosomal-recessive microcephaly and chorioretinopathy. Role in maintenance of retinal homeostasis, through participating in assembly of γ -TuRC (PMID: 31209365).	PMID: 29514869, PMID: 21725292, PMID: 31209365, PMID: 32317396	Basal Body, Centriole, Cytosol	x		x		x

ENSG00000275835	tubulin gamma complex associated protein 5(TUBGCP5)	TUBGCP5		Also known as GCP5. Required for the initial assembly of γ -tubulin ring complexes and for their stabilization after assembly. These complexes are used as templates for the longitudinal association with α/β -tubulin dimers during the nucleation process (PMID: 29514869, PMID: 32317396). Associates with γ -tubulin small complexes, GCP4 and GCP6 (PMID: 21725292).	PMID: 29514869, PMID: 32317396, PMID: 21725292	Basal Body, Centriole, Cytosol	x				
ENSG00000128159	tubulin gamma complex associated protein 6(TUBGCP6)	TUBGCP6		Also known as GCP6. Required for the initial assembly of γ -tubulin ring complexes and for their stabilization after assembly. These complexes are used as templates for the longitudinal association with α/β -tubulin dimers during the nucleation process (PMID: 29514869, PMID: 32317396). Associates with γ -tubulin small complexes, GCP4 and GCP5 (PMID: 21725292).	PMID: 29514869, PMID: 32317396, PMID: 21725292	Basal Body, Centriole, Cytosol	x		x		x
ENSG00000112041	tubby like protein 1(TULP1)	TULP1	602280	Role in intracellular transport, cell differentiation, signaling, and motility. Mutations result in progressive photoreceptor degeneration in mice and retinitis pigmentosa in humans (PMID: 33187986, PMID: 25074776, PMID: 9462750).	PMID: 33187986, PMID: 25074776, PMID: 9462750	Plasma Membrane, inner segment of photoreceptors,		x	x		x
ENSG00000078246	tubby like protein 3(TULP3)	TULP3		Role in intracellular transport, cell differentiation, signaling, and motility (PMID: 33187986). Associates with IFT-A in a complex to regulate trafficking and localisation of G protein-coupled receptors into cilia. Negative regulator of Hedgehog signaling (PMID: 20889716). Adapter for ciliary trafficking of membrane cargo; regulates PC1/2 complex trafficking (PMID:	PMID: 33187986, PMID: 20889716, PMID: 28154160, PMID: 19286674, PMID: 30799240, PMID: 30583862	Nucleus, cytoplasm, plasma membrane, ciliary tip	x		x		x

				28154160). Required for proper Shh-dependent limb patterning, as it inhibits Gli2 activity (PMID: 19286674). Essential for trafficking of Arl13b and INPP5E into cilia. Loss of function causes ciliopathy-associated polycystic kidney disease, downregulates the formation of cilia and their length (PMID: 30799240, PMID: 30583862).							
ENSG00000113621	thioredoxin domain containing 15(TXNDC15)	TXNDC15		Mutated in patients with Meckel-Gruber syndrome (PMID: 27894351, PMID: 30851085, PMID: 31411728). Loss of TXNDC15 prevented correct localization of the TMEM67 ciliary receptor to the transition zone (PMID: 27894351).	PMID: 27894351; PMID: 30851085; PMID: 31411728	Not reported					
ENSG00000104517	ubiquitin protein ligase E3 component n-recogin 5(UBR5)	UBR5		Positive regulator of ciliogenesis. Ubiquitylates CSPP1 and is required for cytoplasmic organisation of CSPP1-associated centriolar satellites in centrosomal periphery. Regulates ciliogenesis via maintenance of centriolar satellite stability and CSPP1 localisation (PMID: 29742019).	PMID: 29742019	Centrosome, centriolar satellites	x				
ENSG00000162543	UBX domain protein 10(UBXN10)	UBXN10		Facilitates recruitment of VCP to IFT-B. Interaction with VCP is required for UBXN10 ciliary localisation. Depletion of UBXN10 in zebrafish causes defects in left-right asymmetry (PMID: 26389662).	PMID: 26389662	Cilia	x				x

ENSG00000168038	unc-51 like kinase 4(ULK4)	ULK4		Hydrocephalus due to dysfunctional motile cilia in KO mouse, suggesting it may be involved in the development and/or function of motile cilia in the respiratory tract and ependyma (PMID: 21746835). Essential for Ciliogenesis and CSF Flow. Modulates expression of the master regulator of ciliogenesis, Foxj1, its target genes and other ciliogenesis molecules, influencing axoneme structure, basal body alignment, cilia number and beating. Implicated in ciliopathy-related disorders (PMID: 27445138).	PMID: 21746835; PMID: 27445138	Not reported			x		x
ENSG00000169344	uromodulin(UMOD)	UMOD	191845	Mutations are responsible for medullary cystic kidney disease type 2 (PMID: 15029248), familial juvenile hyperuricemic nephropathy and glomerulocystic kidney disease. Expressed in renal primary cilia. UMOD mutations result in decreased ciliary uromodulin expression and delayed ER to Golgi trafficking (PMID: 20172860).	PMID: 15029248, PMID: 20172860	Base of cilium and along the entire length of the cilium in renal cells.	x				x
ENSG00000006611	USH1 protein network component harmonin(USH1C)	USH1C	605242	Interacts with DOCK4 to regulate actin cytoskeleton organisation in stereocilia. Mutation are associated with Usher syndrome type 1c (PMID: 16464467). Binds to myosin VIIa, cadherin 23 and F-actin (PMID: 12588794).	PMID: 16464467, PMID: 12588794	Stereocilium	x		x		x

ENSG00000182040	USH1 protein network component sans(USH1G)	USH1G	607696	Ciliopathy gene OMIM: 607696. Mutation are associated with Usher syndrome type 1g. Associates with harmonin and is involved in the functional network formed through it (PMID: 12588794). Required for outer segment differentiation, formation and maintenance of retina and photoreceptor cell polarity. Scaffold protein in the USH network during ciliogenesis, as dysfunction leads to retinal degeneration (PMID: 17923142). Form a complex with myomegalin, cooperating in the vesicle transport towards the ciliary base of photoreceptor cells (PMID: 21767579). Interacts directly with IFT-B proteins, IFT20, IFT52, IFT57, and IFT74; ISHG1 may control the transport of these molecules to the ciliary base (PMID: 31637240).	PMID: 12588794, PMID: 17923142, PMID: 21767579, PMID: 31637240	Plasma Membrane, Centrosome, basal body, connecting cilium of photoreceptor cells.	x		x		x
ENSG00000042781	usherin(USH2A)	USH2A	608400	MIM: 608400 (PMID: 10738000) Required for the long-term maintenance of retinal photoreceptors (PMID: 17360538). Interacts and colocalises with Nlp and lebercilin at the centrosome (PMID: 18826961). Interacts with SPAG5 and NINL to regulate microtubule-based cytoplasmic trafficking of proteins required for ciliogenesis, maintenance and/or function, to the apical inner segment and basal body of photoreceptor. Mutations are associated with non-syndromic retinitis pigmentosa and Usher syndrome (PMID: 23351521).	PMID: 10738000, PMID: 17360538, PMID: 18826961, PMID: 23351521	Cilium, Centrosome, connecting cilium, stereocilia, ciliary membrane,	x		x		x

ENSG00000101557	ubiquitin specific peptidase 14(USP14)	USP14		Deubiquitinating enzyme. Regulator of the ubiquitin proteasome system that controls ciliogenesis and cilium elongation; it positively affects Hh signalling through the degradation of Kif7. Controls Gli2 ciliary localisation (PMID: 30388222).	PMID: 30388222	Cytosol, plasma membrane		x		x	
ENSG00000138592	ubiquitin specific peptidase 8(USP8)	USP8		Binds to, stabilises and deubiquitinates HIF1 α to control ciliogenesis, maintaining a basal expression of HIF1 α (PMID: 24378640). Recruited by SUMOylated Smo to antagonise Smo ubiquitination and degradation, leading to increased Hh pathway activity (PMID: 27746045). USP8 phosphorylation by EGFR is essential for suppression of ciliogenesis, indicating that mutations or lack of USP8 function may cause ciliopathies through elongation of primary cilia (PMID: 29472535).	PMID: 24378640; PMID: 27746045, PMID: 29472535	Cytosol, early endosome, plasma membrane, midbody		x			
ENSG00000124486	ubiquitin specific peptidase 9, X-linked(USP9X)	USP9X	300072	Deubiquitinating enzyme localised to centrosome, involved in context-dependent ciliogenesis. Regulates level of ubiquitination of NPHP5 to control ciliogenesis. In the G0/G1/S phase, USP9X is recruited to the centrosome by NPHP5, and later protects NPHP5 from ubiquitination, promoting cilia assembly. In the G2/M phase, USP9X dissociates from the centrosome allowing BBS11 to K63 ubiquitinate NPHP5, triggering loss of cilia. Works with two E3 ubiquitin ligases BBS11/TRIM32 and MARCHF7 (PMID: 28498859). Binds and stabilises PCM1 to prevent its degradation and regulate	PMID: 28498859, PMID: 30584065, PMID: 31671755	Cytosol, golgi complex, centrosome, centriolar satellite, cilia.		x			

				centriolar satellite integrity. Regulates centrosome duplication (PMID: 30584065). During Serum-Starvation-Induced Ciliogenesis, it is recruited by SNX17 to antagonize MIB1-Mediated Ubiquitination and Degradation of PCM1 (PMID: 31671755).							
ENSG00000162738	VANGL planar cell polarity protein 2(VANGL2)	VANGL2	600533	Planar cell polarity protein, required for Kupffer's vesicle function and establishment of asymmetry. Interacts with BBS8 to establish and maintain the left-right asymmetry during early embryogenesis in zebrafish. Required for nuclear positioning, centrosome migration and regulation of the number and length of cilia (PMID: 20643117). Cooperates with cofilin to target Rab11+ vesicles containing planar planar cell polarity proteins to the apical membrane (PMID: 23406901). Interacts with Arl13b to regulate cilia and photoreceptor outer segment length in Zebrafish. Mutations cause neural tube defects (PMID: 27571019).	PMID: 20643117, PMID: 23406901, PMID: 27571019	COPII-coated ER to Golgi vesicles, basal body, proximal ciliary axoneme.	x				x
ENSG00000165280	valosin containing protein(VCP)	VCP		Regulator of protein quality control and ubiquitin-mediated signaling. Interacts with UBXN10 to control its localisation to cilia. VCP and UBXN10 interaction is also required for ciliogenesis and maintenance of cilia (PMID: 26389662).	PMID: 26389662	Cytosol, ER, cilia.	x				

ENSG00000078668	voltage dependent anion channel 3(VDAC3)	VDAC3		Required for the maintenance of axonemes of sperm; absence of VDAC3 in the mouse leads to defective axonemal structures in sperm and immotility (PMID: 11507092). Modulates centriole assembly by recruits Mps1 to centrosomes. VDAC3-Mps1 interaction promotes ciliary disassembly during cell cycle entry and prevents cilia assembly in proliferating cells (PMID: 23388454, PMID: 26264029).	PMID: 11507092, PMID: 23388454, PMID: 26264029	Mitochondria, centrosome (mother centriole).	x		x		x
ENSG00000134086	von Hippel-Lindau tumor suppressor(VHL)	VHL	608537	MIM: 193300, MIM: 144700, MIM: 171300, MIM: 263400; Somatic inactivation of VHL causes renal and pancreatic cysts associated with cilia loss. Stabilises microtubule dynamics in the axoneme (PMID: 17450132). Interacts with KIF3A and kinesin-2 to bind microtubules. Mutations cause von Hippel-Lindau disease (PMID: 17825299). Depletion or reduction of VHL leads to fewer ciliated cells, which suggests a role in ciliary formation and/or maintenance (PMID: 17912253, PMID: 16775032). Influences epithelial cell polarity and induces primary cilium formation in renal cells (PMID: 16849532). Required for oriented growth of microtubules toward the cell periphery, which is critical for ciliogenesis (PMID: 17101696). Phosphorylated by Nek1, which promotes its degradation to regulate ciliary homeostasis (PMID: 23255108).	PMID: 17450132, PMID: 17825299, PMID: 17912253, PMID: 16775032, PMID: 16849532, PMID: 17101696, PMID: 23255108	Cilium, cytosol.	x		x		x

ENSG00000136631	vacuolar protein sorting 45 homolog(VPS45)	VPS45		Regulates cilium length and morphology, ciliary and preciliary membrane volume. Interacts with RABS-5 to control periciliary vesicle number, levels of polycystin-2, fusion and maturation of specific cilium-derived endocytic vesicles and early endosomes (PMID: 29572244).	PMID: 29572244	Early endosome		x				
ENSG00000132612	vacuolar protein sorting 4 homolog A(VPS4A)	VPS4A		Dynamic component of both centrioles. Regulates centrosome localisation and levels of γ -tubulin, centriolar satellite stability and ciliogenesis. Essential for activation of membrane remodeling during cilium extension. Depletion alters centrosome numbers and size (PMID: 29463826). De Novo VPS4A Mutations Cause Multisystem Disease with Abnormal Neurodevelopment, with mutations affecting Mutations affected other ESCRT-dependent cellular processes, including regulation of centrosome number, primary cilium morphology, nuclear membrane morphology, chromosome segregation, mitotic spindle formation, and cell cycle progression (PMID: 33186545)	PMID: 29463826; PMID: 33186545	Centrosome, transition zone		x				
ENSG00000132970	WASP family member 3(WASF3)	WASF3		Arp2/3 complex activator. PCARE and WASF3 regulate ciliary F-actin assembly that is required for the initiation of photoreceptor outer segment disk formation (PMID: 32312818)	PMID: 32312818	Actin		x				

ENSG00000143951	WD repeat containing planar cell polarity effector(WDPCP)	WDPCP	613580	MIM: 613580. Also known as C2ORF86, BBS15 or FRITZ. Regulates planar cell polarity through the modulation of the actin cytoskeleton. Required for recruitment of proteins essential for ciliogenesis to the transition zone, including Sept2, Nphp1, and Mks1, and facilitates the assembly of multiprotein complexes. Mutations have been associated with Bardet-Biedl and and Meckel-Gruber syndromes (PMID: 24302887; PMID: 32055034). Knockdown leads to fewer and shorter cilia and decreases the expression of Septin7 in human nasal epithelium, indication that Septin7 could be a target of WDPCP (PMID: 28001338). Regulates cilia function and beating through activation of MAPK/ERK pathway in the mitochondria of sinonasal epithelial cells (PMID: 33598458).	PMID: 24302887, PMID: 28001338, PMID: 33598458; PMID: 32055034	Cytosol, axoneme, transition zone, basal body.	x		x		x
ENSG00000120008	WD repeat domain 11(WDR11)	WDR11	606417	Modulates the Hedgehog (Hh) signalling pathway and is essential for ciliogenesis. It may be required for cilia extension and growth. Interacts with ciliary components involved in human ciliopathies such as BBSomes and IFT proteins. Shuttles between cilium and nucleus in response to Shh signalling. Implicated in congenital hypogonadotropic hypogonadism (CHH) and Kallmann syndrome (KS), characterised by delayed puberty and infertility (PMID: 29263200).	PMID: 29263200	Ciliary axoneme, nucleus	x				

ENSG00000157796	WD repeat domain 19(WDR19)	WDR19	608151	Also known as IFT144. Retrograde IFT-A protein. Ortholog of <i>C.elegans</i> DYF-2 (PMID: 16957054). Mutations are associated with a broad spectrum of ciliopathies (PMID: 24504730) including nephronophthisis, Jeune syndrome (PMID: 22019273), Sensenbrenner syndrome (PMID: 29134781), asphyxiating thoracic dysplasia (PMID: 23339108), retinal dystrophies and Caroli disease (PMID: 23559409). Mutations can also cause disorganization of microtubules in sperm flagella and nonsyndromic asthenoteratospermia (PMID: 32323121)	PMID: 16957054, PMID: 22019273, PMID: 24504730, PMID: 29134781, PMID: 23339108, PMID: 23559409; PMID: 32323121	Cilium, IFT	x		x		x
ENSG00000119333	WD repeat domain 34(WDR34)	WDR34	613363	Dynein intermediate chain associated with the retrograde intraflagellar transport motor dynein 2 (PMID: 25205765). Orthologous to Chlamydomonas FAP133 and Trypanosome DIC5 (PMID: 24989795). Homozygous missense mutations cause short-rib polydactyly syndrome type III/severe asphyxiating thoracic dysplasia (OMIM:; PMID: 24183449; PMID: 24183451). Acts as a TAK1-associated suppressor of the IL-1R/TLR3/TLR4-induced NF-κB activation pathway. Patient fibroblast cilia are short with a bulbous tip (PMID: 24183449). Localises around basal bodies, centrioles and along the axoneme (PMID: 24183451). Interacts with DYNLL1 and DYNLL2 (PMID: 25294941), TCTEX1D2 (PMID: 25830415). Required for retrograde IFT and Hedgehog signalling in mouse (PMID: 28379358). WDR34 KO cells can assemble a	PMID: 24183451; PMID: 24183449; PMID: 30649997; PMID: 25205765; PMID: 24183451; PMID: 28379358; PMID: 30320547; PMID: 30649997	Basal bodies, centrioles, axoneme	x				x

				dynein-2 motor complex that binds IFT proteins yet fails to extend an axoneme (PMID: 30320547). Incorporation of DYNLL1/DYNLL2 and DYNLRB1/DYNLRB2 into the dynein-2 complex via interactions with the WDR34 intermediate chain is crucial for dynein-2 function in retrograde ciliary protein trafficking (PMID: 30649997)							
ENSG00000118965	WD repeat domain 35(WDR35)	WDR35	613602	IFT121. IFT complex A component, endogenous WDR35 localizes to cilia and centrosomes throughout the developing embryo and human and mouse fibroblasts lacking the protein fail to produce cilia. Human and mouse homozygous mutations in WDR35 cause ciliopathy short-rib polydactyly syndrome (PMID: 22987818). Heterozygous mutations shown to cause cranioectodermal dysplasia-2. OMIM:613602. Interacts with RagA, mTORC1 activator (PMID: 30570184)	PMID: 20817137; PMID: 21473986; PMID: 22987818; PMID: 30570184	Along ciliary axoneme, IFT complex A	x		x		x
ENSG00000131725	WD repeat domain 44(WDR44)	WDR44		Rab11 binding protein, negatively regulates ciliogenesis. Akt phosphorylates WDR44 at Ser342/344 to stabilise it and increase its binding to Rab11, which prevents ciliogenesis. WDR44 depletion promotes binding of Rab11a to Rabin8, which leads to preciliary trafficking and ciliogenesis-initiating events at the mother centriole. Disruption of Akt signaling causes a switch from Rab11-WDR44 to the Rab11-FIP3-Rabin8 complex which promotes ciliogenesis(PMID: 31204173) Endosomal protein phosphorylated by Akt at Ser346 (PMID: 31665227)	PMID: 31204173; PMID: 31665227	Rab11-positive membrane vesicles. Endosome. Plasma membrane?		x			

ENSG00000196363	WD repeat domain 5(WDR5)	WDR5		Stabilizes actin architecture to promote multiciliated cell formation (PMID: 30205038). Regulates FOXJ1 expression to control ciliogenesis at the node and LR patterning. De novo change in patient with CHD and heterotaxy (PMID: 30377171)	PMID: 30205038; PMID: 30377171	Nucleus and base of motile cilia	x				x
ENSG00000126870	WD repeat domain 60(WDR60)	WDR60	615462	Dynein 2 intermediate chain. Biallelic truncating, frameshift or missense mutations cause autosomal recessive short-rib thoracic dysplasia type 8 with or without polydactyly (OMIM:615462) (PMID: 23910462). Localizes to the base of the cilium in human chondrocytes (PMID: 23910462), RPE1 cells (PMID: 25205765) and HEK293 cells (PMID: 25830415). Ciliogenesis is inhibited in fibroblasts from affected patients, GLI2 accumulates at basal body (PMID: 23910462). Stability of WDR34 and WDR60 is interdependent (PMID: 25205765). Localises to the centrosome in interphase, the spindle poles during mitosis and cytokinesis, and near the cytokinetic bridge during cytokinesis in HeLa cells (PMID: 25830415). Interaction with TCTEX1D2 light chain of the dynein-2 complex is crucial for ciliary protein trafficking. (PMID: 29742051).WDR60 KO cells extend axonemes but show reduced assembly of dynein-2 and binding to IFT proteins. Required to maintain a functional transition zone and for efficient bidirectional intraflagellar transport (PMID: 30320547; PMID: 30649997). Structure resolved in PMID:	PMID: 22326026; PMID: 25205765; PMID: 25830415; PMID: 29742051; PMID: 30320547; PMID: 30649997; PMID: 31451806	Base of cilium, centrosome, spindle poles, cytokinetic bridge	x		x	x	

				31451806							
ENSG00000075702	WD repeat domain 62(WDR62)	WDR62	613583	Truncating, frameshift or missense mutations cause primary autosomal recessive microcephaly type 2 (OMIM:613583). A WDR62-CEP170-KIF2A pathway promotes cilium disassembly in neural progenitors (PMID: 31197141). WDR62 ablation leads to retarded cilium disassembly, long cilium, and delayed cell cycle progression leading to decreased proliferation and premature differentiation of neural progenitor cells (PMID: 31197141). Localises to mitotic spindle poles with paralog MAPKBP1 (PMID: 28089251). Localises to the basal body. Missense mutants localise to basal body but fail to recruit CPAP and IFT88 (PMID: 31816041)	PMID: 31197141; PMID: 31816041	Mitotic spindle poles. Basal bodies	x				
ENSG00000162643	WD repeat domain 63(WDR63)	WDR63		Vertebrate orthologue of Chlamydomonas intermediate chain inner dynein arm protein IC140 which localises to Chlamydomonas flagellar axoneme (PMID: 9843573). In Chlamydomonas interacts with IC138 (WDR78 in vertebrates) and directly interacts with tubulin (PMID: 24080090). Associates with inner dynein arm subunit Wdr78 in mice (PMID: 30060180). Rare intragenic heterozygous in-frame WDR63 deletion identified as likely cause of occipital	PMID: 9843573; PMID: 24080090; PMID: 30060180; PMID: 29285825; PMID: 29285825, PMID: 32128961	Chlamydomonas flagellar axoneme	x			x	x

				encephalocele in one human fetus (PMID: 29285825). Similar mutations in zebrafish caused similar malformations (PMID: 29285825). Also interacts with Arp2/3 to regulate actin polymerisation and inhibit cell migration, invasion, and metastasis (PMID: 32128961)							
ENSG00000158023	WD repeat domain 66(WDR66)	WDR66	618146	Encodes CFAP251, suggested to play structural role during biogenesis of the spermatozoon flagellum in vertebrates.. Biallelic loss-of-function mutations in CFAP251 cause spermatogenic failure type 33 in human males (OMIM: 618146) (PMID: 30122541). Normally located along the sperm flagellum but absent in men carrying WDR66 mutations (PMID: 30122541). Component of the calmodulin- and radial-spoke- associated complex, located adjacent to DNAH1, on the inner surface of the peripheral microtubule doublets of the axoneme (PMID: 30122541). <i>Tetrahymena</i> ortholog necessary for efficient coordinated ciliary beating. (PMID: 30122541). Involved in central pair complex assembly/stability PMID: 30686508	PMID: 30122540; PMID: 30310178; PMID: 30686508	Inner surface of the peripheral microtubule doublets of the axoneme of the sperm flagellum	x			x	x
ENSG00000152763	WD repeat domain 78(WDR78)	WDR78		Identified as a putative cilia gene by proteomics and comparative genomics (PMID: 15937072), characterised as a cilia protein via siRNA knockdown of Wdr78 in 3 different mouse cell lines and high content imaging which showed perturbed receptor transport to the ciliary membrane and HH signalling (PMID: 21289087). Vertebrate orthologue of	PMID: 15937072; PMID: 21289087; PMID: 30060180	Axoneme of motile cilia	x		x		x

				Chlamydomonas inner dynein arm protein IC138. Associates with inner dynein arm subunits such as Dnah2 (a heavy chain) and Wdr63 (Chlamydomonas IC140 orthologue) in mice. Localises specifically to motile cilia in mice and knockdown leads to immotility of ependymal cilia with failure of heavy chains and intermediate chains of inner dynein arms to localise to motile cilia PMID: 30060180 . Zebrafish Wdr78 morphants have ciliopathy-type phenotypes (PMID: 30060180).							
ENSG00000161996	WD repeat domain 90(WDR90)	WDR90		POC16 in <i>Chlamydomonas</i> , localises to the central core region of the centriole in Chlamydomonas, which correlates with the region of tubulin glutamylation in humans. Recruited after the cartwheel protein CrSAS-6 and before tubulin glutamylation. Localizes to mother and daughter centrioles in human RPE1 cells and is required for ciliogenesis (PMID: 28781053)	PMID: 28781053	Mother and daughter centrioles in human RPE1 cells	x				
ENSG00000243667	WD repeat domain 92(WDR92)	WDR92		Highly conserved WD-repeat protein which acts as part of a chaperone complex involved in pre-assembly of axonemal dynein in the cytoplasm. Required for motile ciliogenesis in planarian <i>Schmidtea mediterranea</i> , knockdowns have partial loss of dynein arms (PMID: 26912790). Works with the DNAAF Spag1 at a distinct stage in dynein preassembly to regulate cytoplasmic pre-assembly of ciliary dynein in Drosophila (PMID: 29743191). Functions with the HSP90 co-chaperone R2TP-like complex and other	PMID: 26912790; PMID: 29743191; PMID: 30428028; PMID: 31116681	Cytoplasm	x				

				DNAAFs in axonemal dynein preassembly in the cytoplasm of Chlamydomonas (PMID: 30428028). Stabilises axonemal dynein heavy chains in cytoplasm in Chlamydomonas (PMID: 31116681)							
ENSG00000095397	whirlin(WHRN)	WHRN	607928	DFNB31/whirlin. PDZ scaffold protein. Mutated in autosomal recessive non-syndromic deafness type 31 (PMID: 12833159) and Usher syndrome type 2D (PMID:17171570) in humans (OMIM 607928). Only the whirlin long isoform (with an additional N-terminal PDZ domain) is expressed in the retina, the retina is affected if this extra PDZ domain is mutated (PMID: 20502675). Localises to the tip of the ssterocilium of the cochlear hair cell (PMID: 15590698). Interacts directly with USH2A and VLGR1 in the connecting cilium of photoreceptor cells (PMID: 16434480). Colocalizes with MPP1 in the retina at the outer limiting membrane, at the outer synaptic layer and at the basal bodies and the ciliary axoneme of the connecting cilium of the photoreceptors (PMID: 17584769). Interacts with L-type calcium channel subunit Ca(v)1.3 (alpha(1D) at synpase in retina (PMID: 19959638). Interacts with RPGR(ORF15) in the photoreceptors (PMID: 22323458)	PMID: 12833159; PMID: 16434480; PMID: 17584769; PMID: 19959638; PMID: 20502675; PMID: 22323458	Basal body and axoneme of photoreceptor connecting cilium. Outer limiting membrane and outer synaptic layer of retina. Tip of sterocilium of cochlear hair cell.	x		x		x

ENSG00000116213	WD repeat containing, antisense to TP73(WRAP73)	WRAP73		WDR8. Centriolar satellite and centriole-associated protein. Interacts with centriolar satellite proteins SSX2IP and PCM1 and centriolar proximal end component Cep135. Promotes ciliary vesicle docking during ciliogenesis. (PMID: 26675238).	PMID: 26675238	Centriole, centriolar satellite	x					x
ENSG00000196236	X-prolyl aminopeptidase 3(XPNPEP3)	XPNPEP3	613553	Mitochondrial protein expressed specifically in distal convoluted tubule and cortical collecting duct cells. Homozygous frameshift and splice-site mutations cause an autosomal recessive nephronophthisis-like nephropathy in humans (OMIM:613553), PMID: 20179356. targeted to mitochondria by an amino-terminal mitochondrial signal sequence. Has N-terminal proline cleavage activity. E.coli ortholog can efficiently cleave CEP290/NPHP6, ALMS1, and LRRC50 N-terminus. Cleavage of N-terminus of LRRC50 is essential for normal gastrulation in zebrafish (PMID: 20179356)	PMID: 20179356	Mitochondria (processed form), cytosol (unprocessed form). Specifically in distal convoluted tubule and cortical collecting duct cells		x	x			x
ENSG00000137693	Yes associated protein 1(YAP1)	YAP1	606608	Transcriptional co-activator with TAZ. Knockdown of YAP does not inhibit ciliogenesis(PMID: 25367221) but is sufficient to induce ciliogenesis, whereas YAP/TAZ hyperactivation suppresses serum starvation-mediated ciliogenesis (PMID: 25849865). Cytoplasmic retention of YAP/TAZ correlates with active ciliogenesis either in spatially confined cells or in cells treated with an actin filament destabilizer (PMID: 25849865). Mutations in NEK8/NPHP9 affect YAP	PMID: 25849865; PMID: 26967905	Cytosol, membrane, nucleus		x				

				nuclear localisation, with different mutations having different effects, leading to major organ developmental defects due to altered ciliogenesis and cell differentiation/proliferation (PMID: 26967905)							
ENSG00000167645	Yip1 interacting factor homolog B, membrane trafficking protein(YIF1B)	YIF1B		YIF1B mutations cause a post-natal neurodevelopmental syndrome associated with Golgi and primary cilium alterations (PMID: 33103737)	PMID: 33103737	Golgi. ER, cell membrane		x			
ENSG00000043355	Zic family member 2(ZIC2)	ZIC2	603073	Transcription factor. Directs the formation and function of node cilia to control cardiac <i>situs</i> in mice. Component of the gene regulatory network that drives ciliation of node cells during gastrulation in mice. PMID: 24585447 Contributes to establishment of LR asymmetry by regulating left-sided nodal expression in mice PMID: 29992973. Involved in dorsal neural tube formation in zebrafish and mice, downregulated in zebrafish Rfx4 mutants (PMID: 29243319). Heterozygous frameshift mutations cause holoprosencephaly in humans (OMIM: 603073).	PMID: 24585447; PMID: 29992973	Nucleus		x			

ENSG00000004838	zinc finger MYND-type containing 10(ZMYND10)	ZMYND10	603395	Recessive biallelic mutations in ZMYND10 cause a form of PCD in which patients lack inner dynein arms and outer dynein arms and lack axonemal protein components DNAH5 and DNALI1 from respiratory cilia (CILD22, OMIM:63395) (PMID: 23891469). Interacts with LRCC6, also a cause of PCD (CILD19, OMIM:614930). LRRC6 mutations abrogate the interaction between the LRRC6 CS domain and the ZMYND10 C-terminal domain (PMID: 23891469). Colocalizes with SAS6 and PCM at the centriole in rat trachea (PMID: 23891469). Stabilizes intermediate chain proteins in the cytoplasmic pre-assembly of dynein arms (PMID: 29601588). Functions in a chaperone relay during axonemal dynein assembly (PMID: 29916806)	PMID: 23891471; PMID: 23891469; PMID: 29601588; PMID: 29916806	Centrioles of motile cilia	x					x
ENSG00000102935	zinc finger protein 423(ZNF423)	ZNF423	604557	30-zinc-finger transcription factor involved in key developmental pathways (PMID:17151198). Interacts with CEP290 and colocalizes with SDCCAG8 and CEP164 at nuclear foci that are positive for the DDR signaling proteins SC35, TIP60 and Chk1 in hTERT-RPE cells (PMID: 22863007). Thought to play a role in the DNA damage response (PMID: 22863007). Patients homozygous for ZNF423 mutations manifest as a form of Joubert syndrome with infantile-onset kidney disease, cerebellar vermis hypoplasia, and situs inversus (JBTS19, OMIM:604557) (PMID: 22863007). In Zfp23 mouse mutants cell cycle progression is remarkably delayed and DDR markers are upregulated	PMID: 22863007; PMID: 17151198; PMID: 28893945	Nuclear foci which are positive for DDR proteins		x		x		x

				in cerebellar ventricular zone progenitor (PMID: 28893945)							
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